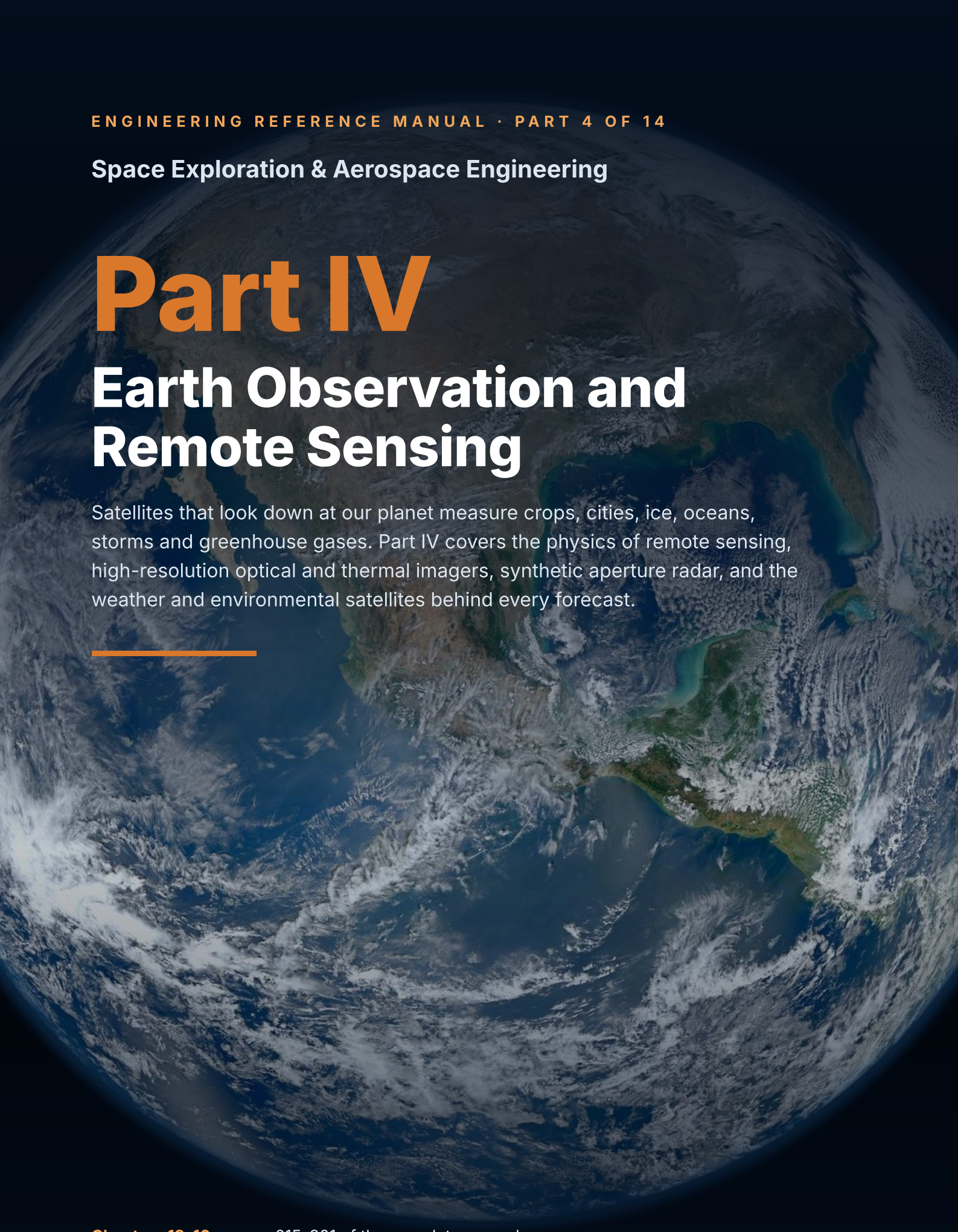
Space Exploration & Aerospace Engineering

Part IV

Earth Observation and Remote Sensing

Satellites that look down at our planet measure crops, cities, ice, oceans, storms and greenhouse gases. Part IV covers the physics of remote sensing, high-resolution optical and thermal imagers, synthetic aperture radar, and the weather and environmental satellites behind every forecast.



Chapters 13–16 · pages 215–301 of the complete manual
First Edition · Current to October 2026 · Written in plain language with deep technical detail

ENGINEERING REFERENCE MANUAL

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Version 1.0 · October 2026

About this edition

Space Exploration & Aerospace Engineering — An Engineering Reference Manual. First Edition, Version 1.0, October 2026.

This manual was written to be read by students, engineers, analysts, managers and curious readers. Each topic starts with plain-language explanations and builds up to the equations, numbers and design trade-offs that practising engineers use. Sentences are kept short and every technical term is explained in plain words when it first appears; the glossary (Appendix H) and the list of acronyms (Appendix G) collect them. Facts, mission status and programme milestones are current to October 2026; space programmes change quickly, so dates of future events should be read as plans, not certainties.

Figures, photographs and data

Diagrams, charts and graphs were drawn specifically for this manual from published physical relationships and public data. Photographs are from the NASA Image and Video Library and are used under NASA's media usage guidelines (NASA material is generally not subject to copyright in the United States); individual credits appear with each image and in Appendix J. NASA does not endorse this publication. Numerical values (masses, power levels, performance) are representative published figures and may differ slightly between sources and mission phases.

Safety and responsibility

This is an educational reference. It is not a design standard, a flight rule, or a substitute for the requirements, analyses and verification processes of a qualified engineering organisation. Chapters dealing with military and intelligence systems use only openly published information and describe principles, not operational or classified details.

How numbers are written

SI units are used throughout, with common aerospace units (nautical miles, feet, pounds-force) shown where the industry uses them. Large numbers use thin spacing (e.g., 35 786 km). "≈" means approximately, "~" means order-of-magnitude. Dates follow the ISO format (YYYY-MM-DD) in tables.

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This volume is one Part of *Space Exploration & Aerospace Engineering — An Engineering Reference Manual*. Page numbers, chapter numbers and figure numbers are the same as in the complete manual, so cross-references to other chapters (for example "Chapter 24") and to the appendices refer to the complete book.



PART IV

Earth Observation and Remote Sensing

Satellites that look down at our planet measure crops, cities, ice, oceans, storms and greenhouse gases. Part IV covers the physics of remote sensing, high-resolution optical and thermal imagers, synthetic aperture radar, and the weather and environmental satellites behind every forecast.

13 Remote Sensing Fundamentals

14 Electro-Optical and Infrared Imaging Satellites

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Image: NASA/GSFC

13 CHAPTER Remote Sensing Fundamentals



Image: NASA/GSFC

Remote sensing is the science of measuring things without touching them. From orbit, satellites measure the light, heat and radio waves leaving the Earth and turn them into maps of crops, forests, cities, oceans, ice and air. This chapter explains the physics that makes this possible - electromagnetic radiation, its interaction with the atmosphere and the surface, and the spectral signatures that identify materials - and the engineering concepts used to describe sensors: spatial, spectral, radiometric and temporal resolution, image quality, geometric correction, processing levels and analysis-ready data. It ends with the methods used to turn images into information and the open-data ecosystem that has made satellite data a global public good.

IN THIS CHAPTER

13.1 What Remote Sensing Is
13.2 Electromagnetic Radiation
13.3 Interaction with the Atmosphere
13.4 Interaction with the Surface: Spectral Signatures
13.5 The Four Resolutions
13.6 Image Quality
13.7 Geometry: Putting Pixels on the Map
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13.9 Turning Images into Information
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13.1 What Remote Sensing Is

Remote sensing means acquiring information about an object from a distance, usually by measuring electromagnetic radiation that the object reflects or emits. Satellites have made it possible to observe the whole planet repeatedly, consistently and objectively. The first weather satellite, TIROS-1, returned its first television images of clouds in April 1960. The US CORONA reconnaissance programme returned film capsules from 1960. **Landsat 1**, launched in July 1972,

began the longest continuous record of Earth's land surface — a record that continues today with Landsat 8 and 9. Since 2008, all Landsat data have been free. Since 2014 the European Union's **Copernicus** programme has provided free and open data from its Sentinel satellites. Together they have transformed Earth observation (EO) into a global public good used by millions.

Remote sensing systems are classified by the source of the energy they measure:

- **Passive optical** systems measure reflected sunlight (visible, near-infrared and short-wave infrared, about 0.4–2.5 µm). They only work in daylight and are blocked by clouds.
- **Passive thermal** systems measure radiation emitted by the surface and atmosphere (mid-wave and long-wave infrared, about 3–14 µm). They work day and night but are blocked by clouds.
- **Passive microwave** systems measure natural microwave emission. They see through most clouds but have coarse resolution.
- **Active** systems — radar and lidar — provide their own illumination. Radar works day and night through clouds (Chapter 15); lidar measures heights and atmospheric profiles precisely.

The complete remote-sensing chain (Figure 13.1) links a source of energy, its passage through the atmosphere and its interaction with the surface. It continues with the return trip through the atmosphere, the sensor, and the processing and analysis that turn measurements into decisions.

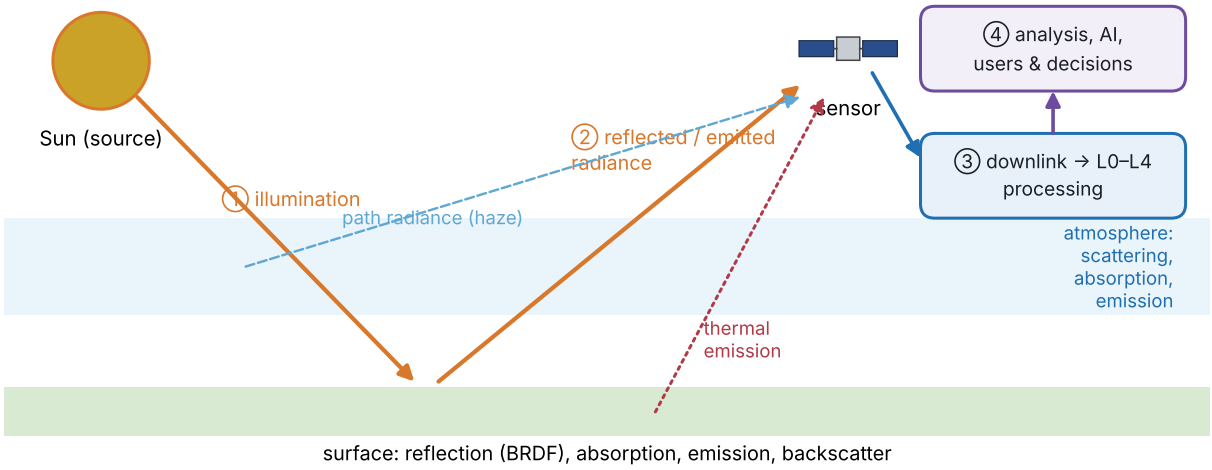


FIGURE 13.1 The remote-sensing chain: illumination, interaction with the atmosphere and surface, measurement by the sensor, and processing into information for users.

13.2 Electromagnetic Radiation

13.2.1 Waves, Photons and the Spectrum

Electromagnetic radiation travels at the speed of light $c = 299\,792\,458$ m/s. Its wavelength λ and frequency ν are related by $c = \lambda\nu$, and each photon carries an energy $E = h\nu = hc/\lambda$, where $h = 6.626 \times 10^{-34}$ J s is Planck's constant. Remote-sensing wavelengths span from about 0.3 µm (ultraviolet) to tens of centimetres (microwave radar) — a range of a million to one (Chapter 12, Figure 12.3).

13.2.2 Thermal Emission: Planck's Law

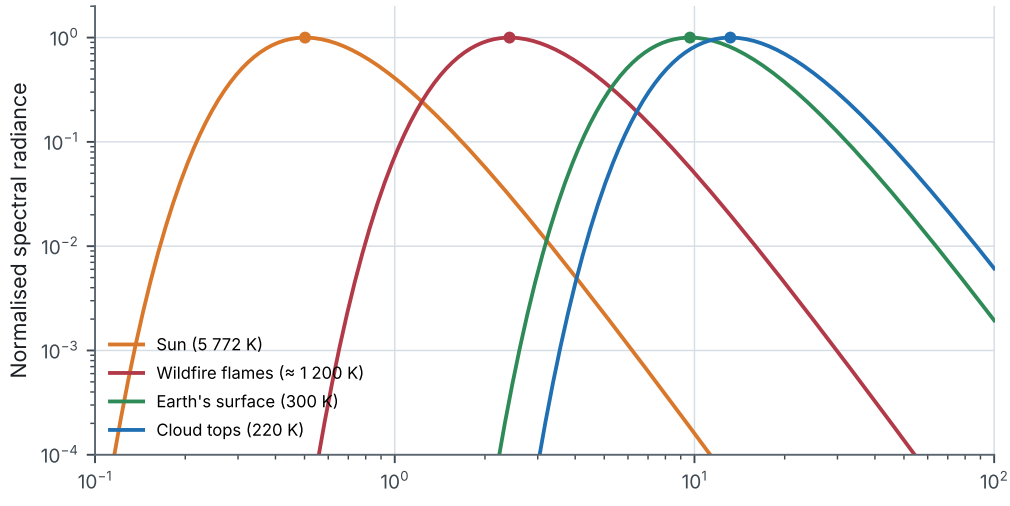
Every object above absolute zero emits radiation. An ideal emitter — a **blackbody** — emits a spectral radiance given by **Planck's law**:

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5}\,\frac{1}{e^{hc/\lambda k_B T} - 1}$$

Two consequences are central to remote sensing (Figure 13.2):

- **Wien's displacement law**: the peak wavelength is $\lambda_{max} = 2\,898\ \mu\text{m·K}/T$. The Sun (≈ 5 800 K) peaks near 0.5 µm — which is why our eyes evolved to see there. The Earth's surface (≈ 300 K) peaks near 10 µm. A forest fire (≈ 1 000–1 200 K) peaks at 2.4–2.9 µm, which is why fires are detected in the mid-wave infrared around 4 µm.
- **Stefan–Boltzmann law**: the total emitted power per unit area is $M = \varepsilon\sigma T^4$; hot objects radiate enormously more energy than cool ones.

Real surfaces emit less than a blackbody by a factor called the **emissivity** $\varepsilon(\lambda)$ (0–1). Water and vegetation have thermal-infrared emissivities of 0.97–0.99; bare rocks and soils vary from 0.85 to 0.98, and their emissivity spectra identify minerals.



Planck curves, each normalised to its peak. Wien's law: λ_max = 2 898 µm·K / T — sunlight peaks at 0.5 µm, Earth's emission at ≈ 10 µm, fires at ≈ 2.4 µm.

FIGURE 13.2 Planck curves for sunlight, a wildfire, the Earth's surface and cold cloud tops, each normalised to its peak.

13.2.3 Radiometric Quantities

Engineers use precise quantities to describe radiation:

TABLE 13.1 Radiometric quantities used in remote sensing.

Quantity	Symbol	Unit	Meaning
Radiant flux	Φ	W	Power
Irradiance	E	W m^{-2}	Power arriving per unit area
Radiant exitance	M	W m^{-2}	Power leaving per unit area
Radiance	L	W m^{-2} sr^{-1}	Power per unit area per unit solid angle — what a sensor pixel measures

Quantity	Symbol	Unit	Meaning
Spectral radiance	L_λ	W m^{-2} sr^{-1} µm^{-1}	Radiance per unit wavelength
Reflectance	ρ	dimensionless	Fraction of incident radiation reflected
Brightness temperature	T_B	K	Temperature of a blackbody producing the measured radiance

A sensor measures **radiance** at the top of the atmosphere. Converting it to **top-of-atmosphere (TOA) reflectance** removes the effect of solar illumination:

$$\rho_{TOA} = \frac{\pi \, L_\lambda \, d^2}{E_{sun,\lambda} \cos\theta_s}$$

where d is the Earth–Sun distance in AU, $E_{sun,\lambda}$ the exo-atmospheric solar irradiance and θ_s the solar zenith angle. Removing the effects of the atmosphere then gives **surface reflectance** — the property of the ground itself.

13.3 Interaction with the Atmosphere

Between the Sun, the ground and the sensor, the atmosphere **scatters** and **absorbs** radiation:

- **Rayleigh scattering** by gas molecules (much smaller than the wavelength) is proportional to λ^{-4}. Blue light (450 nm) scatters roughly four to five times more than red light (650 nm). This is why the sky is blue and why images in the blue band look hazy (Figure 13.3).
- **Mie scattering** by aerosols (dust, smoke, sea salt, pollution particles of similar size to the wavelength) depends more weakly on wavelength and produces the whitish haze over cities and deserts. The **aerosol optical depth (AOD)** at 550 nm ranges from about 0.05 in clean air to more than 2 in heavy smoke or dust.
- **Absorption** by water vapour, carbon dioxide, ozone and oxygen removes radiation in specific bands (Chapter 12, Figure 12.4). Sensors observe the surface through the **atmospheric windows**.
- **Clouds** block optical and infrared radiation completely; on average about 67 % of the Earth is cloud-covered at any time — a fundamental limitation that makes all-weather radar so valuable.

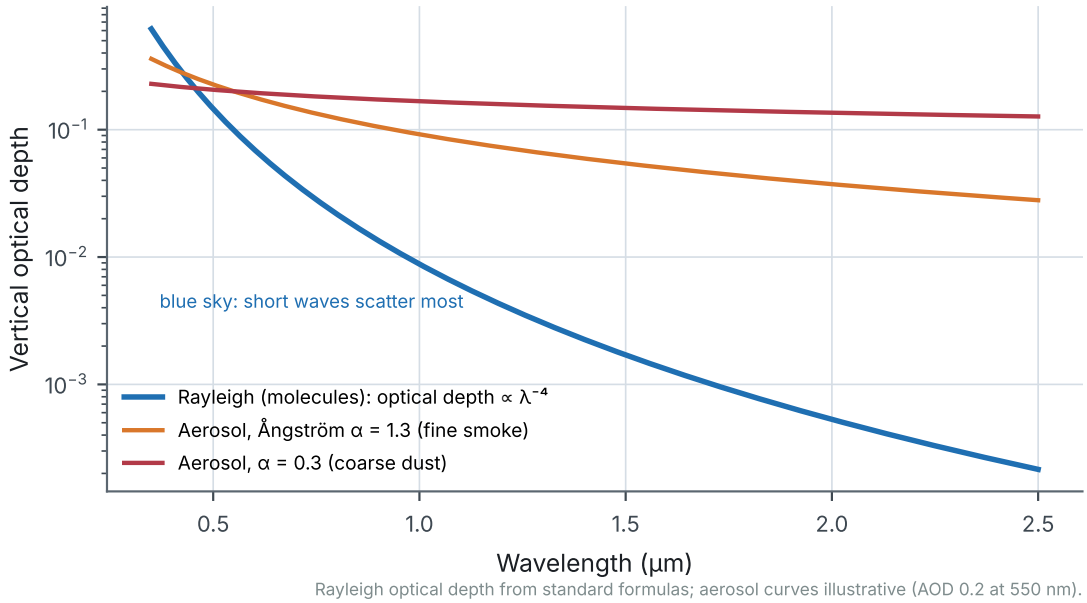


FIGURE 13.3 Vertical optical depth for Rayleigh scattering by air molecules and for two aerosol types as a function of wavelength. Short wavelengths are attenuated far more.

The radiance reaching a sensor over a dark surface may be more than half **path radiance** — light scattered into the sensor by the atmosphere without ever touching the ground. **Atmospheric correction** removes these effects using radiative-transfer models (e.g., 6S, MODTRAN, libRadtran) and information on aerosols, water vapour and ozone, often retrieved from the image itself (e.g., using dark-target or "dense dark vegetation" methods). Operational processors such as ESA's Sen2Cor and NASA's LaSRC/LEDAPS produce surface-reflectance products for Sentinel-2 and Landsat.

13.4 Interaction with the Surface: Spectral Signatures

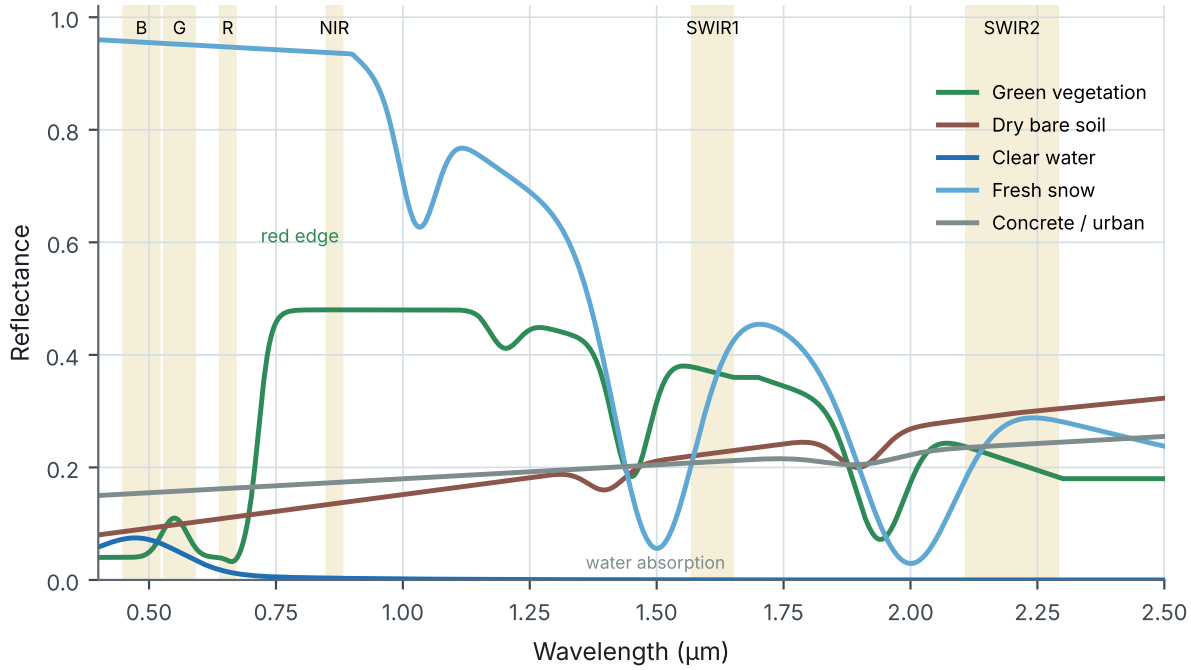
13.4.1 Why Things Look Different

When radiation strikes a surface it may be reflected, absorbed or transmitted. The fraction reflected depends on wavelength in ways that are characteristic of each material — its **spectral signature** (Figure 13.4):

- **Healthy vegetation** absorbs strongly in the blue and red (by chlorophyll, for photosynthesis) and reflects a little more in the green (which is why leaves look green). It reflects strongly in the near-infrared (40–50 %) because of scattering in the internal cell structure of leaves. The sharp rise between red and near-infrared at about 0.70–0.75 µm is the **red edge**; its position shifts with chlorophyll content and plant stress. In the short-wave infrared, reflectance falls in the water absorption bands at 1.45 and 1.94 µm; drier leaves reflect more.
- **Soils and rocks** generally show gently increasing reflectance with wavelength, modified by moisture (wetter soils are darker) and organic matter. Iron oxides (absorption near 0.9 µm) and clay and carbonate minerals (absorption features near 2.2 and 2.3 µm) also modify it — the basis of geological mapping.
- **Water** reflects little except in the blue and green, and absorbs almost everything beyond about 0.75 µm — so water appears black in near-infrared images, making shorelines and floods easy to map. Suspended sediment and phytoplankton change the visible reflectance of

water, the basis of **ocean-colour** remote sensing.

- **Snow** is highly reflective in the visible but much darker in the short-wave infrared, whereas clouds stay bright — a property used to distinguish snow from cloud.
- **Built surfaces** (concrete, asphalt, roofs) have relatively flat, moderate spectra.



Schematic spectra after USGS/ECOSTRESS spectral libraries; shaded: Landsat 8/9 OLI bands.

FIGURE 13.4 Schematic spectral reflectance signatures of common surface materials, with Landsat 8/9 OLI bands shaded. Differences in these curves allow materials to be distinguished from orbit.

13.4.2 Bidirectional Reflectance

Real surfaces are not perfect diffuse (Lambertian) reflectors: their brightness depends on the directions of illumination and viewing, described by the **bidirectional reflectance distribution function (BRDF)**. Forests appear brightest when viewed with the Sun behind the observer (the "hot spot", where shadows are hidden); water shows sharp **sun glint** in the specular direction. Wide-swath sensors and agile satellites that view at different angles must correct for BRDF effects to compare observations.

13.4.3 Spectral Indices

Simple combinations of bands — **spectral indices** — capture key surface properties and are widely used because they are easy to compute and interpret:

TABLE 13.2 Widely used spectral indices. NIR = near-infrared; SWIR1 ≈ 1.6 µm; SWIR2 ≈ 2.2 µm.

Index	Formula	Use
Normalised Difference Vegetation Index (NDVI)	$(\rho_{NIR} - \rho_{red})/(\rho_{NIR} + \rho_{red})$	Vegetation greenness and density
Enhanced Vegetation Index (EVI)	$2.5(\rho_{NIR} - \rho_{red})/(\rho_{NIR} + 6\rho_{red} - 7.5\rho_{blue} + 1)$	Vegetation in dense canopies, less atmospheric effect
Normalised Difference Water Index (NDWI, McFeeters)	$(\rho_{green} - \rho_{NIR})/(\rho_{green} + \rho_{NIR})$	Open water
Normalised Difference Moisture Index (NDMI)	$(\rho_{NIR} - \rho_{SWIR1})/(\rho_{NIR} + \rho_{SWIR1})$	Vegetation water content

Index	Formula	Use
Normalised Burn Ratio (NBR)	$(\rho_{NIR} - \rho_{SWIR2})/(\rho_{NIR} + \rho_{SWIR2})$	Burn severity (difference before/after fire, dNBR)
Normalised Difference Snow Index (NDSI)	$(\rho_{green} - \rho_{SWIR1})/(\rho_{green} + \rho_{SWIR1})$	Snow cover
Normalised Difference Built-up Index (NDBI)	$(\rho_{SWIR1} - \rho_{NIR})/(\rho_{SWIR1} + \rho_{NIR})$	Urban areas

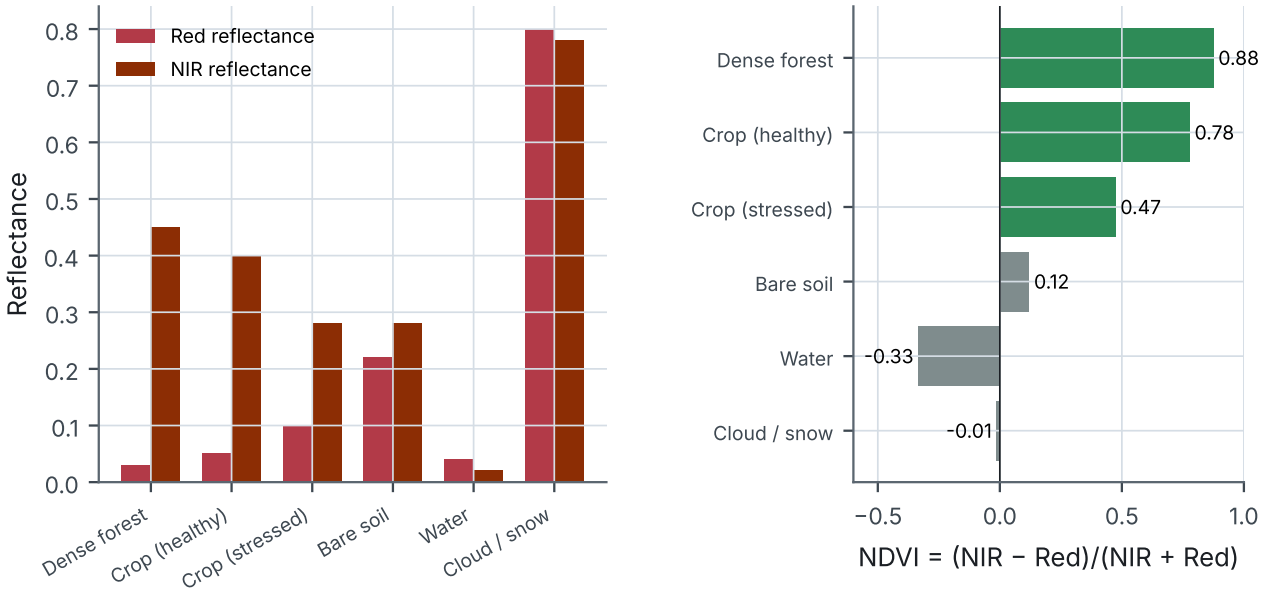


FIGURE 13.5 Red and near-infrared reflectance of six surface types (left) and the resulting NDVI (right). Healthy vegetation gives NDVI values of 0.6–0.9; bare soil 0.1–0.2; water and clouds near zero or negative.

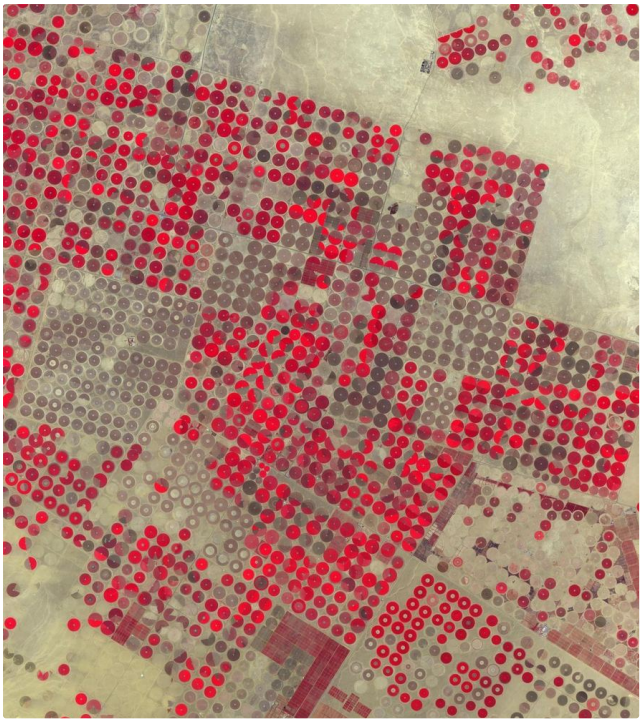


FIGURE 13.6 Centre-pivot irrigated fields in the desert of Al Jowf, Saudi Arabia, imaged by NASA's ASTER instrument in false colour, where vigorous vegetation appears red because it reflects strongly in the near-infrared. Image: NASA/JPL.



FIGURE 13.7 The San Francisco Bay Area in an ASTER false-colour composite. Vegetation is red; urban areas appear blue-grey; water is dark. Image: NASA/JPL.

13.5 The Four Resolutions

A remote-sensing system is characterised by four kinds of resolution (Figure 13.8):

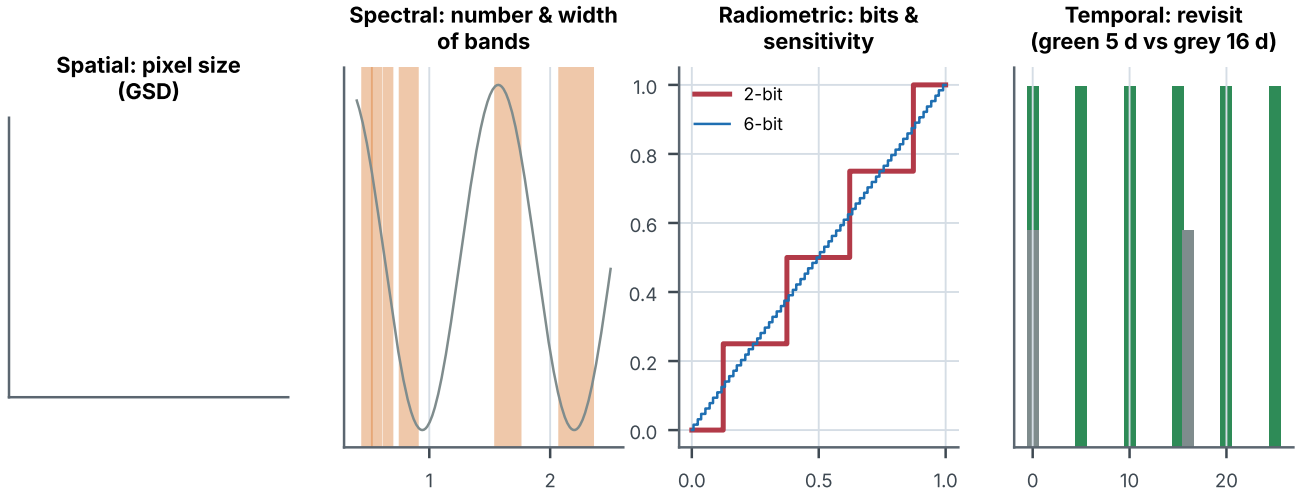


FIGURE 13.8 The four resolutions of a remote-sensing system: spatial, spectral, radiometric and temporal.

Spatial resolution is the size of the smallest feature that can be distinguished. It is commonly expressed by the **ground sampling distance (GSD)** — the distance between pixel centres on the ground — although true resolving power also depends on optical quality and motion (see MTF

below). EO systems range from about 30 cm (very-high-resolution commercial imagers) to 1–2 km (global weather imagers).

Spectral resolution is the number, position and width of the spectral bands. **Panchromatic** sensors have one broad band; **multispectral** sensors have 4–20 bands; **hyperspectral** (imaging spectroscopy) sensors have 100–400 contiguous narrow bands (5–10 nm wide) that record a near-continuous spectrum for every pixel (Chapter 14).

Radiometric resolution is the sensor's ability to distinguish small differences in brightness. It is set by the number of bits used to record each pixel (8-bit = 256 levels; Landsat 9 uses 14 bits = 16 384 levels) and, more fundamentally, by its signal-to-noise ratio.

Temporal resolution is how often the same place is observed (revisit time), from 30 seconds (geostationary "rapid scan" over storms) to 16 days (a single Landsat) or longer.

These resolutions trade against one another and against swath width, data volume and cost. A satellite cannot image a wide area at very fine resolution in hundreds of bands every day, all at the same time. Figure 13.9 shows how today's systems span the trade space.

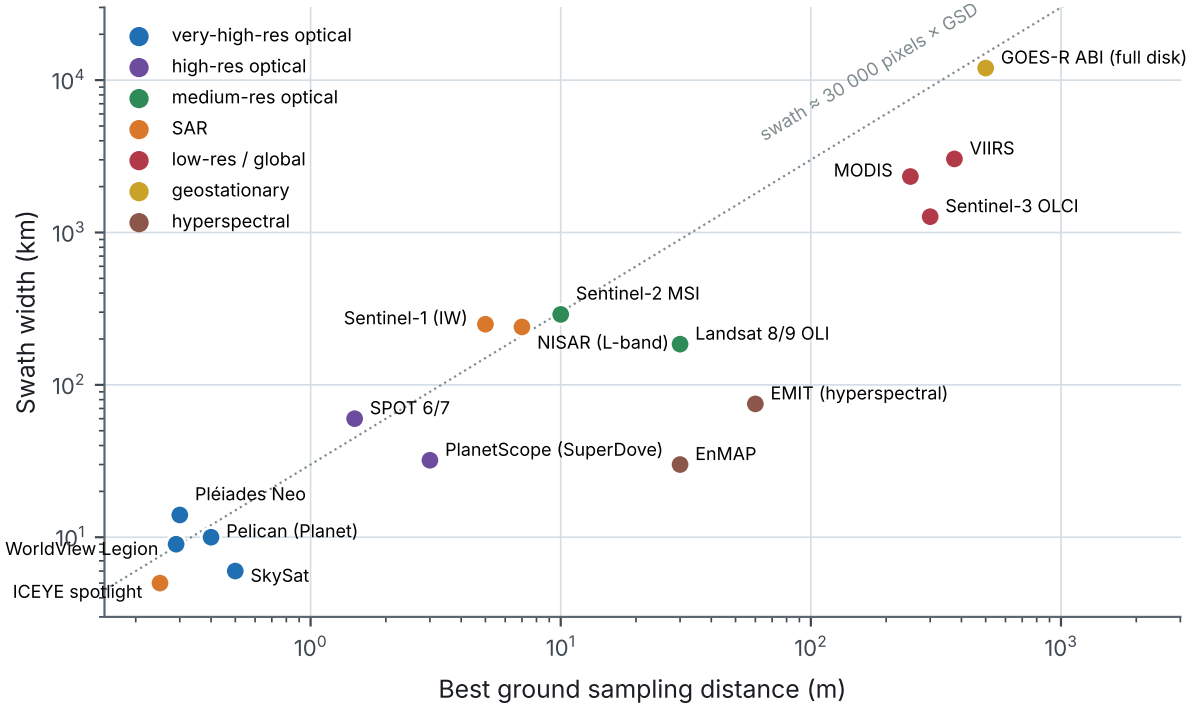


FIGURE 13.9 Ground sampling distance versus swath width for representative EO systems. Most sensors lie near a line corresponding to a few tens of thousands of pixels across the swath, the practical limit of detector arrays and data rates.

13.6 Image Quality

13.6.1 Modulation Transfer Function

GSD alone does not tell how sharp an image is. A sensor blurs the scene through optical diffraction and aberrations, the finite size of each detector pixel, motion of the satellite and pointing jitter during the exposure, and the atmosphere. The **modulation transfer function (MTF)** describes how much contrast the system preserves as a function of spatial frequency; the

system MTF is the product of the MTFs of each contributor (Figure 13.10). A common specification is the MTF at the **Nyquist frequency** (one cycle per two pixels): values of 0.1–0.3 are typical. Too low and images look soft; too high and fine patterns produce **aliasing** artefacts.

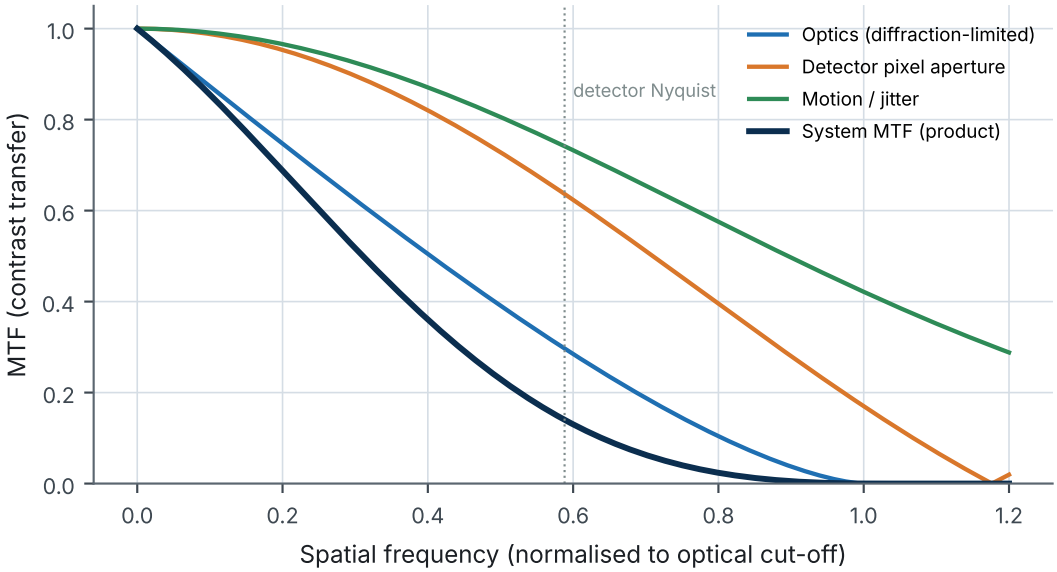


FIGURE 13.10 Contributions to the modulation transfer function of a typical EO imager. The system MTF is the product of optics, detector and motion terms; its value at the detector Nyquist frequency is a key image-quality specification.

13.6.2 Signal-to-Noise Ratio

The **signal-to-noise ratio (SNR)** expresses how well the signal stands above random noise from photon statistics (shot noise, $\sqrt{N}$ for N photo-electrons), detector dark current, read-out electronics and quantisation. A related figure of merit is the **noise-equivalent change in reflectance (NEΔρ)** or, for thermal sensors, the **noise-equivalent temperature difference (NEΔT)** — typically 0.05–0.1 K for modern thermal imagers. This figure states the smallest change that can be detected. SNR improves with larger apertures, wider bands, longer integration time (using time-delay integration, TDI) and cooler detectors (Chapter 12).

13.6.3 Interpretability: NIIRS and GIQE

For high-resolution imagery, users care about what tasks an image supports. The US **National Imagery Interpretability Rating Scale (NIIRS)** rates images from 0 to 9 by the objects that can be identified. NIIRS 5 allows individual rail cars to be identified by type, and NIIRS 6 distinguishes between car types (sedans and station wagons). NIIRS 7 allows individual rail ties to be identified. The **General Image Quality Equation (GIQE)** estimates NIIRS from GSD, the relative edge response (a measure of sharpness), image processing gain and SNR. Halving the GSD raises NIIRS by about one level.

13.7 Geometry: Putting Pixels on the Map

13.7.1 Sources of Geometric Distortion

A raw image is not a map. Its geometry is distorted by:

- the **viewing geometry** — off-nadir viewing stretches pixels and makes them oblique;

- **Earth's curvature and rotation** during a push-broom scan (the image is skewed westward as the Earth turns beneath it);
- **terrain relief** — elevated points appear displaced away from the nadir point (Figure 13.11);
- **platform position and attitude errors** from GNSS and star trackers; and
- **sensor geometry** — lens distortion and focal-plane alignment.

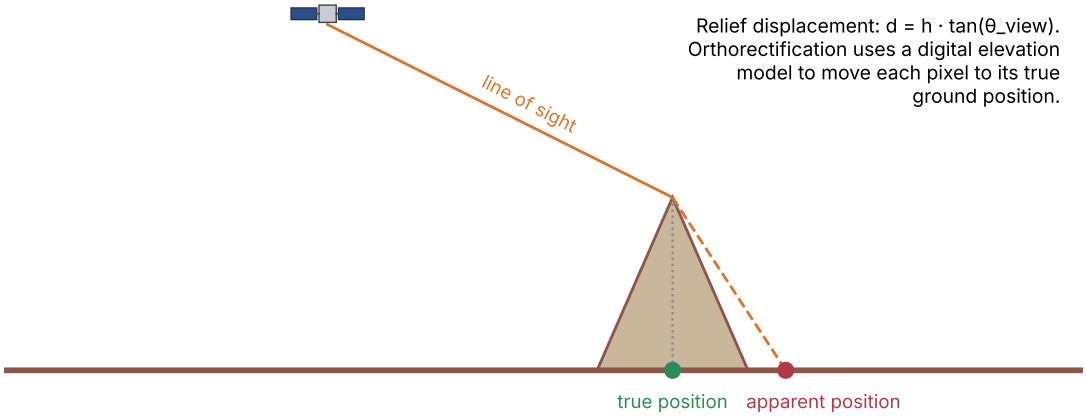


FIGURE 13.11 Relief displacement: the top of a tall object (or mountain) appears shifted away from nadir by d = h·tan θ. Without correction, a 1 000 m mountain imaged 20° off nadir would be displaced by 364 m.

13.7.2 Geolocation and Orthorectification

Geolocation computes the ground position of each pixel by projecting its line of sight from the satellite's position and attitude onto the Earth's surface model. Before ground control, modern satellites achieve geolocation accuracy of better than 5 m CE90 for very-high-resolution systems and about 10–15 m for Landsat-class sensors. With **ground control points (GCPs)** and precision processing, the Landsat and Sentinel-2 archives are co-registered to better than one pixel.

Orthorectification removes the effects of terrain and viewing geometry using a **digital elevation model (DEM)** — a gridded map of surface heights. The result is an **orthoimage** in which every pixel is at its true map position, like a map. Global DEMs widely used for this purpose include the SRTM (Shuttle Radar Topography Mission, 2000, 30 m) and the Copernicus DEM (derived from TanDEM-X, 30 m and 90 m global). Others are ALOS World 3D (30 m) and NASADEM.

For commercial satellites, the camera model is often distributed as **rational polynomial coefficients (RPCs)**. These are ratios of cubic polynomials that map ground coordinates (latitude, longitude, height) to image coordinates. They let users orthorectify images without knowing the proprietary sensor model.

13.7.3 Map Projections and Coordinate Systems

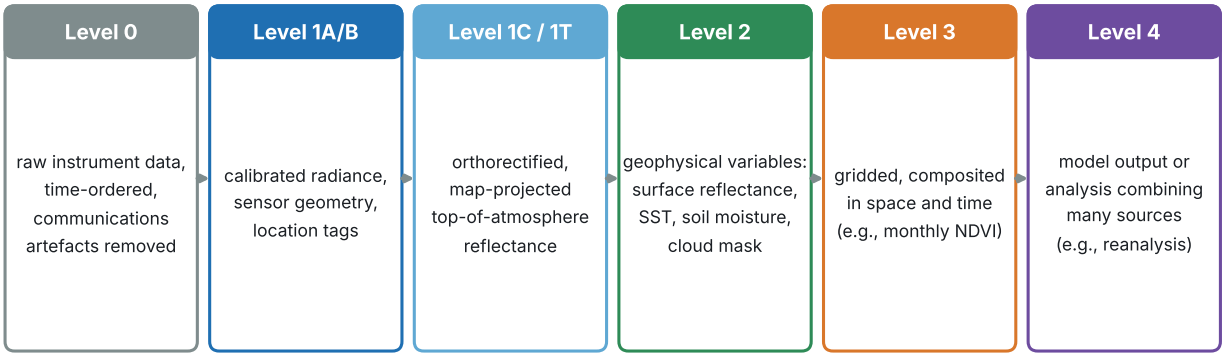
Orthoimages are delivered in a **map projection** — a mathematical transformation of the curved Earth onto a flat plane. The most common for EO is the **Universal Transverse Mercator (UTM)** system, which divides the Earth into 60 zones 6° wide; Sentinel-2 and Landsat products are delivered in UTM tiles. Global products often use a geographic latitude–longitude grid, sinusoidal (MODIS) or equal-area grids (EASE-Grid). All coordinates refer to a **datum**, today almost always WGS 84 (Chapter 18).

13.8 Processing Levels and Analysis-Ready Data

13.8.1 From Raw Bits to Information

Space agencies classify data products by **processing level** (Figure 13.12):

- **Level 0** — raw instrument data at full resolution, time-ordered, with communications artefacts removed.
- **Level 1A/1B** — data with radiometric calibration applied (Level 1B in radiance units) and geolocation information attached.
- **Level 1C/1T** — orthorectified, map-projected TOA reflectance or radiance (e.g., Sentinel-2 L1C, Landsat L1TP).
- **Level 2** — derived geophysical variables at the same resolution: surface reflectance, land-surface temperature, sea-surface temperature, cloud masks.
- **Level 3** — variables mapped onto uniform space–time grids, often composited over days or months (e.g., monthly global NDVI).
- **Level 4** — model output or results from analysis of lower-level data (e.g., gross primary productivity, crop yield, carbon flux).



CEOS Analysis Ready Data (CARD4L) defines Level-2-type products that users can analyse directly.

FIGURE 13.12 Data processing levels from raw telemetry (Level 0) to model-based information products (Level 4).

13.8.2 Calibration and Validation

Calibration establishes the link between the sensor's digital numbers and physical units. It begins with pre-flight laboratory calibration against standards traceable to national metrology institutes. In orbit, it continues with on-board calibrators (solar diffusers, lamps, blackbodies) and observations of the Moon (whose reflectance is extremely stable). It also uses **vicarious calibration** sites with stable bright surfaces (such as Railroad Valley in Nevada, La Crau in France and Gobabeb in Namibia, part of the **RadCalNet** network). Finally, there is **cross-calibration** between satellites viewing the same place at nearly the same time. The radiometric accuracy of good modern sensors is about 3–5 % absolute and better than 1 % relative stability. Upcoming climate-benchmark missions, such as the UK-led ESA mission **TRUTHS**, aim for SI-traceable accuracy of 0.3 % or better so that they can serve as "standards in space" for other sensors.

Validation checks derived products against independent reference data — field measurements, airborne campaigns, in-situ networks (such as AERONET for aerosols, FLUXNET for carbon fluxes and Argo floats for the ocean) — and reports their uncertainty. The international **Committee on Earth Observation Satellites (CEOS)** coordinates calibration and validation among agencies.

13.8.3 Analysis-Ready Data

Most users do not want to correct each image themselves. **Analysis-ready data (ARD)** are satellite data processed to a minimum set of requirements — calibrated, geometrically corrected, atmospherically corrected to surface reflectance, cloud-masked, with per-pixel quality information — so that users can analyse them directly. CEOS has defined the **CEOS Analysis Ready Data (CEOS-ARD)** framework (formerly CARD4L) for optical, thermal and radar data; Landsat Collection 2, Sentinel-2 L2A and the harmonised Landsat–Sentinel (HLS) dataset are examples. HLS combines Landsat 8/9 and the Sentinel-2 satellites into a single 30 m surface-reflectance record with a revisit of 2–3 days.

13.8.4 Cloud Platforms, Data Cubes and Standards

Earth-observation archives now grow by many petabytes per year. The Copernicus Sentinel satellites alone produce more than 15 TB of data per day, and a single radar mission such as NISAR adds tens of terabytes per day. Instead of downloading data, users increasingly bring their algorithms to the data on **cloud computing platforms**:

- **Google Earth Engine** — a planetary-scale analysis platform with a multi-petabyte catalogue of Landsat, Sentinel, MODIS and many other datasets, free for research and non-commercial use;
- the **Copernicus Data Space Ecosystem** — the EU's platform providing free access to all Sentinel data with processing services;
- **Microsoft Planetary Computer**, **Amazon Web Services Open Data** and the **NASA Earthdata Cloud**, which hosts NASA's archive in AWS;
- national **data cubes** such as Digital Earth Australia and Digital Earth Africa, which organise ARD into time series for every pixel of a country or continent.

Common open standards make this possible. **Cloud-Optimized GeoTIFF (COG)** files are organised so that a program can read just the part of an image it needs over the internet. **Zarr** stores multidimensional arrays in chunks. The **SpatioTemporal Asset Catalog (STAC)** specification provides a common way to describe and search EO data across providers.

13.9 Turning Images into Information

13.9.1 Visual Interpretation and Colour Composites

Humans interpret images using tone, colour, texture, pattern, shape, size, shadow, context and association. Multispectral data are displayed by assigning three bands to the red, green and blue channels of a display. A **true-colour** composite (red, green, blue bands) looks like a photograph. A **false-colour infrared** composite (near-infrared, red, green displayed as red, green, blue) makes healthy vegetation bright red, as in Figure 13.6 and Figure 13.13, and is the traditional way to assess vegetation. A **SWIR composite** (SWIR2, NIR, red) highlights burned areas and active fires while seeing through smoke, as in Figure 13.14.

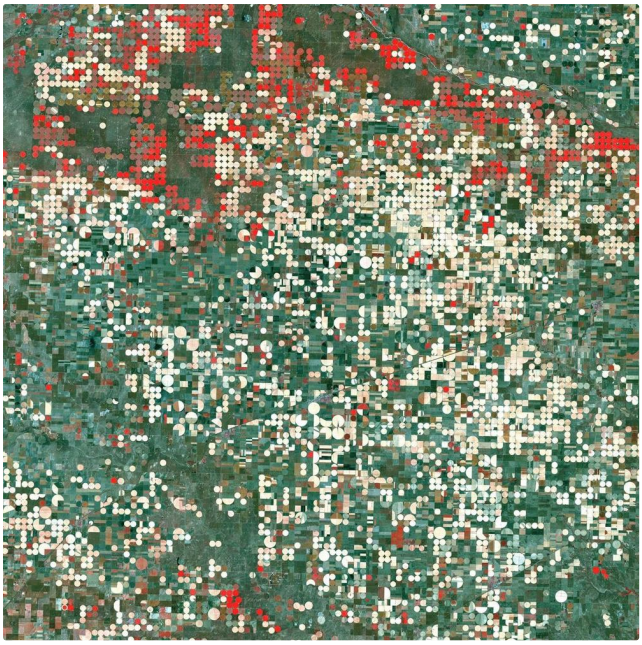


FIGURE 13.13 Centre-pivot irrigated fields near Garden City, Kansas, in a Landsat 7 false-colour composite. Each circle is typically 800 m or 1.6 km in diameter; colour shows crop vigour and growth stage. Image: NASA/GSFC.

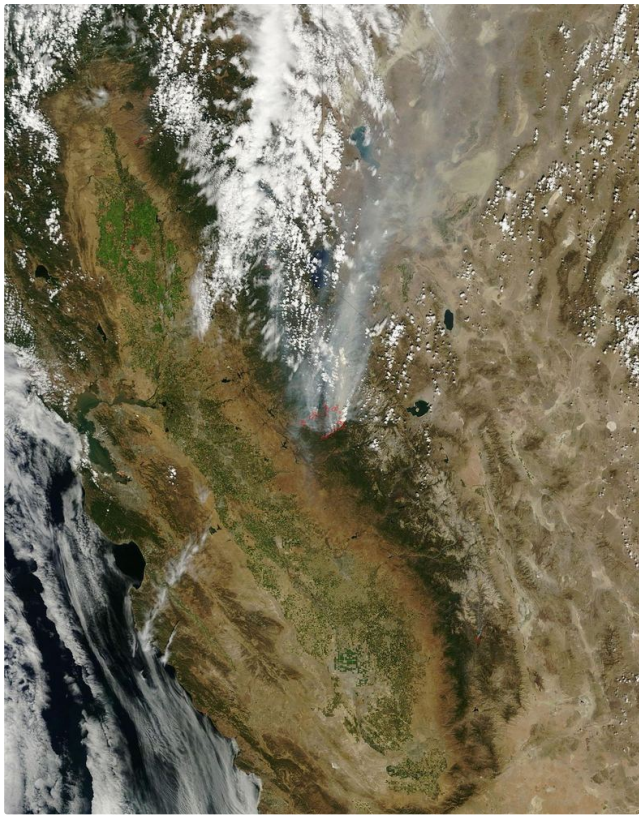


FIGURE 13.14 The 2013 Rim Fire near Yosemite, California, imaged by MODIS on NASA's Terra satellite. The smoke plume streams north-east; thermal and short-wave infrared bands map the active fire front through the smoke. Image: NASA/GSFC.

13.9.2 Classification

Image classification assigns each pixel (or object) to a class such as forest, cropland, water or urban:

- In **unsupervised classification** (e.g., k-means, ISODATA), an algorithm groups pixels into spectral clusters, which the analyst then labels.
- In **supervised classification**, the analyst provides labelled training samples, and a classifier learns to recognise each class. Traditional classifiers include maximum likelihood; modern ones include **random forests**, support vector machines, gradient-boosted trees and **deep learning** (convolutional neural networks and transformers — see Part XIII).
- **Object-based image analysis (OBIA)** first segments the image into homogeneous objects (fields, buildings) and then classifies the objects using their spectral, shape and context attributes — useful for high-resolution imagery.

Global land-cover maps are now produced at 10 m resolution: ESA WorldCover (2020, 2021), Esri/Impact Observatory's annual land cover and Google's **Dynamic World**, which classifies every Sentinel-2 image in near-real time using deep learning.

13.9.3 Accuracy Assessment

A map is only useful if its accuracy is known. Accuracy is assessed by comparing the map with independent **reference data** collected by a statistically valid sample and tabulating the result in a **confusion matrix** (also called an error matrix):

TABLE 13.3 Example confusion matrix for a four-class land-cover map with 550 validation samples (illustrative values).

Map class ↓ / Reference →	Forest	Crop	Water	Urban	Total	User's accuracy
Forest	182	12	1	2	197	92.4 %
Crop	15	141	0	9	165	85.5 %
Water	1	0	88	1	90	97.8 %
Urban	2	11	1	84	98	85.7 %
Total	200	164	90	96	550	
Producer's accuracy	91.0 %	86.0 %	97.8 %	87.5 %		**Overall 90.0 %**

From the confusion matrix (Table 13.3), **overall accuracy** is the fraction of samples correctly classified (here 495/550 = 90.0 %). **User's accuracy** (the complement of commission error) tells a map user how often a pixel labelled "crop" really is crop. By contrast, **producer's accuracy** (the complement of omission error) tells the map producer how much of the real crop was found. Best practice (Olofsson et al., 2014) also uses the matrix to correct area estimates for map errors and to report confidence intervals.

13.9.4 Change Detection and Time Series

Many of the most important applications are about **change**: deforestation, urban growth, flood extent, crop development, glacier retreat. Methods include:

- **image differencing and ratioing** of bands or indices between two dates (e.g., differenced NBR for burn severity);
- **post-classification comparison** of two classified maps;
- **time-series analysis** of every observation of a pixel over years or decades, using algorithms such as LandTrendr, CCDC (Continuous Change Detection and Classification) and BFAST, which model seasonal cycles and detect breaks;
- **deep-learning change detection**, which learns to recognise meaningful changes (new buildings, damage) while ignoring irrelevant ones (shadows, season).

Dense time series are possible because of free data and combined constellations. The Global Forest Watch annual tree-cover-loss maps (University of Maryland) are based on the full Landsat archive since 2000. These maps, and the GLAD and RADD near-real-time deforestation alerts (using Landsat, Sentinel-2 and Sentinel-1 radar), are examples used by governments and companies to enforce laws and supply-chain commitments.

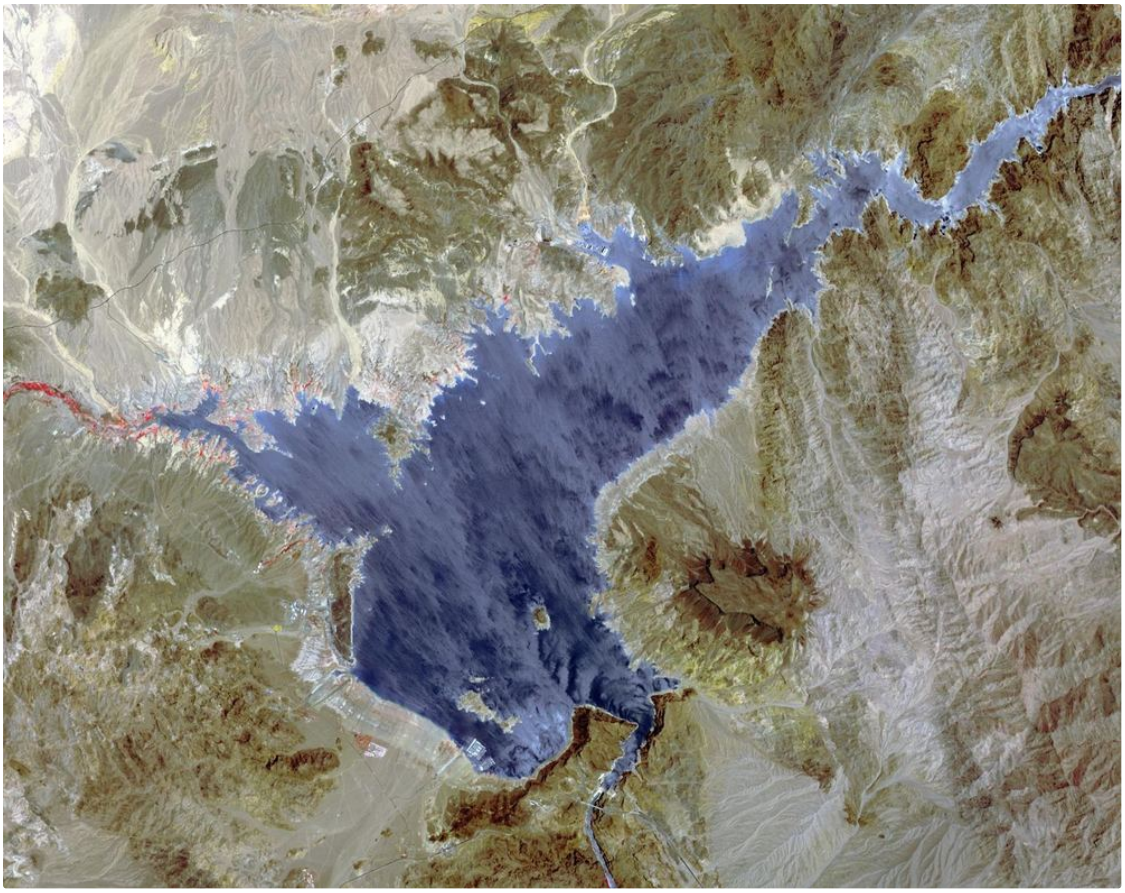


FIGURE 13.15 Lake Mead on the Colorado River seen by ASTER on NASA's Terra satellite. Comparison of images from 2000 and 2015 reveals the light-coloured "bathtub ring" of exposed shoreline left as the water level fell by about 40 m to a record low. Image: NASA/JPL.

13.9.5 From Pixels to Physical Variables

Beyond classification, remote sensing retrieves quantitative **geophysical variables** by inverting physical models: leaf-area index, soil moisture, land-surface temperature, sea-surface height, ocean chlorophyll concentration, aerosol optical depth, trace-gas columns. The Global Climate Observing System (GCOS) has defined 55 **Essential Climate Variables (ECVs)**, more than half of which can be observed globally only from space. ESA's Climate Change Initiative and the Copernicus Climate Change Service produce long, consistent satellite records of many of them.

13.10 The Open-Data Revolution and the Users

13.10.1 Data Policy

For its first decades, satellite data were expensive: a Landsat Thematic Mapper scene cost about US $4 400 in the 1990s, and most data were never used. In 2008 the US Geological Survey made the entire Landsat archive free to all users, and downloads increased from about 25 000 scenes per year to millions. In 2014 the EU adopted a **full, free and open** data policy for Copernicus. Studies have estimated the annual economic benefit of open Landsat data at several billion US dollars. Other agencies, including JAXA, INPE (Brazil), CSA and ISRO, have followed with open policies for many of their datasets.

Commercial data, by contrast, are licensed. Very-high-resolution optical and SAR imagery from companies such as Maxar (Vantor), Airbus, Planet, BlackSky, ICEYE, Capella and Umbra is sold under licences that restrict redistribution. Governments are often anchor customers through programmes such as the US National Reconnaissance Office's Electro-Optical Commercial Layer. Another is NASA's Commercial SmallSat Data Acquisition (CSDA) programme, which buys data for use by NASA-funded scientists.

National regulations also govern remote sensing. In the USA, commercial remote-sensing systems are licensed by NOAA's Commercial Remote Sensing Regulatory Affairs office under a tiered system introduced in 2020; Germany has the SatDSiG data security law; other countries have similar rules. Internationally, the **UN Principles Relating to Remote Sensing of the Earth from Outer Space** (1986) state that remote sensing shall be carried out for the benefit of all countries. They also state that a sensed state shall have access to data concerning its territory on a non-discriminatory basis. Privacy concerns grow as resolution improves, although at 30 cm individuals cannot be identified.

13.10.2 Applications

TABLE 13.4 Major application areas of Earth observation.

Sector	Typical applications	Typical data
Agriculture	Crop type and area, crop health, yield forecasting, irrigation, insurance	Sentinel-2, Landsat, Planet, Sentinel-1, MODIS/VIIRS
Forestry and carbon	Deforestation alerts, forest biomass and height, carbon credits verification	Landsat, Sentinel-1/2, GEDI lidar, BIOMASS radar, ICESat-2
Water resources	Reservoir levels, irrigation water use, snow cover, groundwater change	SWOT, Landsat thermal, GRACE-FO, MODIS snow
Disasters	Flood, fire, earthquake and storm mapping, damage assessment	SAR constellations, VHR optical, VIIRS fire, GOES/Himawari
Urban and infrastructure	Urban growth, informal settlements, subsidence, building footprints	VHR optical, InSAR (Sentinel-1, ICEYE), night lights
Oceans and coasts	Sea-surface temperature, ocean colour, oil spills, ship detection, coastal erosion	Sentinel-3, PACE, SAR, AIS from space
Climate and atmosphere	Greenhouse gases, aerosols, ozone, ice sheets, sea-level rise	OCO-2/3, Sentinel-5P, Copernicus Sentinel-6, ICESat-2, CryoSat-2
Defence and security	Monitoring of facilities, borders, maritime domain awareness	VHR optical, SAR, RF geolocation (Chapter 19)
Finance and insurance	Commodity monitoring, parametric insurance, supply-chain verification	Optical and SAR constellations, analytics platforms
Public health	Disease vector habitats, heat stress, air quality	Thermal, aerosol and land-cover data

The **International Charter "Space and Major Disasters"** was founded in 2000 by ESA, CNES and CSA. It now unites some 17 space agencies and operators who provide satellite data free of charge to countries affected by major disasters. It has been activated more than 900 times. The Copernicus Emergency Management Service and UN-SPIDER/UNOSAT provide rapid mapping services.

EXAMPLE: ESTIMATING THE IRRIGATED AREA OF A DISTRICT

An analyst wants the area of irrigated cropland in a 5 000 km² district during the dry season.

1. Select all Sentinel-2 L2A (surface reflectance) images for the dry season and mask clouds using the scene classification layer.
2. Compute NDVI for each image and take the maximum value composite per pixel (to capture peak green-up).
3. Collect 400 training samples (irrigated, rain-fed, natural vegetation, bare, water, urban) from field visits and very-high-resolution images.
4. Train a random-forest classifier using the dry-season NDVI time series plus selected bands.
5. Classify the district and assess accuracy with 600 independent stratified-random samples; suppose the overall accuracy is 88 % and the user's and producer's accuracies for "irrigated" are 84 % and 90 %.
6. Use the error matrix to adjust the mapped area: if 610 km² is mapped as irrigated, the bias-corrected estimate (Olofsson method) might be 640 ± 45 km² at 95 % confidence.

The entire workflow can run in a few minutes on a cloud platform at zero data cost — a task that would have taken months and thousands of dollars in data purchases in the 1990s.

AI IN REMOTE SENSING (PREVIEW)

Machine learning now underpins most large-scale EO mapping. Deep convolutional networks and vision transformers segment buildings, roads and fields; detect ships, aircraft and vehicles; and classify land cover. **Geospatial foundation models** pre-trained on millions of unlabelled images — such as NASA–IBM's Prithvi, Clay, SatMAE and Google's AlphaEarth Foundations embeddings — can be fine-tuned for new tasks with few labels. Part XIII (Chapters 43–53) covers these methods in depth, including on-board AI that decides which images to keep and downlink.

KEY TAKEAWAYS

- Remote sensing measures electromagnetic radiation reflected or emitted from the Earth. Planck's law and Wien's law explain why reflected sunlight dominates at 0.4–2.5 µm and thermal emission at 3–14 µm.
- The atmosphere scatters (strongly at short wavelengths) and absorbs (in gas bands) radiation; sensors observe through atmospheric windows and use atmospheric correction to retrieve surface reflectance.
- Each material has a characteristic spectral signature. Vegetation's red-edge contrast underlies NDVI and many other indices.
- Spatial, spectral, radiometric and temporal resolution trade against each other and against swath, data volume and cost.
- Image quality depends on MTF and SNR, not just GSD; NIIRS and GIQE relate them to interpretability.
- Geometric correction (geolocation and orthorectification with a DEM) turns images into maps; calibration and validation make measurements trustworthy.
- Data progress from Level 0 to Level 4; analysis-ready data, open standards (COG, STAC, Zarr) and cloud platforms let users analyse petabytes without downloading them.
- Classification, change detection and time-series analysis, with rigorous accuracy assessment, turn images into information. Free and open Landsat and Copernicus data have made EO a global public good.

REVIEW QUESTIONS

1. Using Wien's law, find the wavelength of peak emission for the Sun (5 772 K) and for the Earth's surface (300 K). What does this imply for optical and thermal sensors?
2. Why does the sky look blue, and why do Earth-observation sensors avoid the strong water-vapour and CO_2 absorption bands for surface imaging?
3. Calculate the NDVI for a vegetated pixel (red reflectance 0.05, near-infrared 0.45) and for bare soil (red 0.25, near-infrared 0.30).
4. Describe the four resolutions of a remote-sensing system and give an example of a trade-off between two of them.
5. Why can two imagers with the same ground sample distance produce images of very different quality?
6. Why does orthorectification need a digital elevation model? What happens to geolocation accuracy in mountainous terrain without one?
7. A classification is checked at 500 points: 180 forest correctly mapped, 20 forest mapped as non-forest, 30 non-forest mapped as forest and 270 non-forest correct. Calculate the overall accuracy and the user's and producer's accuracies for forest.
8. What are analysis-ready data, and how do cloud platforms and standards such as COG and STAC change the way users work with satellite archives?

CHAPTER

14 Electro-Optical and Infrared Imaging Satellites

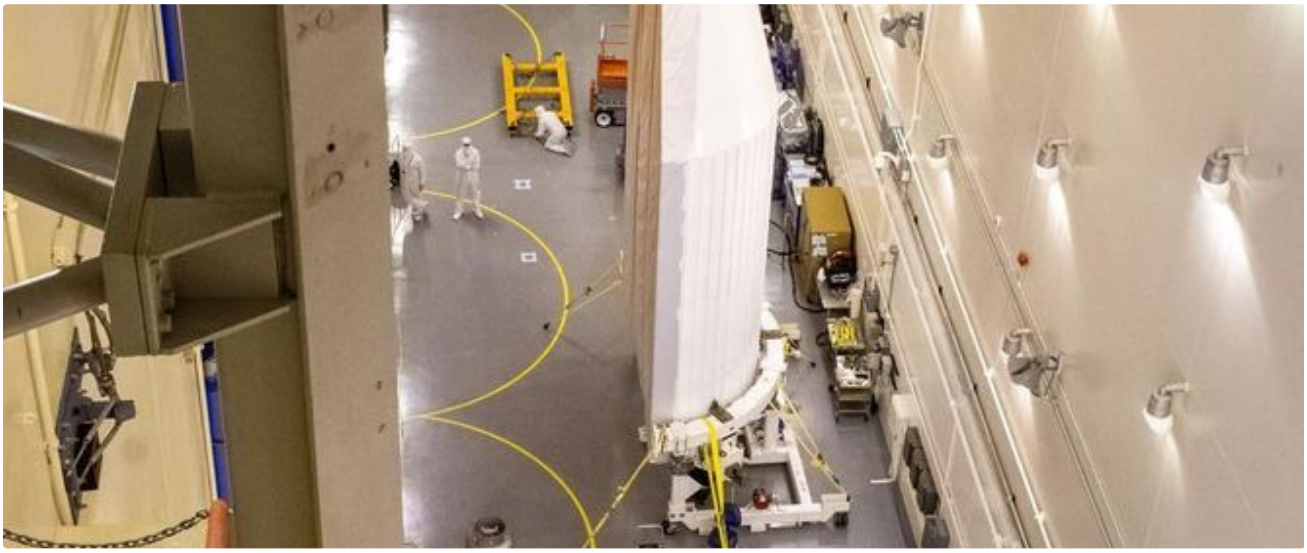


Image: NASA/KSC (NASA/Randy Beaudoin)

Electro-optical and infrared (EO/IR) satellites are cameras in orbit. They range from government "workhorse" missions such as Landsat and Sentinel-2, which map the whole land surface every few days at 10–30 m, through commercial constellations of hundreds of shoebox-sized satellites that image the planet daily, to very-high-resolution satellites that can see objects 30 cm across. This chapter explains how these systems are designed — telescopes, focal planes, time-delay integration, agility and data handling — and why the laws of optics and photon statistics set their limits. It then covers thermal infrared imaging for temperature and fire, imaging spectroscopy (hyperspectral) for minerals, vegetation chemistry and methane, global moderate-resolution imagers, and the state of the industry in 2026.

IN THIS CHAPTER

14.1 The Family of EO/IR Imagers
14.2 Landsat and Sentinel-2: The Workhorses
14.3 Very-High-Resolution Imaging
14.4 Constellations: Trading Resolution for Revisit
14.5 Thermal Infrared Imaging
14.6 Imaging Spectroscopy (Hyperspectral)
14.7 Global Imagers: MODIS, VIIRS and Their Successors
14.8 Design Example: A Small VHR Imaging Satellite

14.1 The Family of EO/IR Imagers

Electro-optical (EO) sensors convert visible and near-infrared light into electrical signals; **infrared (IR)** sensors extend this to the short-wave, mid-wave and long-wave infrared. Together they form the largest and most familiar class of Earth-observation satellites. Their principal

parameters — resolution, spectral bands, swath and revisit — were introduced in Chapter 13; their instrument building blocks (telescopes, detectors, scanning methods) in Chapter 12. This chapter looks at complete systems.

EO/IR satellites are usually grouped by their spatial resolution, which largely determines their applications, size and cost (Table 14.1).

TABLE 14.1 Classes of EO/IR imaging satellites.

Class	GSD	Swath	Typical missions	Typical uses
Low resolution	250 m – 5 km	1 000 – 3 000 km	MODIS, VIIRS, Sentinel-3, geostationary imagers	Weather, climate, ocean colour, fires, global vegetation
Medium resolution	10 – 100 m	100 – 300 km	Landsat 8/9, Sentinel-2, CBERS, Resourcesat	Land cover, agriculture, forestry, water, change monitoring
High resolution	1 – 10 m	10 – 60 km	PlanetScope, SPOT 6/7, Gaofen-1, RapidEye (retired)	Field-level agriculture, mapping, monitoring
Very high resolution (VHR)	0.25 – 1 m	5 – 20 km	WorldView, Pléiades Neo, SkySat, BlackSky, Jilin-1, Cartosat-3	Urban mapping, infrastructure, defence and intelligence, disaster damage
Extremely high resolution	< 0.25 m	< 5 km	Military reconnaissance; very-low-orbit commercial concepts	Detailed object identification

There is also a sharp division between **government science and operational missions**, which emphasise calibration accuracy, long-term continuity and free data, and **commercial missions**, which emphasise resolution, revisit, responsiveness and fast delivery to paying customers. The two are increasingly complementary: scientists use commercial data under licence, and companies build value-added products on top of free government data.

14.2 Landsat and Sentinel-2: The Workhorses

14.2.1 Landsat — Fifty Years of Land Observation

The Landsat programme is managed jointly by NASA (which builds and launches the satellites) and the US Geological Survey (which operates them and archives and distributes the data). It has observed the Earth's land surface continuously since 1972 (Figure 14.1). Its value comes from **continuity**: the same places observed in consistent, carefully cross-calibrated ways for more than five decades, providing the longest record of human and natural change on the land surface.

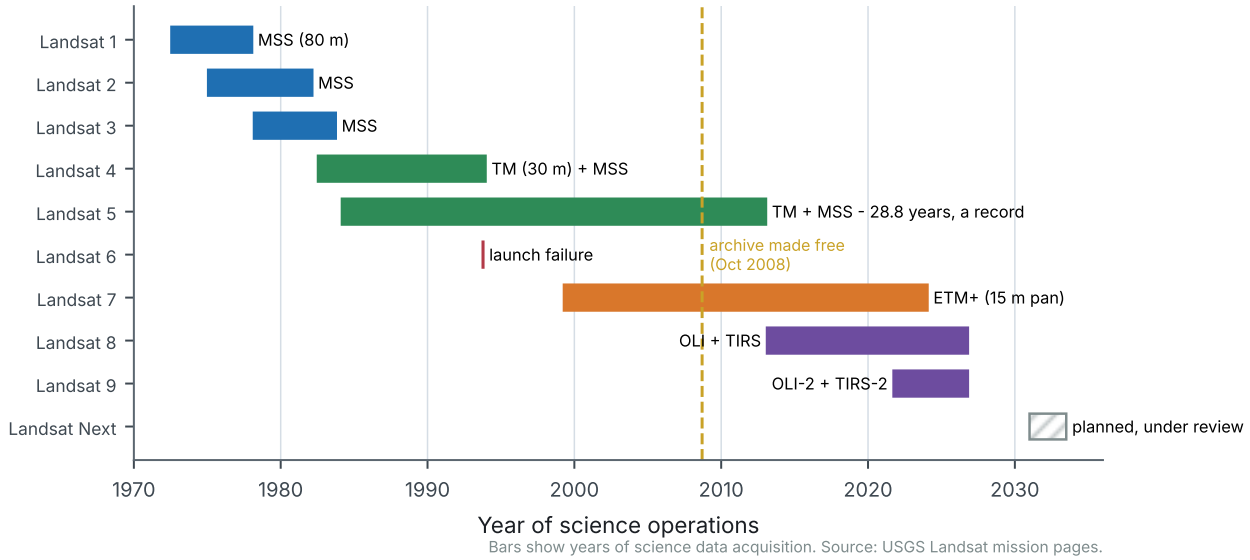


FIGURE 14.1 The Landsat missions. Landsat 5 holds the record for the longest-operating Earth-observation satellite (28.8 years). The free-data decision of 2008 transformed the use of the archive.

Landsat 8 (launched 2013) and Landsat 9 (2021) fly in the same 705 km sun-synchronous orbit with a 10:00 a.m. descending-node crossing. Each repeats its ground track every 16 days. They are spaced half a cycle apart, so together they revisit every point every **8 days**. Each carries two instruments:

- The **Operational Land Imager (OLI / OLI-2)**, built by Ball Aerospace, is a **push-broom** imager with a four-mirror telescope. Its focal plane of 14 modules carries more than 6 500 detectors per spectral band across the 185 km swath (twice as many for the panchromatic band). It has nine bands — including a coastal-aerosol band (0.43–0.45 µm) and a cirrus band (1.36–1.39 µm) for detecting thin high clouds — at 30 m (15 m panchromatic). Landsat 9's OLI-2 records data with 14-bit quantisation, giving more brightness levels than Landsat 8's 12 bits.
- The **Thermal Infrared Sensor (TIRS / TIRS-2)**, built by NASA Goddard, images in two long-wave bands (10.6–11.2 and 11.5–12.5 µm) at 100 m. It uses **quantum-well infrared photodetectors (QWIPs)** cooled to about 43 K by a cryocooler. TIRS-2 corrected a stray-light problem discovered in TIRS on Landsat 8.

Landsat 7, launched in 1999, suffered a failure of its scan-line corrector in 2003 that left wedge-shaped gaps in its images. Even so, it continued to provide useful data until January 2024 and was decommissioned in June 2025. The planned successor, **Landsat Next**, was originally conceived as a three-satellite "superspectral" constellation with 26 bands, 10–20 m resolution and a 6-day revisit, launching around 2030–31. It was restructured in the US administration's 2026 budget proposal, and its final architecture was still under review in 2026.

14.2.2 Sentinel-2 — The Copernicus High-Resolution Imager

The European Union's **Sentinel-2** mission, developed by ESA with Airbus as prime contractor, has become the most widely used medium-resolution optical mission. Each satellite carries the **Multispectral Instrument (MSI)**, a push-broom imager with a 15 cm three-mirror anastigmat telescope and two focal planes (visible/near-infrared silicon and short-wave infrared HgCdTe). It

records **13 spectral bands**: four at 10 m, six at 20 m and three atmospheric bands at 60 m, across a **290 km swath**. Its three **red-edge** bands at 705, 740 and 783 nm are especially valuable for vegetation monitoring (Figure 14.2).

The satellites fly at 786 km with a 10:30 a.m. descending node; each repeats every 10 days and two together give a **5-day revisit** at the equator — 2–3 days at mid-latitudes, where swaths overlap. Sentinel-2A was launched in 2015, 2B in 2017 and **2C in September 2024** on the last flight of Europe's Vega rocket. Sentinel-2C took over from 2A in early 2025, and 2A continues in an extended campaign. Sentinel-2D is planned for about 2028, and a **Sentinel-2 Next Generation** with finer resolution and more bands is planned for the 2030s.

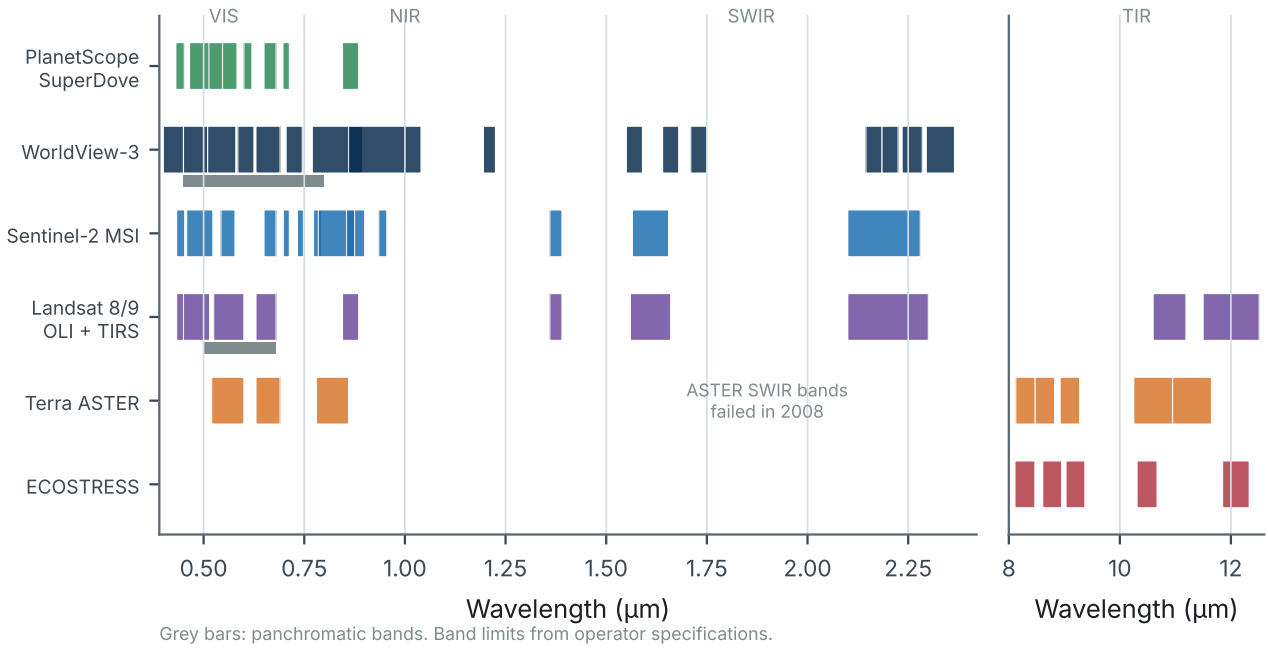


FIGURE 14.2 Spectral bands of representative EO/IR sensors. Commercial imagers concentrate bands in the visible and near-infrared; science missions add short-wave and thermal infrared bands for atmospheric correction, geology, fire and temperature.

TABLE 14.2 Comparison of the two leading medium-resolution land-imaging missions.

Parameter	Landsat 8/9	Sentinel-2A/B/C
Operators	NASA (development) / USGS (operations)	EU Copernicus programme; developed and operated by ESA
Orbit	705 km SSO, 10:00 descending	786 km SSO, 10:30 descending
Revisit (one / two satellites)	16 / 8 days	10 / 5 days
Swath	185 km	290 km
Bands	9 reflective + 2 thermal	13 reflective
Resolution	15 m pan, 30 m MS, 100 m thermal	10, 20 and 60 m
Quantisation	12 bit (L8), 14 bit (L9)	12 bit
Data volume	≈ 1 500 scenes per day (two satellites)	≈ 1.6 TB of compressed data per satellite per day
Data policy	Free and open (since 2008)	Free, full and open (since 2014)

The two missions are complementary. Landsat has the longer record, thermal bands and stringent calibration; Sentinel-2 has finer resolution, red-edge bands and more frequent revisit. NASA's **Harmonized Landsat and Sentinel-2 (HLS)** product combines them into a single 30 m surface-reflectance record with observations every 2–3 days.

Other national programmes contribute medium-resolution data: the China–Brazil **CBERS** series, India's **Resourcesat** (LISS-III and AWiFS) and China's **Gaofen-1/6** wide-field cameras. Commercial wide-area constellations also contribute, such as **EarthDaily**, which is building a constellation designed for scientific-grade daily global coverage.

14.3 Very-High-Resolution Imaging

14.3.1 The Optics of Seeing Small Things

For a push-broom imager, the ground sampling distance is set by the detector pixel pitch p, the telescope focal length f and the altitude h:

$$\mathrm{GSD} = \frac{p\,h}{f}$$

A satellite at 600 km with 5 µm detectors therefore needs a **10 m focal length** to achieve 30 cm GSD. Such long focal lengths are folded into compact telescopes with three or four curved mirrors (Chapter 12).

A long focal length alone is not enough: the aperture must be large enough that diffraction does not blur the image. The smallest resolvable separation on the ground is roughly given by the Rayleigh criterion:

$$r \approx 1.22\,\frac{\lambda\,h}{D}$$

For λ = 0.6 µm and h = 600 km, resolving 30 cm requires an aperture of at least 1.5 m by this criterion. In practice, however, systems sample somewhat finer than the Rayleigh limit and recover detail through processing (Figure 14.3). This is the fundamental reason why very-high-resolution satellites are large. WorldView-3 has a 1.1 m telescope and weighs about 2 800 kg, while a 3U CubeSat with a 9 cm aperture cannot do better than about 3 m. It also explains the military reconnaissance satellites believed to have 2.4 m mirrors, and the commercial ventures aiming for 10 cm. These either use very large telescopes or fly much lower (see "Very low Earth orbit" below).

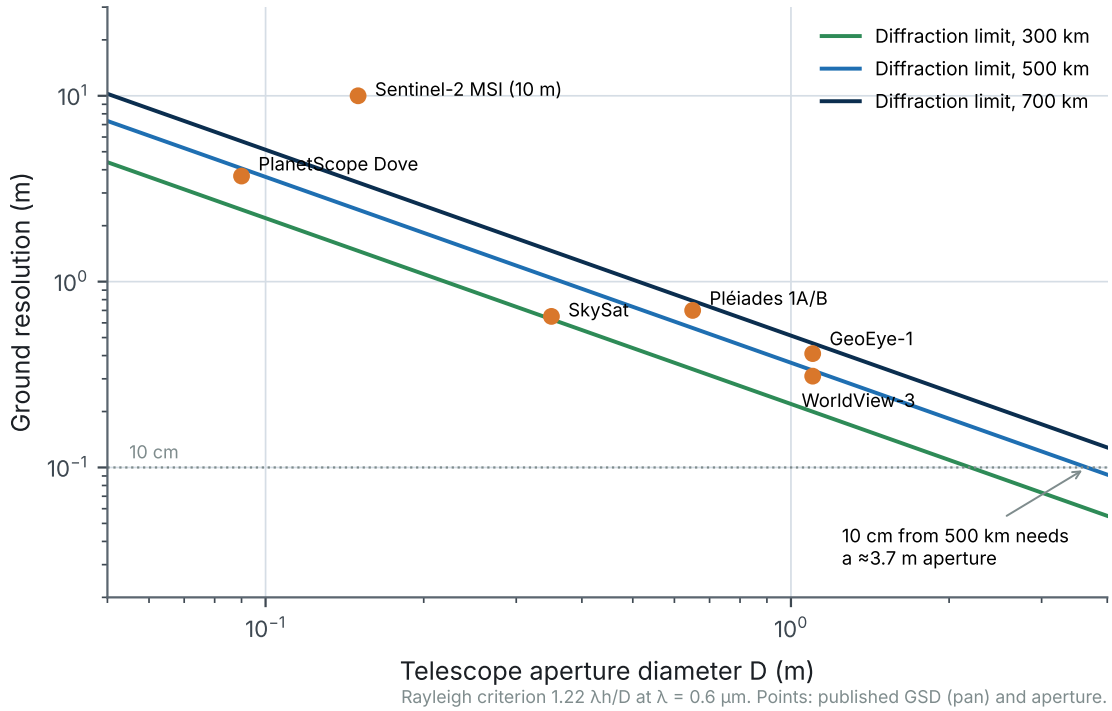
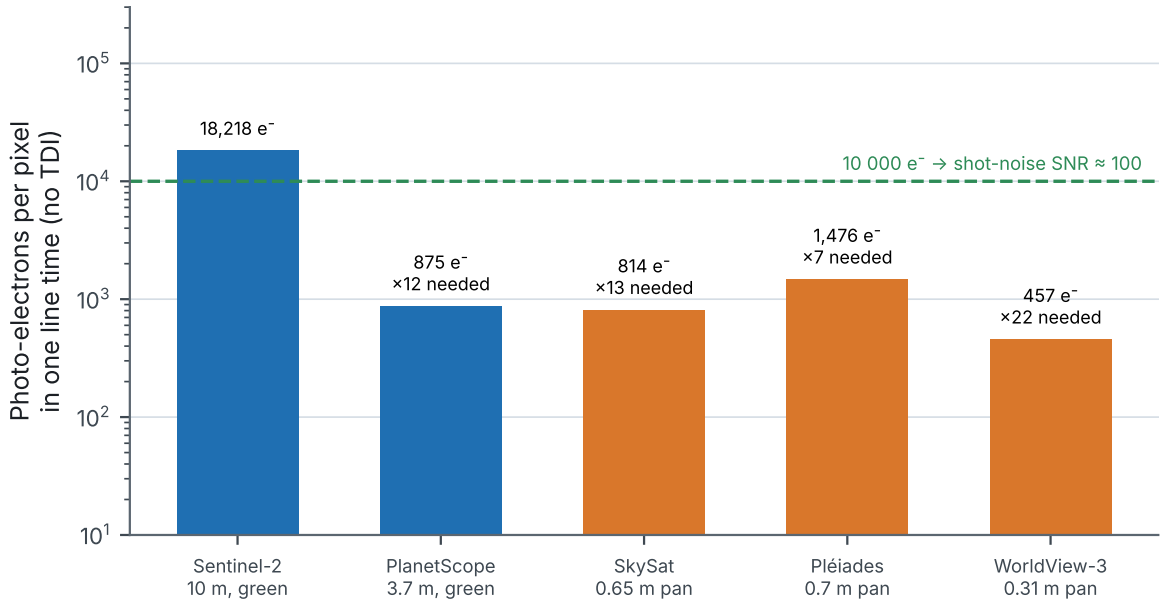


FIGURE 14.3 Diffraction-limited ground resolution versus telescope aperture at three altitudes, with the published panchromatic GSD of representative systems. Small apertures (Sentinel-2) are deliberately not diffraction-limited; VHR systems sample at or slightly beyond the diffraction limit.

14.3.2 Not Enough Photons: Time-Delay Integration

The second limit is light. As the GSD shrinks, each pixel collects light from a smaller area, for a shorter time. The satellite moves 7 km/s over the ground, so a 31 cm pixel is crossed in about 45 µs. Figure 14.4 estimates the number of photo-electrons collected per pixel in one line time for several systems. A 10 m Sentinel-2 pixel collects tens of thousands of electrons, but a 31 cm WorldView-3 pixel collects only a few hundred. This gives a shot-noise-limited signal-to-noise ratio of about 20, far too low for good imagery.



Assumes L = 100 W m⁻² sr⁻¹ µm⁻¹, optics × detector efficiency 0.4, dwell = GSD / ground speed.
Push-broom imagers make up the shortfall with TDI; frame cameras (Dove, SkySat) stack several frames.

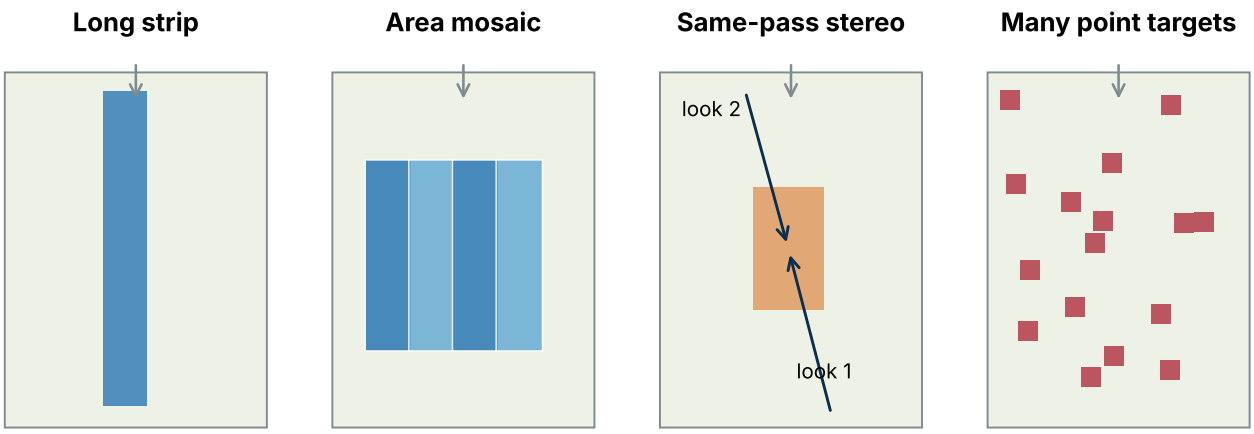
FIGURE 14.4 Estimated photo-electrons per pixel collected in one line time for representative imagers. VHR systems collect far fewer photons per line and rely on time-delay integration (TDI) or frame stacking to reach useful signal-to-noise ratios.

The solution is **time-delay integration (TDI)**. A TDI detector has many rows (stages) of pixels aligned along the direction of motion. As the image of a ground point moves across the detector, the charge collected in one row is shifted to the next row in exact synchrony. In this way the same ground point is integrated by 8, 16, 32, 64 or more rows in succession. The signal grows in proportion to the number of stages N while the shot noise grows only as $\sqrt{N}$. Modern VHR focal planes use CCD or CMOS TDI detectors with up to 64–128 stages, selectable to adapt to illumination — fewer stages for bright desert, more for dark forest or low sun. TDI requires extremely precise synchronisation between the charge transfer and the ground image motion, which depends on altitude, satellite speed, viewing angle and Earth rotation; errors smear the image.

Frame cameras, such as those on PlanetScope Doves and SkySats, use another approach. They take many overlapping short exposures and combine ("stack") them on the ground. This also allows **super-resolution** processing that recovers detail finer than a single frame.

14.3.3 Agility, Pointing and Collection Modes

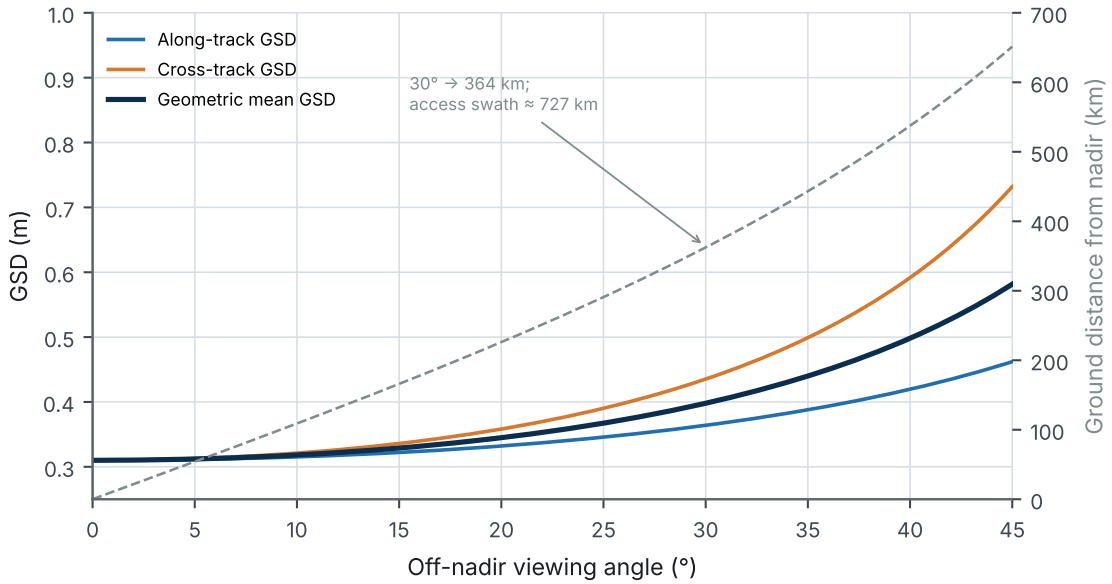
A VHR satellite with a swath of only 10–15 km would take years to map the whole Earth. Instead, VHR satellites are **agile**: they rotate the whole spacecraft to point the telescope at targets up to 30–45° away from the ground track, using control-moment gyroscopes (CMGs) or large reaction wheels (Chapter 6). An agile satellite can typically retarget across hundreds of kilometres in about ten seconds, collecting many separate targets on a single pass or sweeping its view to build an area larger than its swath (Figure 14.5).



Arrows show the direction of flight. Agile satellites slew between targets in seconds.

FIGURE 14.5 Collection modes of an agile VHR satellite: a long strip along the ground track, an area mosaic built from adjacent strips on one pass, a stereo pair from fore and aft views, and many point targets collected in rapid succession.

Off-nadir viewing has a cost: the distance to the ground increases and the ground is viewed obliquely, so the GSD grows (Figure 14.6). At 30° off nadir, a satellite with 31 cm nadir GSD produces imagery with about 40 cm (geometric mean) GSD; at 45°, about 60 cm. Oblique views also suffer more atmospheric haze, longer building shadows and greater layover of tall structures — so customers specify a maximum off-nadir angle when ordering.



WorldView-3-like case: 617 km altitude, 0.31 m GSD at nadir, spherical Earth.

FIGURE 14.6 Ground sampling distance versus off-nadir angle for a WorldView-3-like satellite. The dashed curve gives the distance from the ground track: viewing up to 30° off nadir gives access to a strip about 725 km wide.

14.3.4 Stereo Imaging and 3-D Mapping

By imaging the same area from two different directions on the same pass — looking forward and then backward — an agile satellite collects a **stereo pair**. Tall features appear shifted (parallax) between the two images in proportion to their height. Photogrammetric software measures the parallax by automatically matching features and computes a **digital surface model (DSM)** (Figure 14.7). The precision of height measurement depends on the ratio of the stereo base to the height (B/H) and the precision of image matching. VHR stereo pairs routinely produce DSMs with

1–2 m vertical accuracy. They are used to build 3-D city models, measure volumes of mines and stockpiles, and produce terrain data for defence and aviation. Some missions, such as Japan's ALOS PRISM (2006–2011) and the Chinese ZY-3 series, carry two or three fixed cameras looking forward, nadir and backward to collect stereo continuously.

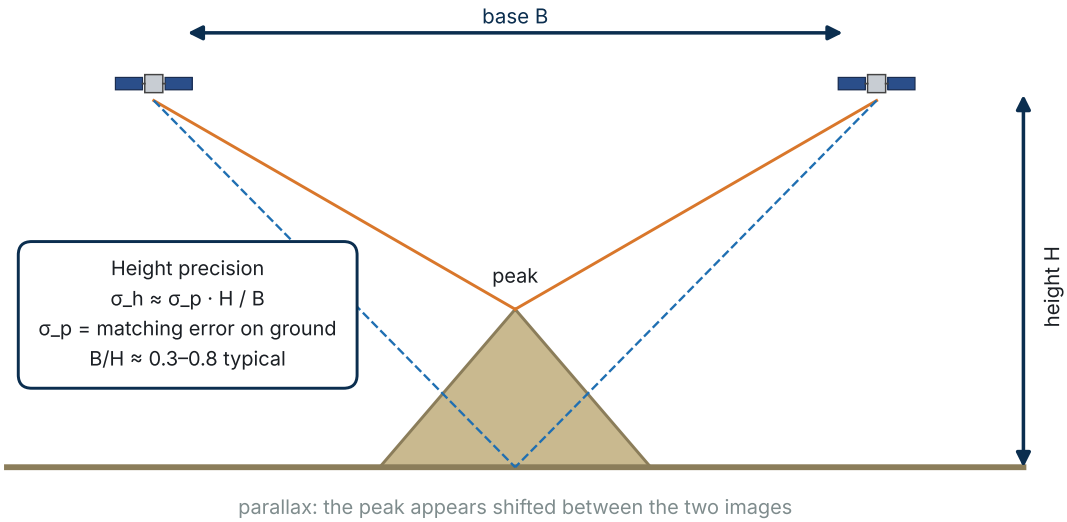


FIGURE 14.7 Stereo imaging geometry. Two views separated by base B from height H see a tall feature displaced by parallax; height precision is approximately the matching precision multiplied by H/B.

14.3.5 Pan-Sharpening

Most VHR satellites collect a high-resolution **panchromatic** band and lower-resolution (typically 4×) **multispectral** bands. The panchromatic band collects more light because it is wide, so its pixels can be smaller with the same SNR. **Pan-sharpening** combines the two to create a colour image with the panchromatic resolution (Figure 14.8). Methods range from simple ratio (Brovey) and intensity–hue–saturation substitution to Gram–Schmidt, wavelet and, increasingly, deep-learning fusion. Pan-sharpening is cosmetically excellent but can distort the spectral values, so scientists usually prefer the original multispectral data for quantitative analysis.

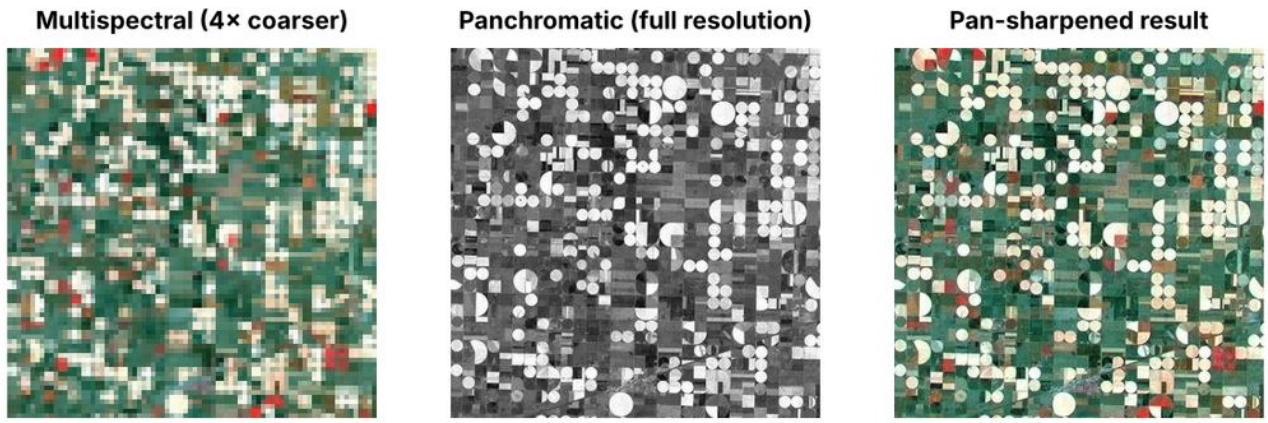


FIGURE 14.8 Pan-sharpening simulated with part of a Landsat scene: coarse multispectral data (left) and a sharp panchromatic image (centre) combine into a sharp colour image (right).

14.3.6 The Commercial VHR Landscape

The commercial VHR industry began in 1999, when **IKONOS** became the first commercial satellite to sell 1 m imagery. Since then, resolution has improved steadily, limited as much by regulation as by technology (Figure 14.9). In the USA, commercial imaging satellites are licensed by NOAA. The resolution permitted for sale was relaxed from 0.5 m to 0.25 m in 2014. In 2020 fixed limits were replaced by a tiered system that considers what is already available on the world market.

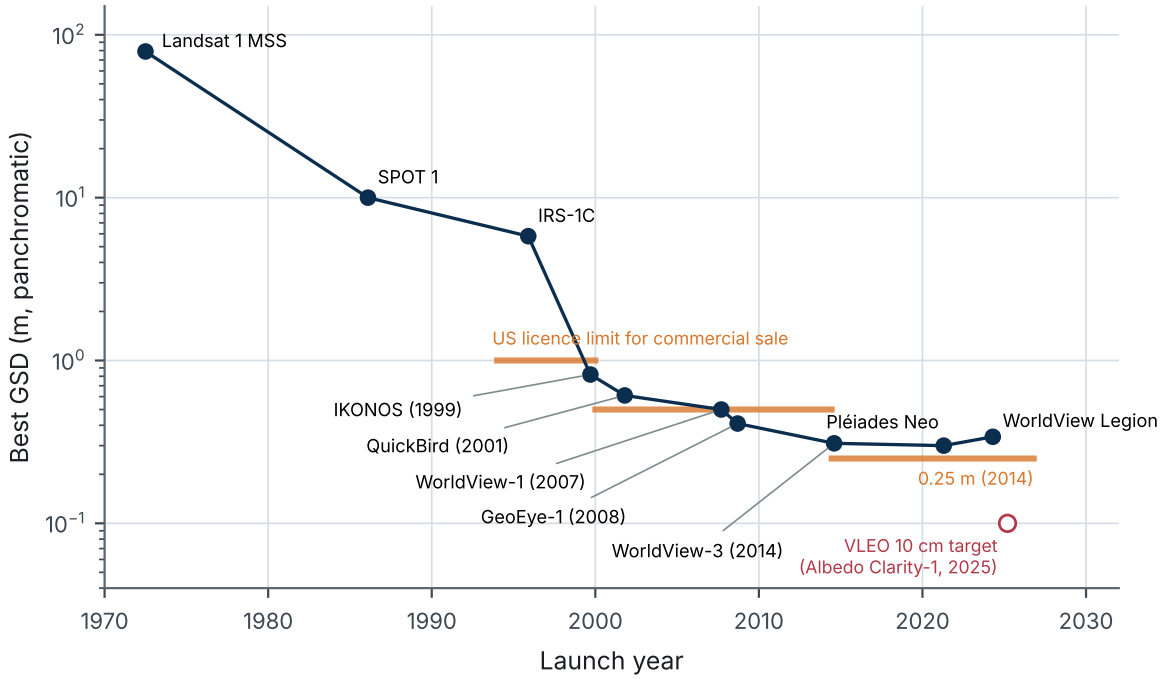


FIGURE 14.9 Best panchromatic GSD of civil and commercial imaging satellites by launch year, and the US licensing limit for commercial sale.

TABLE 14.3 Representative very-high-resolution imaging satellites in 2026 (approximate values from operators' published specifications).

System (operator, country)	First launch	Altitude	Pan / MS GSD	Swath	Notes
WorldView-3 (Vantor, USA)	2014	617 km	0.31 / 1.24 m	13 km	8 VNIR + 8 SWIR bands, CAVIS atmospheric sensor
WorldView Legion (Vantor, USA)	2024	518 km	0.34 / 1.36 m (0.29 m best)	10 km	Six ≈ 630 kg satellites (two SSO, four mid-inclination); rapid revisit
Pléiades Neo (Airbus, France)	2021	620 km	0.30 / 1.2 m	14 km	Two in orbit; two lost in a 2022 Vega C failure
Cartosat-3 (ISRO, India)	2019	≈ 510 km	≈ 0.25 / 1.1 m	16 km	Government system; data sold commercially
SuperView Neo-1 (Siwei/CASC, China)	2022	≈ 530 km	0.3 / 1.2 m	12 km	Part of a larger Chinese commercial fleet

System (operator, country)	First launch	Altitude	Pan / MS GSD	Swath	Notes
Jilin-1 (Chang Guang, China)	2015	500–650 km	0.5–1 m (several types)	10–150 km	Largest Chinese commercial constellation; video and wide-field types
KOMPSAT-3A (KARI, Korea)	2015	528 km	0.55 / 2.2 m	12 km	Also a 5.5 m mid-wave infrared camera
SkySat (Planet, USA)	2013	≈ 450 km	0.5 m (ortho)	≈ 6 km	Frame camera; up to 90 s video; many images per day
BlackSky Gen-3 (BlackSky, USA)	2025	≈ 450 km	0.35 m	≈ 5 km	Small agile satellites; rapid tasking
Pelican (Planet, USA)	2023	≈ 500 km	≈ 0.5 m (Gen 1); 0.3 m class (Gen 2)	—	On-board AI processing; up to 32 planned
NewSat Mark IV (Satellogic, Argentina/USA)	2020	≈ 470 km	≈ 0.7 m / ≈ 25 m hyperspectral	5 km	Low-cost microsatellites

The market is dominated by government customers. In the USA, the National Reconnaissance Office's **Electro-Optical Commercial Layer** contracts (awarded in 2022 to Maxar — now **Vantor** — BlackSky and Planet) buy commercial imagery for defence and intelligence users. Many other governments buy imagery or entire satellites. Since Russia's invasion of Ukraine in 2022, commercial imagery has been used openly by governments, news media and researchers on an unprecedented scale, demonstrating the role of commercial EO as an independent source of information.

14.3.7 Very Low Earth Orbit

Because resolution improves in proportion to altitude for a given aperture, some companies are pursuing **very low Earth orbit (VLEO)** — below about 400 km. Halving the altitude from 600 to 300 km halves the GSD for the same telescope, increases the light received per pixel area, and reduces the power needed for radar or optical communications. The penalties are atmospheric drag and atomic-oxygen erosion (Chapter 2): a satellite at 250–300 km must thrust almost continuously to stay in orbit. Precedents include ESA's GOCE gravity mission (2009–2013), which flew at about 255 km using an ion engine, and JAXA's **SLATS** (Tsubame), which in 2019 set a record by imaging from 167 km. The US company **Albedo** launched its first VLEO satellite, **Clarity-1**, in March 2025 to demonstrate 10 cm visible and 2 m thermal imaging. It operated for several months, but contact was later lost and it re-entered in June 2026. Air-breathing electric propulsion, which collects the thin atmosphere itself as propellant, is being studied as a long-term enabler (Chapter 42).

14.4 Constellations: Trading Resolution for Revisit

14.4.1 The PlanetScope Revolution

In 2013 the start-up **Planet** (then Planet Labs) began launching "Doves" — 3U CubeSats with a 9 cm telescope and a frame camera — in large "Flocks". The company's insight was that many cheap satellites in sun-synchronous orbits, each with a modest 24 km swath, could image **almost the entire landmass of the Earth every day** at about 3–4 m resolution. The current **SuperDove** generation records eight bands, including coastal blue, yellow and red-edge bands designed to be interoperable with Sentinel-2. Each satellite has a lifetime of only a few years, so the fleet is constantly replenished; in October 2026 about 90 Doves were active (Figure 14.10).

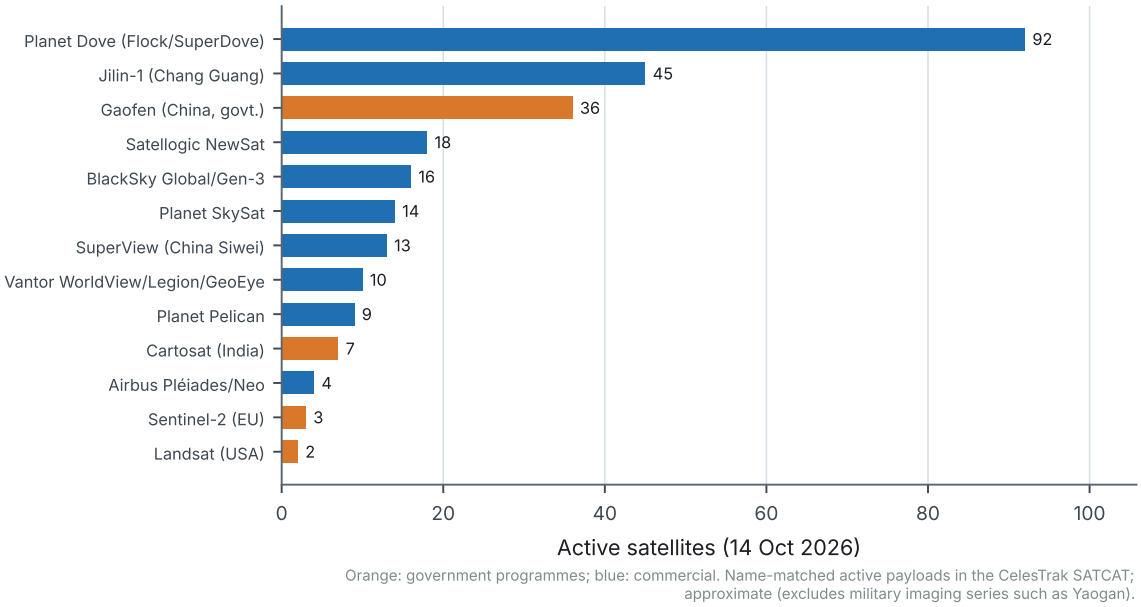


FIGURE 14.10 Active commercial and government optical imaging satellites by operator or series in October 2026 (name-matched in the CelesTrak SATCAT, approximate).

Daily imaging changed the questions that could be asked. Instead of a few cloud-free images per year, analysts can follow a construction site, a crop field, a mine or a refugee camp day by day. The trade-off is radiometric and geometric consistency: hundreds of small satellites with slightly different sensors require careful cross-calibration, often against Sentinel-2 and Landsat.

14.4.2 How Many Satellites for Daily Coverage?

The time to cover the whole equator with nadir-looking imagers can be estimated simply. Each satellite makes about 14.6 orbits per day, and each ascending (or descending) pass covers a strip as wide as its swath. Covering the 40 075 km equator therefore takes about

$$T \approx \frac{40\,075\ \text{km}}{14.6 \times \text{swath} \times N}\ \text{days}$$

for N satellites (Figure 14.11). Two Landsats with 185 km swaths need about 7–8 days, and two Sentinel-2s with 290 km swaths about 5. MODIS, with a 2 330 km swath, covers the Earth every 1–2 days with a single satellite. About 130 Doves with 24 km swaths achieve near-daily coverage. Agile VHR satellites do not cover the Earth systematically: their "revisit" is the time between

opportunities to point at a given target. This is typically less than a day for a constellation of a few satellites at different local times. For the mid-inclination WorldView Legion satellites over populated latitudes, it is down to about an hour.

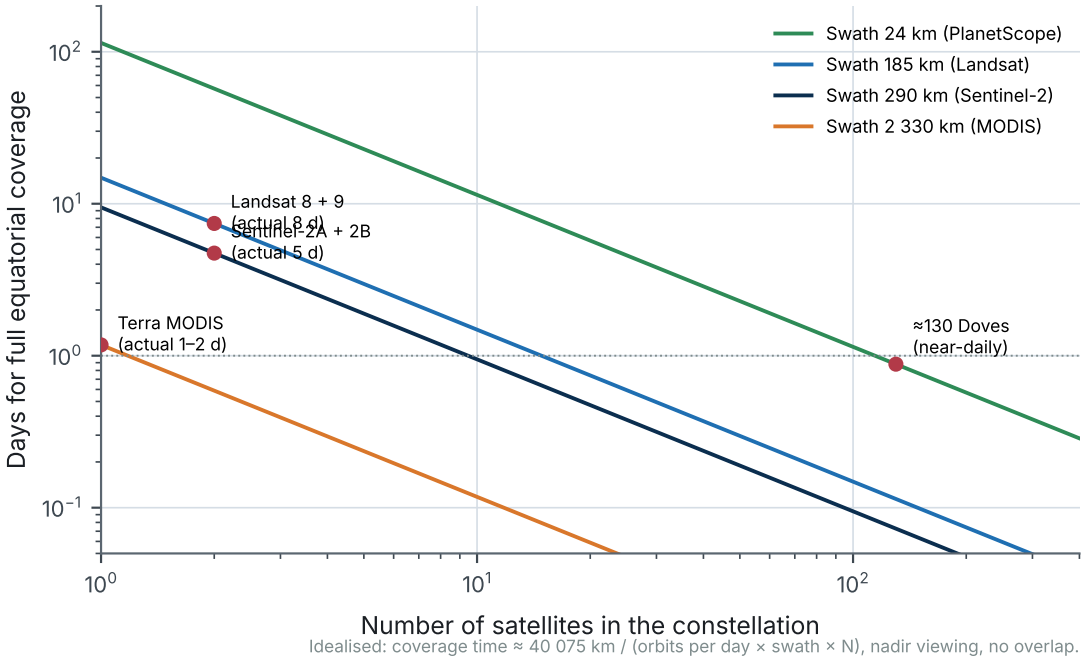


FIGURE 14.11 Idealised days needed for full equatorial coverage versus the number of satellites, for four swath widths, compared with real systems.

14.4.3 Data Volume and Delivery

Imaging satellites produce enormous amounts of data. A push-broom VHR imager with 0.3 m pixels and a 13 km swath generates about 23 000 lines per second, each of about 43 000 pixels — about 1 gigapixel per second before multispectral bands. On-board **compression** (JPEG 2000 or the CCSDS 122 image-compression standard, typically 2:1 to 4:1 with near-lossless quality) and large solid-state recorders (terabits) buffer the data until it can be downlinked. Downlink uses **X-band** (typically 0.5–3 Gbit/s on modern satellites) or **Ka-band** links to networks of ground stations at high latitudes (Svalbard, Inuvik, Troll in Antarctica) and around the world (Chapter 7). Some operators now relay data through **optical inter-satellite links** or data-relay satellites such as the European Data Relay System (EDRS), which relays Sentinel data from LEO via GEO at 1.8 Gbit/s.

For customers, the key metric is the time from **tasking request to image delivery**. Modern operators accept orders through web portals and application programming interfaces (APIs), plan collections automatically, and deliver orthorectified images within hours — sometimes minutes — of collection. Planet's Pelican and BlackSky's Gen-3 satellites include on-board processors that can run AI models to detect objects and send alerts before the full image is downlinked (Part XIII).

14.5 Thermal Infrared Imaging

14.5.1 Why Measure Temperature from Space?

Thermal infrared (TIR) sensors measure radiation emitted by the surface, from which **land-surface temperature (LST)** and **emissivity** are derived (Chapter 13). Surface temperature is one of the most informative variables about the land. It responds immediately to water stress (plants that cannot transpire heat up), drives urban heat islands, and reveals volcanic activity, geothermal resources and industrial heat sources. It is also the key input for mapping **evapotranspiration** — the water used by crops — which is essential for water management in arid regions.

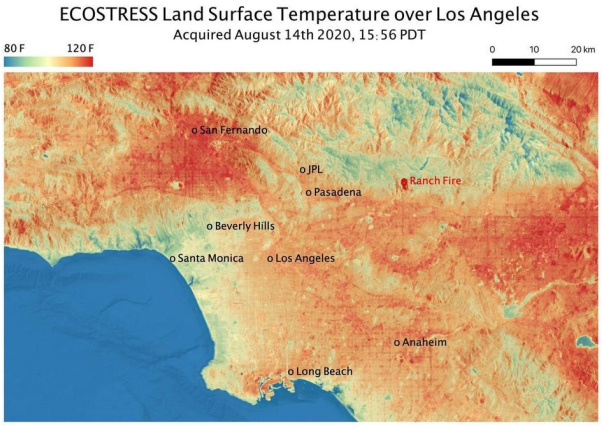


FIGURE 14.12 Land-surface temperature over Los Angeles during an August 2020 heat wave, measured by ECOSTRESS on the International Space Station. Streets and roofs exceeded 50 °C, while parks and the coast stayed far cooler. Image: NASA/JPL (NASA/JPL-Caltech).

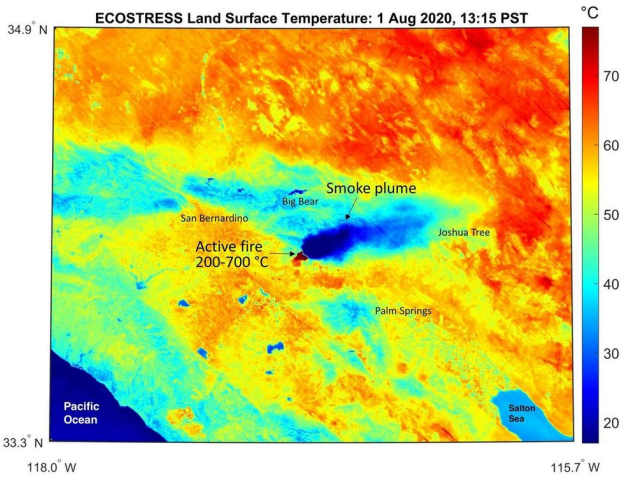


FIGURE 14.13 ECOSTRESS land-surface temperature during the Apple Fire in California, 1 August 2020. The active fire front exceeded 200 °C, and the smoke plume appears cooler than the surrounding ground. Image: NASA/JPL (NASA/JPL-Caltech).

14.5.2 Thermal Instrument Design

Thermal imaging is harder than visible imaging for three reasons:

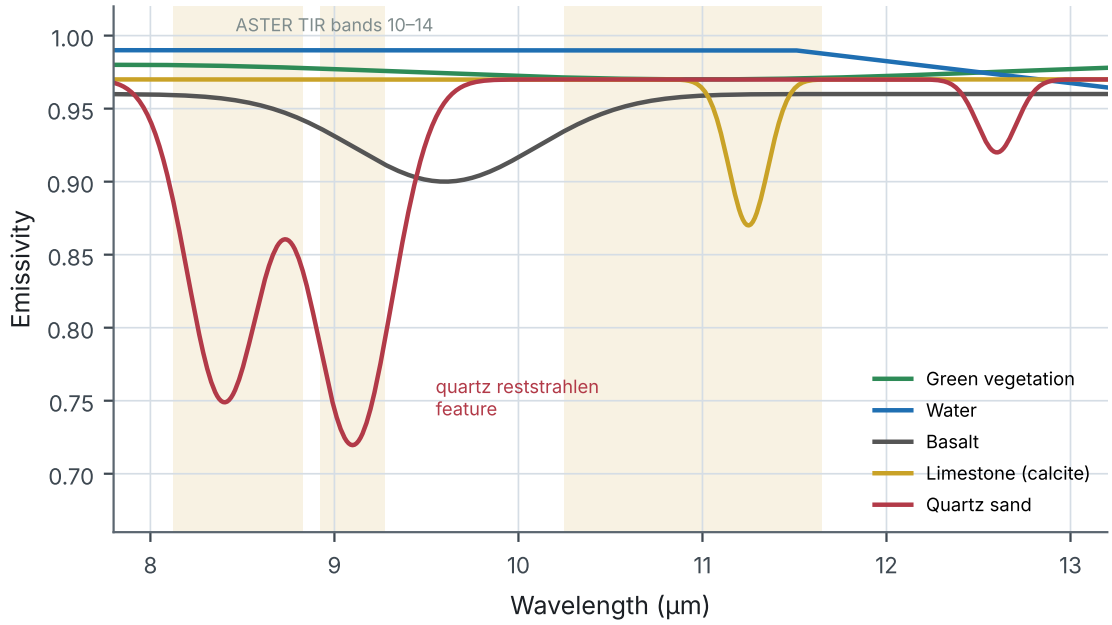
1. **Diffraction:** at 10 µm, the diffraction limit is about 17 times coarser than at 0.6 µm for the same aperture. So thermal pixels are larger (typically 60–100 m from LEO for science missions, 3–10 m from VHR systems with large apertures).
2. **Detectors:** sensitive photon detectors for the long-wave infrared (mercury cadmium telluride, HgCdTe; QWIPs; strained-layer superlattices) must be cooled to 40–80 K by mechanical cryocoolers. Uncooled **microbolometers**, which sense the heating of a tiny suspended membrane, are cheaper and smaller and are used by many small satellites, but are less sensitive and slower.
3. **Self-emission:** at room temperature, the telescope and instrument themselves glow in the thermal infrared. Thermal instruments therefore cool their optics or baffles, use cold stops, and calibrate frequently by viewing an on-board **blackbody** of known temperature and a cold reference (deep space).

The sensitivity of a thermal sensor is expressed as the **noise-equivalent temperature difference (NEΔT)** — 0.03–0.1 K for science instruments at 300 K.

14.5.3 Retrieving Temperature and Emissivity

The radiance measured by a thermal sensor depends on both the temperature and the emissivity of the surface, and on the atmosphere's emission and absorption — mainly by water vapour. Several methods untangle these:

- **Split-window algorithms** use two adjacent bands (around 11 and 12 µm) in which water vapour absorbs differently; the brightness-temperature difference between them corrects for the atmosphere. They are widely used for sea-surface temperature, where emissivity is known.
- **Temperature–emissivity separation (TES)** was developed for ASTER and is used for ECOSTRESS. It uses five or more bands and an empirical relationship between the spectral contrast and minimum emissivity to solve for both temperature and the emissivity spectrum.
- **Emissivity spectra** in the 8–12 µm range identify minerals (Figure 14.14): silicate rocks show strong "reststrahlen" features near 8–10 µm whose positions shift with silica content, while carbonates show features near 11.3 µm. Multispectral thermal data from ASTER, with five TIR bands, have been used extensively for geological mapping.



Schematic after the ECOSTRESS (ASTER) spectral library.

FIGURE 14.14 Thermal-infrared emissivity spectra of common surface materials (schematic) with the ASTER thermal bands. Quartz shows strong reststrahlen features; vegetation and water are nearly black bodies.

14.5.4 Detecting Fires

Active fires are among the most important things to detect from space. Their detection relies on Planck's law (Figure 14.15). A 1 000 K fire emits thousands of times more radiance than a 300 K background at **3.9 µm** (mid-wave infrared), but only about 30 times more at 11 µm. A fire covering just 0.01 % of a 1 km pixel (100 m^2) raises the pixel's 3.9 µm brightness temperature by about 10 K but its 11 µm temperature by only 0.2 K. Fire-detection algorithms therefore look for

pixels that are much hotter at 3.9 µm than at 11 µm and than their neighbours. They use the excess radiance to estimate the **fire radiative power (FRP)**, which is related to the rate at which biomass is burned.

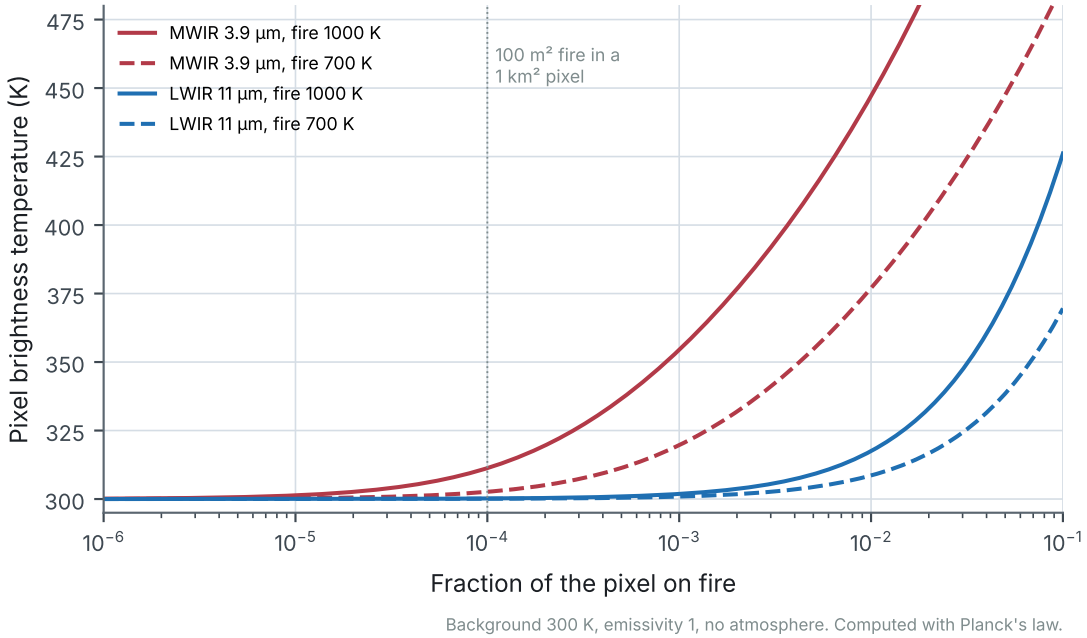


FIGURE 14.15 Brightness temperature of a pixel containing a small fire versus the fraction of the pixel burning. The 3.9 µm band is far more sensitive to small hot fires than the 11 µm band.

NASA's **Fire Information for Resource Management System (FIRMS)** distributes active fire detections from MODIS (1 km) and VIIRS (375 m) within about three hours of observation worldwide. Geostationary imagers (GOES-R, Himawari, Meteosat Third Generation) detect fires every 5–10 minutes (Chapter 16). Dedicated commercial constellations are now adding faster, finer fire detection. The German company **OroraTech** had 20 thermal satellites of its FOREST series in orbit by October 2026, aiming at a global revisit of about 30 minutes. The Earth Fire Alliance's **FireSat** constellation, developed with Muon Space and Google, aims to detect fires as small as 5 m × 5 m.

TABLE 14.4 Thermal infrared imaging missions (resolution and dates approximate).

Mission	Operator	Launch	TIR resolution	TIR bands	Status (2026)
ASTER (Terra)	NASA / Japan	1999	90 m	5	Operating; Terra nearing end of mission
Landsat 8 / 9 TIRS	NASA / USGS	2013 / 2021	100 m	2	Operating
ECOSTRESS (ISS)	NASA JPL	2018	≈ 70 m	5 (3 used since 2019)	Operating
VIIRS	NOAA / NASA	2011, 2017, 2022	375–750 m	7 MWIR/LWIR	Operating (Suomi NPP, NOAA-20, NOAA-21)
FOREST series	OroraTech (Germany)	2022–	≈ 200 m	MWIR + LWIR	20 satellites in orbit
HiVE (SkyBee)	constellr (Germany)	2025–	≈ 30 m class	4	First satellites in orbit
VanZyl	Hydrosat (USA)	2024–	≈ 50–70 m	Multi-band	Early constellation

Mission	Operator	Launch	TIR resolution	TIR bands	Status (2026)
TRISHNA	CNES / ISRO	late 2020s	57 m	4	In development
LSTM	EU Copernicus / ESA	≈ 2029	≈ 50 m	5	In development
SBG-TIR	NASA / ASI	≈ 2029	≈ 60 m	8	In development

14.5.5 Night-Time Lights

The **Day/Night Band (DNB)** of VIIRS is sensitive enough to image city lights, gas flares, fishing boats and moonlit clouds at night (Figure 14.16). NASA's **Black Marble** products correct these images for moonlight, atmosphere and seasonal effects to produce daily, monthly and annual maps of night-time lights. These maps are used to estimate economic activity and population, monitor power outages after disasters and conflicts, and track light pollution.

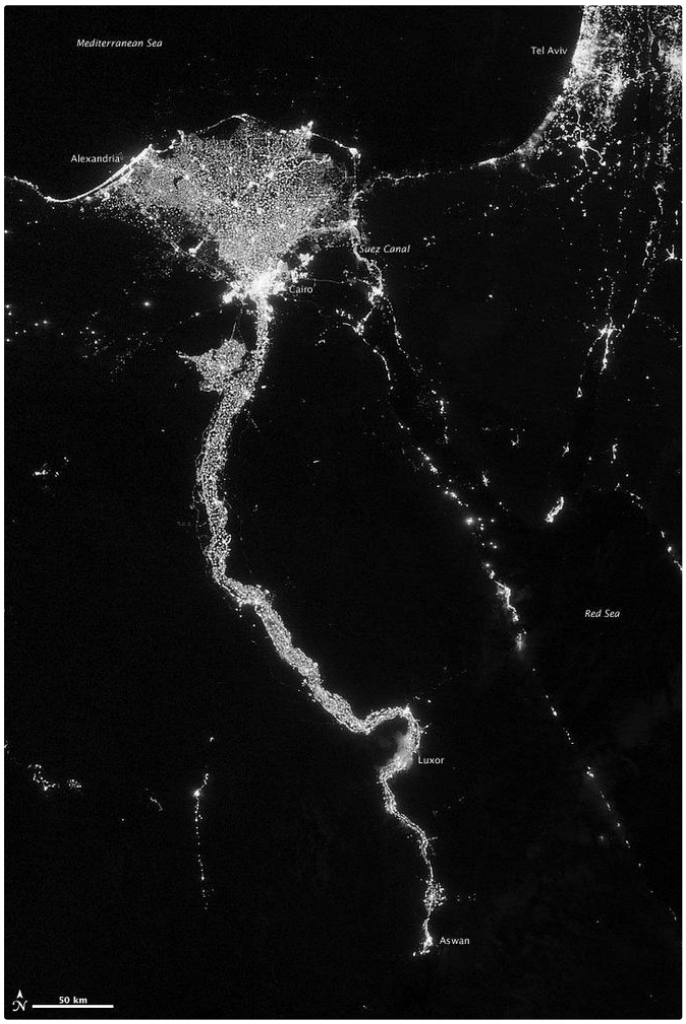


FIGURE 14.16 City lights along the Nile Valley and Delta imaged by the VIIRS Day/Night Band on 13 October 2012. About 97 % of Egypt's population lives on less than 5 % of its land, along the river, which stands out as a ribbon of light through the dark desert. Image: NASA/GSFC.

14.6 Imaging Spectroscopy (Hyperspectral)

14.6.1 The Hypercube

A **hyperspectral imager**, or **imaging spectrometer**, records a continuous spectrum for every pixel — typically 200–300 contiguous bands 5–10 nm wide covering 0.4–2.5 µm. The result is a three-dimensional **hypercube** with two spatial dimensions and one spectral dimension (Figure 14.17). Where a multispectral sensor can tell vegetation from soil, an imaging spectrometer can identify specific minerals (by their narrow absorption features) and estimate leaf chemistry (chlorophyll, water, nitrogen, lignin). It can also map phytoplankton types in water and detect gases such as methane and carbon dioxide by their absorption.

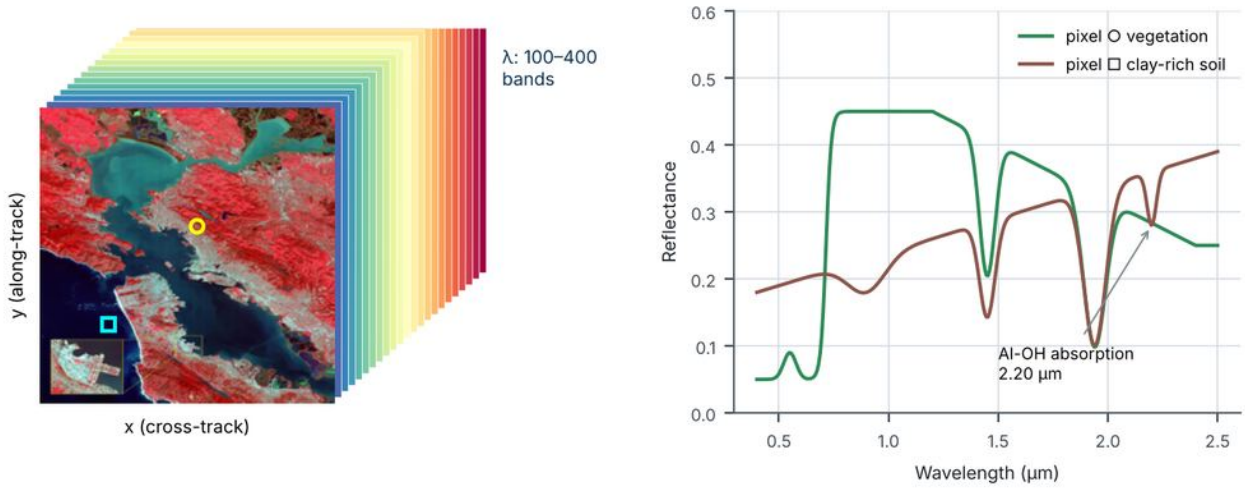


FIGURE 14.17 A hyperspectral data cube (left) records a full spectrum for every pixel (right). Narrow absorption features — here the 2.20 µm aluminium–hydroxyl feature of clay minerals — identify materials that broad-band sensors cannot distinguish.

Most imaging spectrometers are push-broom instruments in which a slit selects one line on the ground. A dispersing element then spreads the light from each point along the slit into a spectrum across a two-dimensional detector. This element is a prism or a diffraction grating, often in compact **Offner** or **Dyson** designs. Each detector frame thus contains one spatial line and its full spectrum. Achieving high SNR in hundreds of narrow bands requires large apertures and long dwell times, which is why space hyperspectral systems usually have 30–60 m pixels and narrow swaths (30–80 km). Calibration is demanding: errors of a fraction of a nanometre in the spectral position of bands ("smile" and "keystone" distortions) can corrupt the retrieval of narrow absorption features.

14.6.2 Missions

NASA's **Hyperion** on EO-1 (2000–2017) was the first civil spaceborne imaging spectrometer, with 220 bands at 30 m. Today, several missions are operating and more are coming (Table 14.5).

TABLE 14.5 Spaceborne imaging spectrometers (approximate characteristics).

Mission	Operator	Launch	GSD	Spectral range / bands	Notes
PRISMA	ASI (Italy)	2019	30 m (5 m pan)	0.4–2.5 µm, ≈ 240	Second generation in development

Mission	Operator	Launch	GSD	Spectral range / bands	Notes
DESIS (ISS)	DLR / Teledyne	2018	30 m	0.4–1.0 µm, 235	Commercial access
HISUI (ISS)	Japan METI	2019	20 × 31 m	0.4–2.5 µm, 185	
EnMAP	DLR (Germany)	2022	30 m	0.42–2.45 µm, > 220	Science mission, open data
EMIT (ISS)	NASA JPL	2022	60 m	0.38–2.5 µm, 285	Mineral dust sources; methane plumes
Tanager-1 / -2	Carbon Mapper / Planet / JPL	2024 / 2026	≈ 30 m	0.4–2.5 µm	Methane and CO_2 point sources
Firefly	Pixxel (India)	2025	≈ 5 m	VNIR, > 150	Commercial constellation
CHIME	EU Copernicus / ESA	≈ 2029	30 m	0.4–2.5 µm	Copernicus Expansion mission
SBG-VSWIR	NASA	2030s	≈ 30 m	0.4–2.5 µm	Recommended by the 2017 Decadal Survey

14.6.3 Seeing Methane

One of the most consequential new uses of imaging spectroscopy is mapping **methane plumes** from individual facilities — leaking oil and gas wells, pipelines, coal mines and landfills. Methane absorbs sunlight in narrow bands near 1.65 and 2.3 µm; a plume of excess methane slightly deepens these absorptions in the pixels it covers, and matched-filter algorithms detect this signature (Figure 14.18). Because methane is a powerful greenhouse gas with a short atmospheric lifetime (about a decade), finding and fixing large leaks — "super-emitters" — is one of the fastest ways to slow global warming. EMIT detected hundreds of methane plumes in its first years; Carbon Mapper's Tanager satellites and the commercial **GHGSat** constellation, which uses Fabry–Pérot interferometers with about 25 m resolution, specialise in this task. Chapter 16 describes broader mapping of methane and carbon dioxide over regions and the globe. This is done by spectrometers such as TROPOMI on Sentinel-5P, NASA's OCO-2/3, Japan's GOSAT series (including GOSAT-GW, launched in June 2025) and Europe's future CO2M mission. The Environmental Defense Fund's MethaneSAT, launched in 2024 to quantify emissions from whole oil and gas basins, was lost in June 2025 after about a year in orbit.

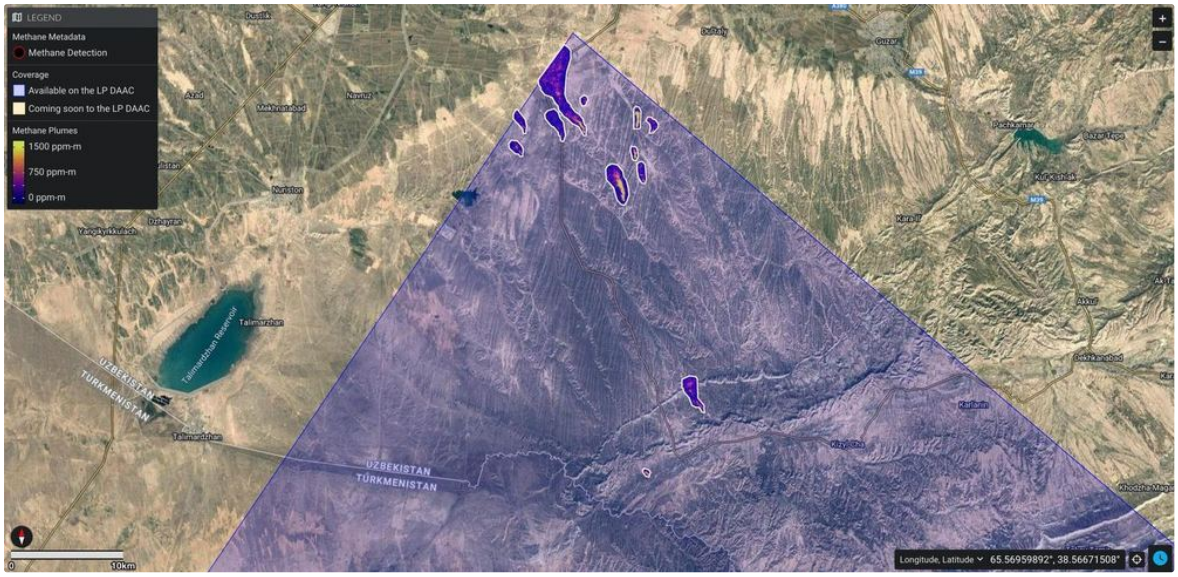


FIGURE 14.18 A cluster of 12 methane plumes detected by NASA's EMIT imaging spectrometer over a 400 km² oil and gas field in southern Uzbekistan on 1 September 2022. Each purple plume marks a source releasing methane at a high rate. Image: NASA/JPL (NASA/JPL-Caltech).

14.7 Global Imagers: MODIS, VIIRS and Their Successors

The **Moderate Resolution Imaging Spectroradiometer (MODIS)**, on NASA's Terra (1999) and Aqua (2002) satellites, has been one of the most productive scientific instruments ever flown. It has 36 spectral bands from 0.41 to 14.4 µm, at 250 m (two bands), 500 m (five bands) and 1 km. Its 2 330 km swath covers the whole Earth every one to two days. Its standard products — surface reflectance, vegetation indices, land cover, snow and ice, fires, aerosols, clouds, ocean colour and sea-surface temperature — have been used in tens of thousands of scientific papers. Terra and Aqua are well beyond their design lives and have run out of the fuel needed to maintain their overpass times, so their observations are drifting and will end in the next few years.

The **Visible Infrared Imaging Radiometer Suite (VIIRS)** continues the MODIS record operationally on the NOAA–NASA Suomi NPP (2011), NOAA-20 (2017) and NOAA-21 (2022) satellites. It has 22 bands at 375 m and 750 m, and a 3 040 km swath that leaves no gaps between orbits even at the equator. In Europe, **Sentinel-3**'s OLCI (ocean and land colour) and SLSTR (sea and land surface temperature) instruments provide similar global observations with an emphasis on climate-quality accuracy. These sensors are described further in Chapter 16.

FIGURE 14.19 The Hyperion imaging spectrometer on NASA's EO-1 satellite views Iceland's Eyjafjallajökull volcano on 2 May 2010. Visible bands (left) show only ash plume and snow; combinations using short-wave infrared bands (centre and right) reveal the hot lava flow extending north from the vent. Image: NASA/JPL (NASA/JPL/EO-1 Mission/GSFC/Ashley Davies).

14.8 Design Example: A Small VHR Imaging Satellite

EXAMPLE: SIZING A 0.5 M SMALL-SATELLITE IMAGER

Requirement: 0.5 m panchromatic GSD from a 500 km orbit, 6 km swath.

1. **Focal length.** With a CMOS TDI detector of 5.5 µm pixels: $f = p\,h/\mathrm{GSD} = 5.5 \times 10^{-6} \times 500\,000/0.5 = 5.5$ m.
2. **Aperture.** The Rayleigh limit at 0.6 µm for 0.5 m is $D = 1.22\lambda h/r = 0.73$ m. Choosing a 0.5 m aperture (f/11) gives an optical sampling ratio $Q = \lambda f/(D\,p) = 1.2$ — slightly under-sampled relative to the optics, a common compromise that keeps the telescope compact. The system MTF at Nyquist will be about 0.1.
3. **Swath and detector.** 6 km / 0.5 m = 12 000 pixels across the swath, for example three staggered 4 096-pixel TDI arrays.
4. **Line rate.** Ground speed at 500 km ≈ 7.06 km/s, so the line rate is 7 060 / 0.5 ≈ 14 100 lines per second.
5. **Signal.** Using the method of Figure 14.4, a 0.5 m aperture with 0.5 m pixels collects roughly 1 000 photo-electrons per line time in the panchromatic band. With 16–32 TDI stages, this rises to 16 000–32 000 electrons and a shot-noise SNR above 100 for bright targets.
6. **Data rate.** 12 000 pixels × 14 100 lines/s × 12 bits ≈ 2.0 Gbit/s for the panchromatic band, plus about 25 % for four multispectral bands at 2 m. With 3:1 compression this is about 0.85 Gbit/s; a 10-second, 70 km long image produces about 1 GB.
7. **Downlink.** An X-band link at 800 Mbit/s over a 6-minute ground-station pass can return about 36 GB per pass — some 36 such images.
8. **Agility.** To collect 20 targets per pass, the satellite must slew roughly 1–2° per second and settle within a few seconds — requiring CMGs or high-torque reaction wheels and a rigid structure with good jitter isolation.

The result is a satellite of roughly 150–250 kg — similar to commercial systems such as BlackSky Gen-3 or Planet's Pelicans.

AI ON EO/IR SATELLITES

Optical imaging is where AI in space is most mature. On-board neural networks screen out cloudy images before downlink (ESA's Φ-sat-1 first demonstrated this in 2020), detect ships, aircraft and vehicles, and generate alerts within minutes of collection. On the ground, deep learning performs super-resolution, pan-sharpening, cloud removal and automatic change detection over entire countries. Chapters 45 and 46 discuss these applications in detail.

KEY TAKEAWAYS

- EO/IR satellites range from global imagers (≈ 1 km) through medium-resolution workhorses (10–30 m) to very-high-resolution systems (≈ 0.3 m).
- Landsat (since 1972) and Sentinel-2 (since 2015) provide free, well-calibrated imagery of the whole land surface every few days and are combined in harmonised products.
- Resolution is limited by diffraction ($r \approx 1.22\lambda h/D$) and by the number of photons per pixel; VHR systems need large apertures, long focal lengths and time-delay integration.
- Agile satellites point off nadir to collect many targets, area mosaics and stereo pairs, at the cost of coarser GSD at large angles.
- Constellations of small satellites trade resolution for revisit: about 90 Doves image the land surface almost daily at 3–4 m.
- Thermal infrared sensors measure surface temperature and emissivity, essential for water use, urban heat, geology and fire detection; fire detection exploits the extreme sensitivity of the 3.9 µm band to small hot sources.
- Imaging spectrometers record hundreds of narrow bands, identifying minerals, plant chemistry and methane plumes from individual facilities.
- MODIS is ending after more than two decades; VIIRS and Sentinel-3 continue the global record.

REVIEW QUESTIONS

1. Calculate the diffraction-limited ground resolution of a telescope with a 1.1 m aperture at 0.55 µm from 500 km altitude.
2. What is time-delay integration, and why do very-high-resolution imagers need it?
3. Why does ground sample distance degrade when an agile satellite images far off nadir?
4. How are stereo pairs collected, and what products are made from them?
5. Explain the trade between resolution and revisit using the PlanetScope constellation as an example.
6. Why is the 3.9 µm band so sensitive to small fires? Use Planck's law in your answer.
7. What is an imaging spectrometer's "hypercube", and how can imaging spectroscopy detect methane from individual facilities?
8. Why are Landsat and Sentinel-2 data often combined, and what must be done to harmonise them?

15 CHAPTER Synthetic Aperture Radar



Image: NASA/JPL (NASA/JPL-Caltech)

Synthetic aperture radar (SAR) satellites carry their own illumination — microwave pulses — and so can image the Earth day and night, through clouds, smoke and rain. By recording the echoes of thousands of pulses as the satellite moves and processing them together, a SAR synthesises an antenna kilometres long and achieves resolutions as fine as 16 cm from hundreds of kilometres away. This chapter explains how SAR works — pulse compression, aperture synthesis, imaging geometry and modes — how radar waves interact with the surface, and why SAR images look so different from photographs. It then covers SAR interferometry, which measures ground movements of millimetres and maps topography, the main applications, and the rapidly growing fleet of government and commercial radar satellites, from Sentinel-1 and NISAR to constellations of 100 kg smallsats.

IN THIS CHAPTER

15.1 Why Radar?
15.2 Radar Fundamentals
15.3 Frequency and Polarisation
15.4 How the Surface Looks to a Radar
15.5 Imaging Modes
15.6 SAR Interferometry
15.7 Applications
15.8 SAR Missions in 2026
15.9 Design Example: An X-band SAR Smallsat

15.1 Why Radar?

Optical and infrared sensors are passive: they need sunlight or thermal emission and they cannot see through clouds, which cover about two-thirds of the Earth at any moment. In the humid tropics, weeks or months can pass without a single clear optical image. A **radar** (radio detection and ranging) sends its own microwave pulses and measures the echoes. Because microwaves

with wavelengths of centimetres to decimetres pass almost unaffected through clouds, rain (except at the shortest wavelengths), smoke, dust and darkness, an imaging radar can observe any place on Earth at any time. This makes SAR indispensable for disaster response, maritime surveillance, sea-ice monitoring, defence and any application that requires guaranteed acquisitions.

Radar also measures things that optical sensors cannot. The echo depends on the **roughness**, **geometry** and **moisture** of the surface rather than its colour, and longer wavelengths **penetrate** vegetation, dry soil and ice. Because the radar records the exact **phase** of the echo, two radar images can be combined to measure distance changes of a few millimetres. This is the basis of **SAR interferometry (InSAR)**.

15.1.1 A Short History

Side-looking airborne radars were developed for military reconnaissance in the 1950s, and the synthetic-aperture principle was invented by Carl Wiley at Goodyear in 1951. The first civilian imaging radar in space was carried by NASA's **Seasat** in 1978, which operated for only 106 days but proved the value of SAR for oceanography and land studies (Figure 15.1). The Shuttle Imaging Radar experiments (SIR-A 1981, SIR-B 1984, and the multi-frequency, multi-polarisation **SIR-C/X-SAR** in 1994, Figure 15.2) and the Soviet **Almaz** missions followed. The era of operational SAR began with ESA's **ERS-1** (1991), Japan's **JERS-1** (1992) and Canada's **RADARSAT-1** (1995). In 2000 the **Shuttle Radar Topography Mission (SRTM)** mapped the topography of 80 % of the Earth's land in 11 days. Today, dozens of radar satellites are in orbit (Figure 15.3).

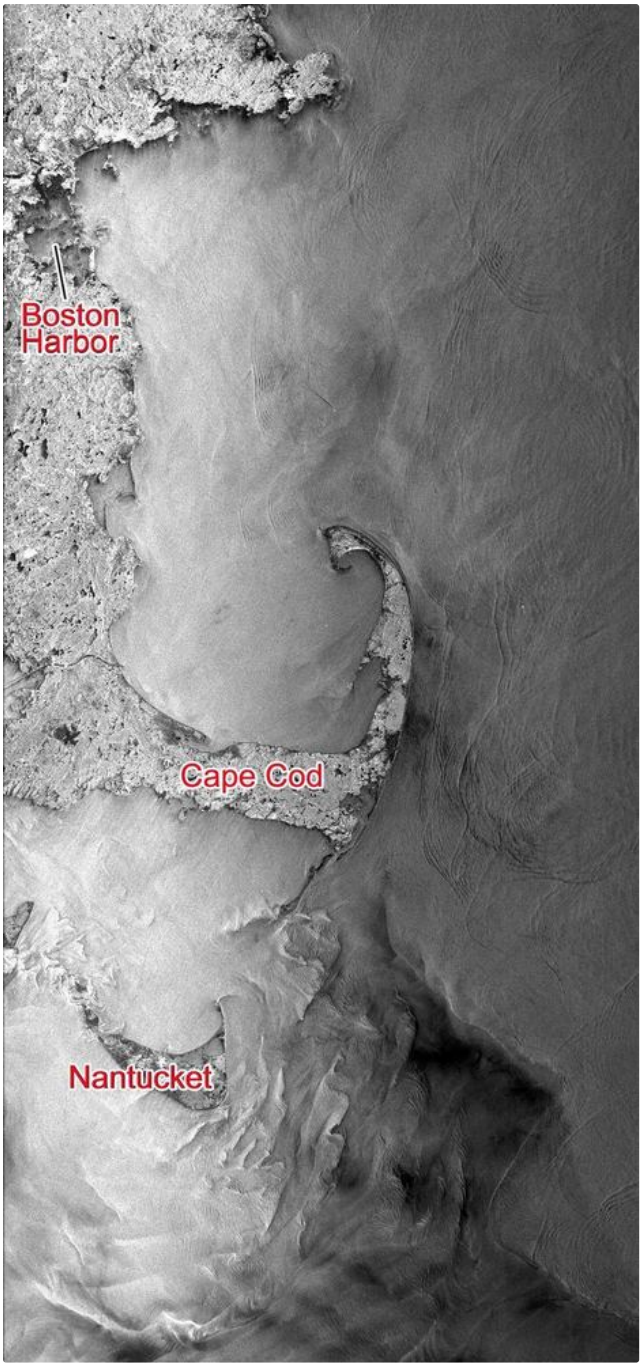


FIGURE 15.1 Seasat's L-band SAR image of the Massachusetts coast and Cape Cod, acquired in 1978. Seasat was the first civilian satellite to carry an imaging radar. Image: NASA/JPL (NASA/JPL-Caltech/Alaska Satellite Facility).



FIGURE 15.2 The SIR-C/X-SAR antenna — 12 m long and weighing 11 tonnes — being prepared for its first Space Shuttle flight in 1994. It was the first spaceborne SAR to image in three frequencies (L, C and X) and multiple polarisations. Image: NASA/JSC.

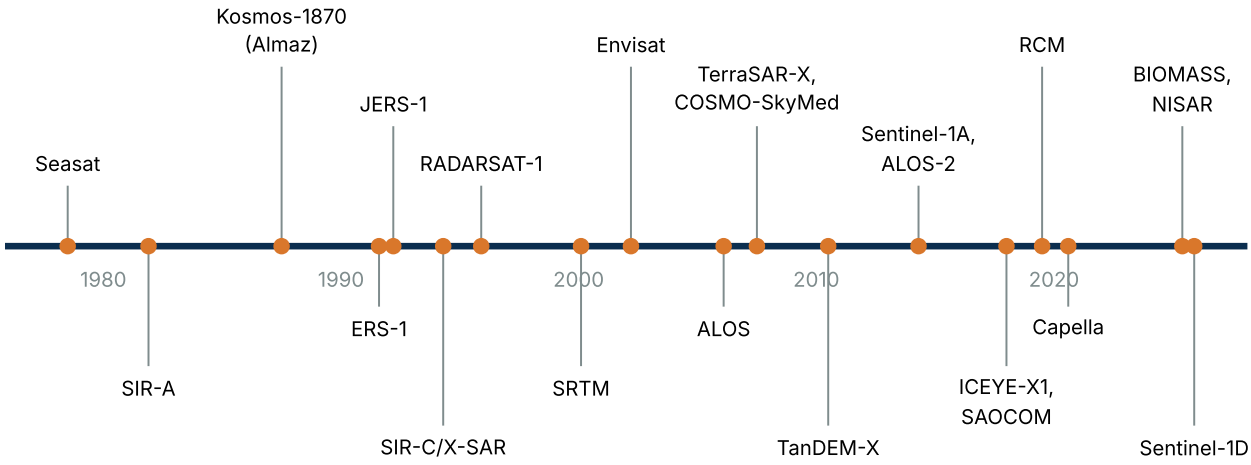


FIGURE 15.3 Milestones in spaceborne SAR, from Seasat to the 2025 launches of NISAR, BIOMASS and Sentinel-1D.

15.2 Radar Fundamentals

15.2.1 Ranging and the Radar Equation

A radar transmits a short pulse and measures the delay τ of its echo; the distance (range) to the target is $R = c\tau/2$. The power of the echo is given by the **radar equation**:

$$P_r = \frac{P_t\, G^2\, \lambda^2\, \sigma}{(4\pi)^3\, R^4\, L}$$

where P_t is the transmitted power, G the antenna gain, λ the wavelength, σ the **radar cross-section** of the target (m^2) and L the system losses. The R^4 term reflects that the signal spreads out on the way to the target and again on the way back. It makes spaceborne radar demanding: an echo from 800 km is 10^{12} times weaker than from 800 m. For extended natural surfaces, the radar cross-section is expressed per unit area as the dimensionless **backscatter coefficient** σ^0 (sigma-nought), usually quoted in decibels. It ranges from about −25 dB for calm water to +5 dB or more for urban areas.

15.2.2 Imaging Geometry

An imaging radar looks **sideways**, perpendicular to its direction of flight (Figure 15.4). Looking straight down would be useless: points equally distant on both sides of the ground track would return their echoes at the same time and could not be separated. The direction of flight is called **azimuth** (or along-track); the direction of the radar beam is called **range**. The angle between the beam and the vertical at the satellite is the **look angle**. The angle at the ground between the beam and the local vertical is the **incidence angle**, which is slightly larger because of the Earth's curvature. The strip of ground illuminated between **near range** and **far range** is the **swath**.

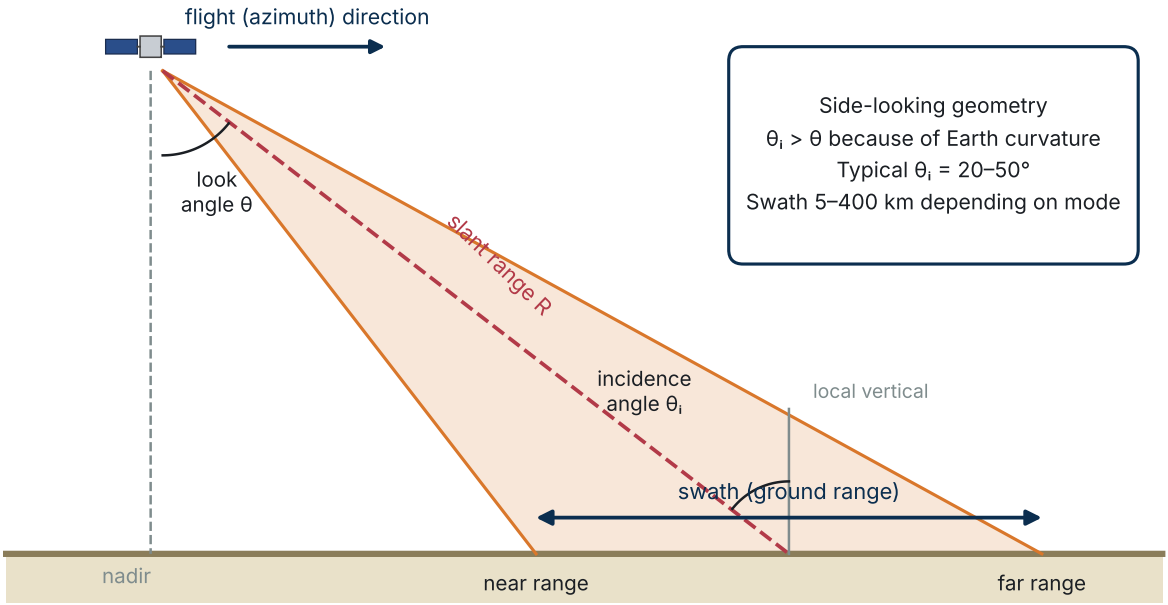


FIGURE 15.4 Side-looking SAR imaging geometry. The radar measures distance along the slant range; images are later projected onto the ground.

15.2.3 Range Resolution and Pulse Compression

Two targets can be separated in range if their echoes do not overlap. For a pulse of duration τ_p, the slant-range resolution is $c\tau_p/2$; a 10 µs pulse would give 1.5 km. To achieve metre-scale resolution with enough energy, radars transmit a long pulse whose frequency changes linearly with time — a **chirp** — and compress it on reception with a **matched filter** (Figure 15.5). The resolution then depends only on the bandwidth B of the chirp:

$$\delta_r = \frac{c}{2B}, \qquad \delta_g = \frac{c}{2B\sin\theta_i}$$

where δ_g is the ground-range resolution. A bandwidth of 50 MHz gives 3 m slant-range resolution; modern X-band systems use up to 600–1 200 MHz for resolutions of 25 cm or better. Pulse compression is a beautiful example of signal processing replacing raw power. The transmitter can operate at modest peak power for a long pulse, while the receiver recovers the resolution of a pulse hundreds of times shorter.

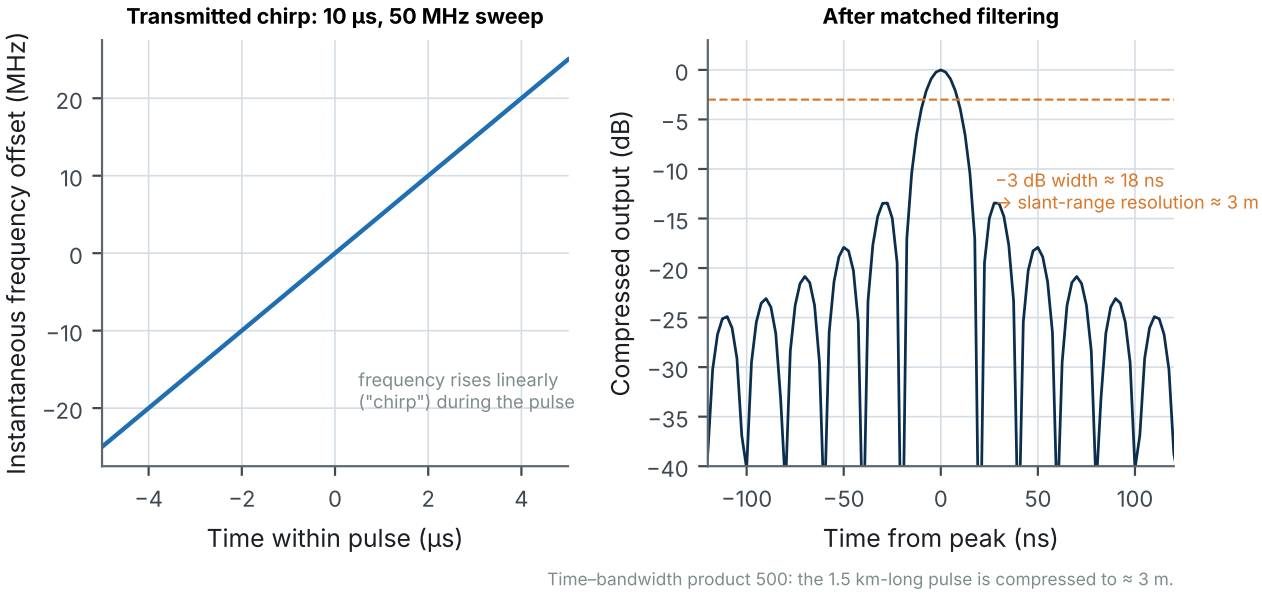


FIGURE 15.5 Pulse compression. A 10 µs chirp sweeping 50 MHz (left) is compressed by matched filtering into a peak about 18 ns wide (right), giving 3 m slant-range resolution. The sidelobes at −13 dB are usually reduced further by weighting.

15.2.4 The Synthetic Aperture

In the azimuth direction, a real antenna of length L forms a beam of angular width about λ/L. At range R the beam footprint — the real-aperture azimuth resolution — is $\lambda R/L$: for a C-band radar (λ = 5.6 cm) with a 10 m antenna at 850 km range, about 4.7 km. To achieve 5 m resolution with a real aperture would require an antenna 10 km long.

The **synthetic aperture** principle solves this (Figure 15.6). As the satellite moves, each point on the ground stays inside the beam for a time during which the satellite travels a distance $L_{sa} = \lambda R/L$ — the length of the **synthetic aperture**. During this time the radar transmits thousands of pulses and records the amplitude and phase of each echo. Because the distance to the target first decreases and then increases, the echo's phase follows a characteristic quadratic history (equivalently, its **Doppler frequency** sweeps from positive to negative). A processor matches this phase history — just as pulse compression matches the chirp in range — and focuses all the echoes from each target into a single sharp point. The resulting azimuth resolution for stripmap imaging is

$$\delta_a \approx \frac{L}{2}$$

a remarkable result: the resolution is **independent of range and wavelength**, and *smaller* antennas give *finer* resolution (because their wider beams illuminate each target for longer). A 10 m antenna gives 5 m resolution whether the satellite is at 500 or 800 km.

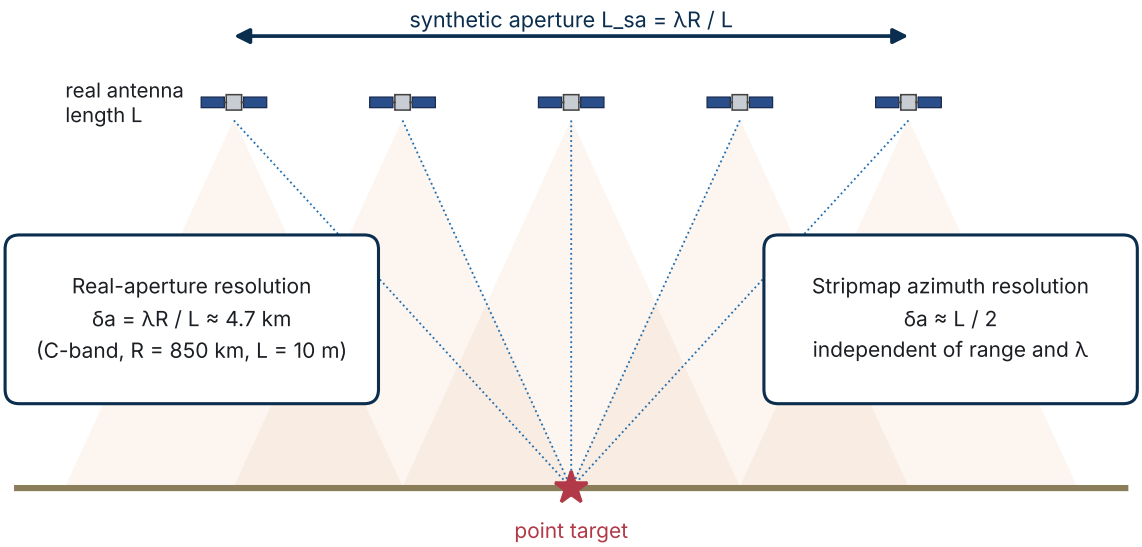


FIGURE 15.6 Aperture synthesis. Each target is illuminated by many successive pulses as the satellite passes; coherent processing combines them into a virtual antenna kilometres long.

Turning raw echoes into an image — **SAR focusing** — requires the processing of enormous amounts of data with precise knowledge of the satellite's orbit. Classical algorithms include the **range–Doppler**, **chirp scaling** and **omega-k** algorithms, which work efficiently in the frequency domain using fast Fourier transforms. The **back-projection** algorithm is computationally heavier but exact for any geometry, and has become practical with GPUs. Focusing a single Sentinel-1 scene takes minutes on a modern server; some new satellites focus images on board.

15.2.5 Constraints: PRF, Ambiguities and Antenna Size

The radar must transmit pulses at a **pulse repetition frequency (PRF)** high enough to sample the Doppler bandwidth of the echoes. This means at least one pulse each time the satellite moves half an antenna length, $\mathrm{PRF} > 2v/L$. But the PRF must be low enough that the echo from the whole swath arrives before the next pulse is sent. Violating these conditions creates **ambiguities**: ghost images displaced in azimuth or range. Together they imply a **minimum antenna area**:

$$A = L\,W > \frac{4\,v\,\lambda\,R\,\tan\theta_i}{c}$$

which grows with wavelength (Figure 15.7). C-band and L-band satellites therefore need antennas of 10–30 m², while X-band smallsats can use antennas of about 1–2 m². Wide swaths, fine resolution and low ambiguities cannot all be achieved simultaneously with a conventional single-beam radar — the central trade-off of SAR design. New techniques overcome it. One is **digital beamforming** with multiple receive channels (as in Germany's planned HRWS mission and Sentinel-1 Next Generation). Another is the **SweepSAR** technique of NASA–ISRO's NISAR, which illuminates a wide swath and uses a large reflector with an array feed to "sweep" a narrow receive beam across it, following the echoes.

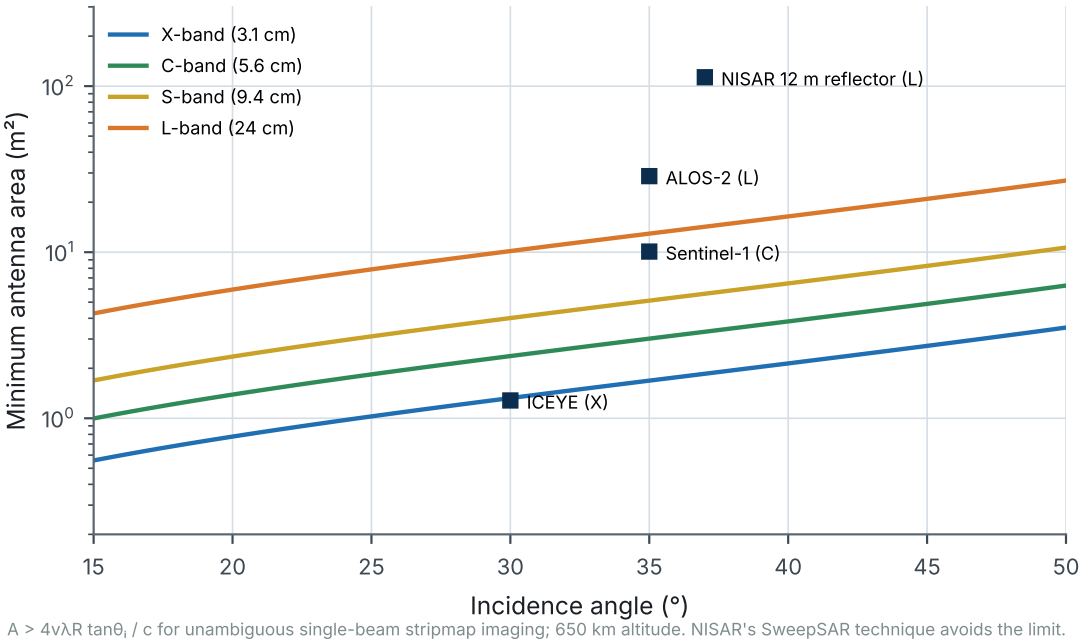


FIGURE 15.7 Minimum antenna area for unambiguous single-beam imaging versus incidence angle, compared with real systems.

The sensitivity of a SAR is described by its **noise-equivalent sigma-zero (NESZ)** — the backscatter coefficient that gives a signal equal to the noise. Values of −20 to −25 dB are typical of large government SARs; smallsats achieve −15 to −20 dB. Surfaces darker than the NESZ, such as calm water, appear as noise.

15.3 Frequency and Polarisation

15.3.1 Radar Bands

The choice of wavelength is the most fundamental decision in SAR design (Table 15.1). Shorter wavelengths give finer resolution with small antennas and respond to small-scale roughness and the tops of vegetation canopies. Longer wavelengths penetrate deeper into vegetation, soils and ice and remain coherent over longer periods for interferometry (Figure 15.8).

TABLE 15.1 Radar frequency bands used for Earth observation (IEEE letter designations).

Band	Frequency	Wavelength	Main characteristics	Example missions
Ka	26.5–40 GHz	0.75–1.1 cm	Very fine resolution; affected by rain; altimetry	SWOT (KaRIn interferometer)
Ku	12–18 GHz	1.7–2.5 cm	Altimeters, scatterometers, precipitation radar	CryoSat-2, Sentinel-3/6 altimeters
X	8–12 GHz	2.4–3.8 cm	High resolution with small antennas; canopy-top scattering	TerraSAR-X, COSMO-SkyMed, ICEYE, Capella, Umbra
C	4–8 GHz	3.8–7.5 cm	Good all-round compromise; oceans, ice, land	ERS, Envisat, RADARSAT, Sentinel-1, Gaofen-3
S	2–4 GHz	7.5–15 cm	Intermediate penetration	NISAR (S-band), NovaSAR-1, Kondor-FKA

Band	Frequency	Wavelength	Main characteristics	Example missions
L	1–2 GHz	15–30 cm	Penetrates vegetation; long-term coherence for InSAR	ALOS-2/4, SAOCOM, NISAR (L-band)
P	420–450 MHz (space allocation)	≈ 69 cm	Penetrates forest canopy to trunks; biomass	BIOMASS

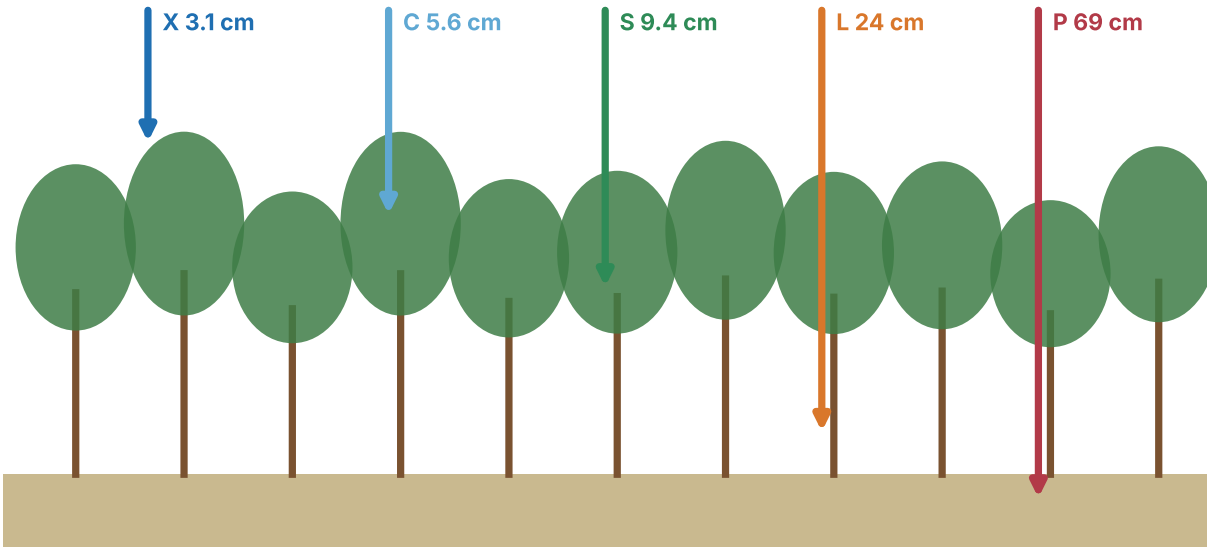


FIGURE 15.8 Schematic penetration of radar waves of different wavelengths into a forest. X-band scatters mainly from leaves at the top of the canopy, while L- and P-band reach the branches, trunks and ground.

15.3.2 Polarisation

Radar waves are polarised: the electric field oscillates in a particular plane. Spaceborne SARs transmit and receive horizontally (H) or vertically (V) polarised waves. An image is described by its transmit and receive polarisations: **HH**, **VV** (co-polarised) and **HV**, **VH** (cross-polarised). Single-polarisation radars record one channel, **dual-pol** radars two (for example VV + VH, Sentinel-1's standard over land), and **quad-pol** (fully polarimetric) radars all four with their relative phases. Some missions use **compact polarimetry**, transmitting circular polarisation and receiving two linear polarisations, as an efficient compromise (RADARSAT Constellation Mission).

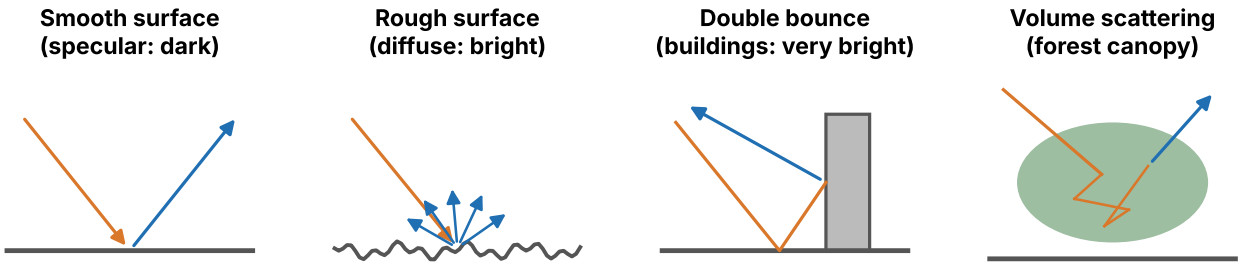
Polarisation reveals the scattering mechanism. Smooth surfaces and the ocean return mostly co-polarised energy; volume scattering by randomly oriented branches and leaves **depolarises** the wave, producing strong cross-polarised (HV) returns — making HV an excellent indicator of vegetation and biomass. **Polarimetric decompositions** (such as the Pauli, Freeman–Durden and Cloude–Pottier entropy/alpha decompositions) separate each pixel's response into surface, double-bounce and volume components, often displayed as red–green–blue colour images.

15.4 How the Surface Looks to a Radar

15.4.1 Brightness: Roughness, Geometry and Moisture

The brightness of a SAR image depends on how much energy is scattered back towards the radar (Figure 15.9):

- **Smooth surfaces** (calm water, roads, runways, smooth sand) reflect the wave away like a mirror and appear **dark**. A surface is "smooth" if its height variations are small compared with the wavelength (the Rayleigh criterion), so the same surface can be smooth at L-band and rough at X-band.
- **Rough surfaces** (ploughed fields, wind-roughened water, rocky terrain) scatter in all directions and appear **brighter**.
- **Double-bounce** reflection — wave bouncing off the ground and then a vertical wall back to the radar — makes buildings, ships, bridges and flooded forests appear **very bright**. Corner reflectors used for calibration exploit this effect.
- **Volume scattering** within vegetation canopies, dry snow or sea ice produces moderate, depolarised returns.
- **Moisture** increases the dielectric constant of soils and vegetation and therefore their reflectivity, so wet soil is brighter than dry soil. This allows SAR to measure soil moisture and to detect flooding under vegetation.



Orange: incident wave from the radar (upper left); blue: scattered energy.

FIGURE 15.9 The main scattering mechanisms that determine the brightness of SAR images.

Backscatter also depends strongly on the incidence angle (Figure 15.10). Calm water reflects strongly only near vertical incidence and is black at normal SAR angles. This is why oil slicks, which damp the small capillary waves on the sea, appear as dark patches and can be detected. Forests are relatively insensitive to angle, while soils and the wind-roughened sea darken steadily with increasing incidence angle. The ocean's backscatter depends on wind speed and direction, which is how SAR and scatterometers measure **ocean winds**.

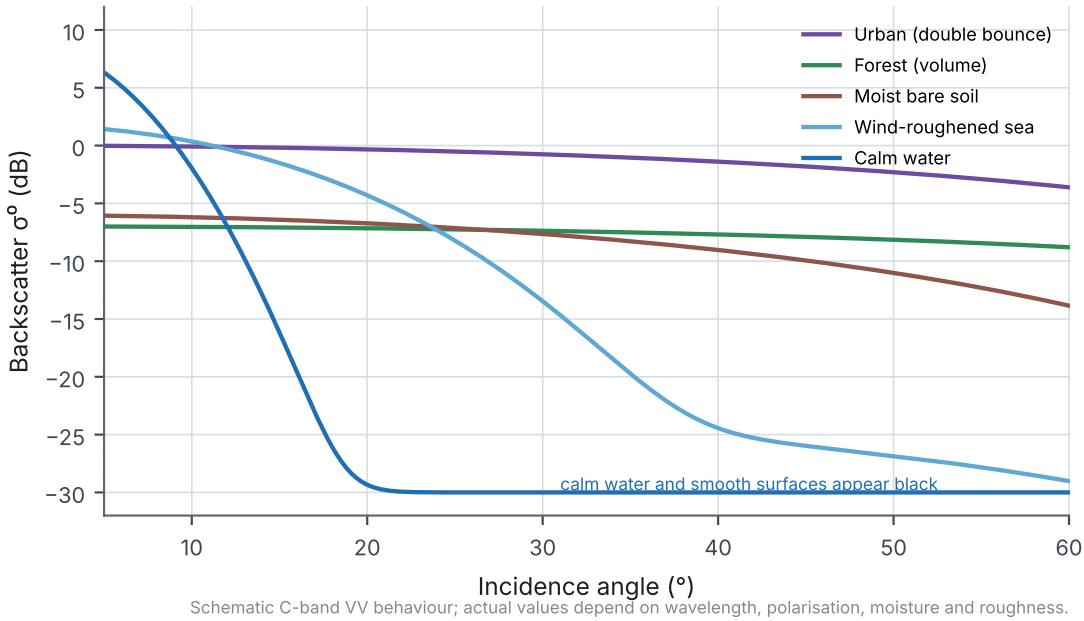


FIGURE 15.10 Schematic variation of backscatter with incidence angle for different surfaces at C-band.

15.4.2 Geometric Distortions

Because a radar measures distances rather than angles, terrain is distorted in characteristic ways (Figure 15.11):

- **Foreshortening**: slopes facing the radar appear compressed and bright.
- **Layover**: when a slope facing the radar is steeper than the incidence angle, its top is closer to the radar than its bottom and is imaged *before* it. Mountains appear to lean towards the radar, and tall buildings lay over onto the street in front of them.
- **Shadow**: slopes facing away from the radar that are steeper than the grazing angle receive no energy and appear black.

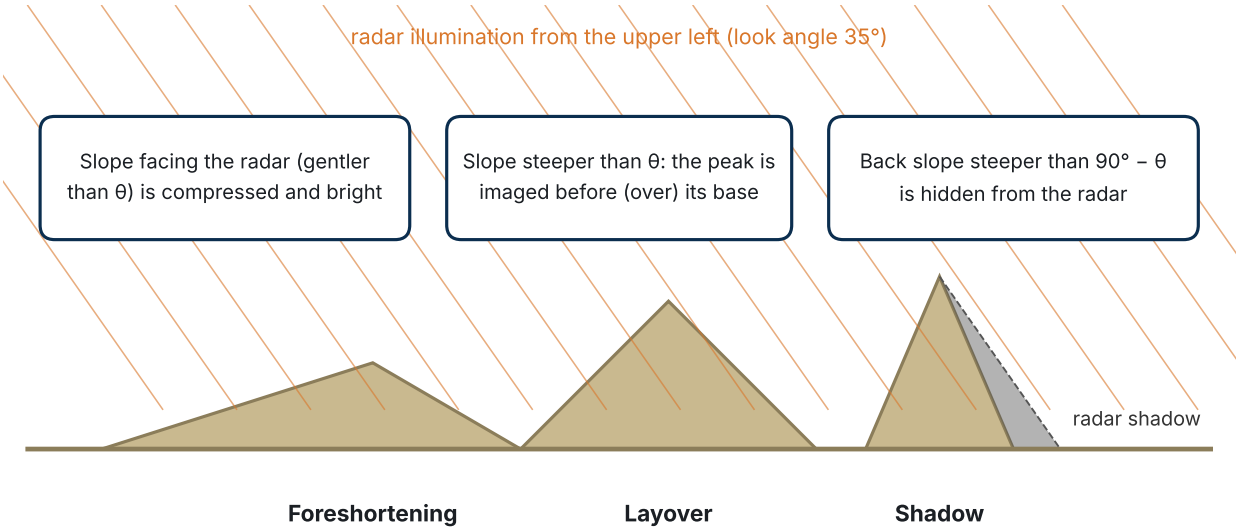


FIGURE 15.11 Radar geometric distortions in mountainous terrain: foreshortening, layover and shadow.

Distortions are corrected by **terrain-corrected geocoding** using a DEM. Brightness variations due to slope can also be corrected — **radiometric terrain correction** converts backscatter to the "terrain-flattened" coefficient γ^0 — but layover and shadow areas contain no recoverable

information. Combining images from ascending and descending passes, which look from opposite directions, fills many gaps.

15.4.3 Speckle

SAR images have a characteristic grainy appearance called **speckle** (Figure 15.12). Each resolution cell contains many individual scatterers whose echoes add coherently with random phases; the sum varies randomly from pixel to pixel even over a uniform surface. Speckle is not noise in the usual sense. It is a real, deterministic consequence of coherent imaging that repeats if the same scene is imaged under identical conditions. But it makes single-look images hard to interpret. Speckle is reduced by **multi-looking** (averaging several independent looks, at the cost of resolution), by averaging images over time, and by adaptive filters (Lee, Frost, non-local means, and deep-learning despeckling).

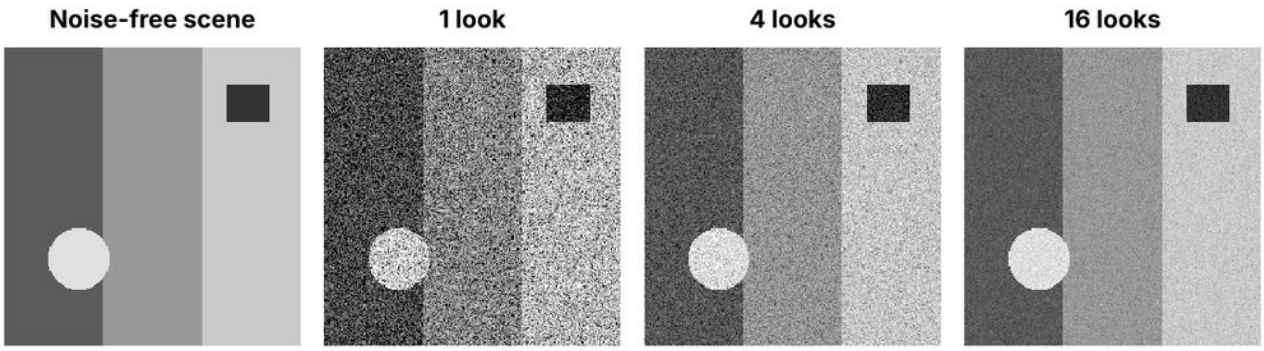


FIGURE 15.12 Speckle in a simulated SAR scene. Averaging 4 or 16 independent looks reduces the speckle but also reduces resolution.

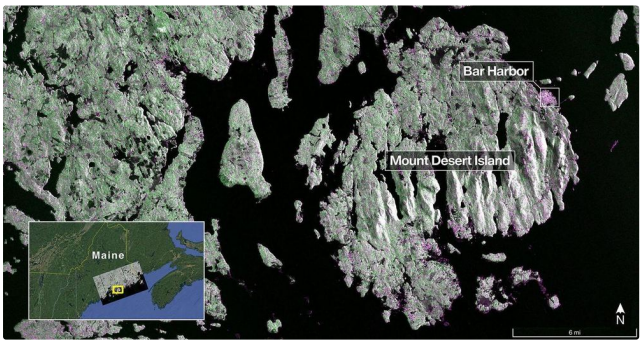


FIGURE 15.13 The first L-band image from NISAR, acquired on 21 August 2025, showing Mount Desert Island on the coast of Maine. Forests appear bright, water dark and urban areas very bright. Image: NASA/JPL (NASA/JPL-Caltech).

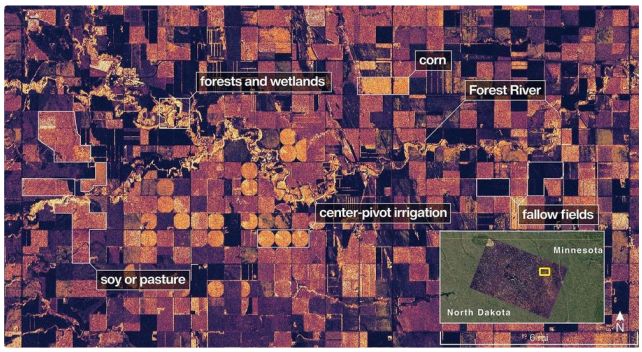
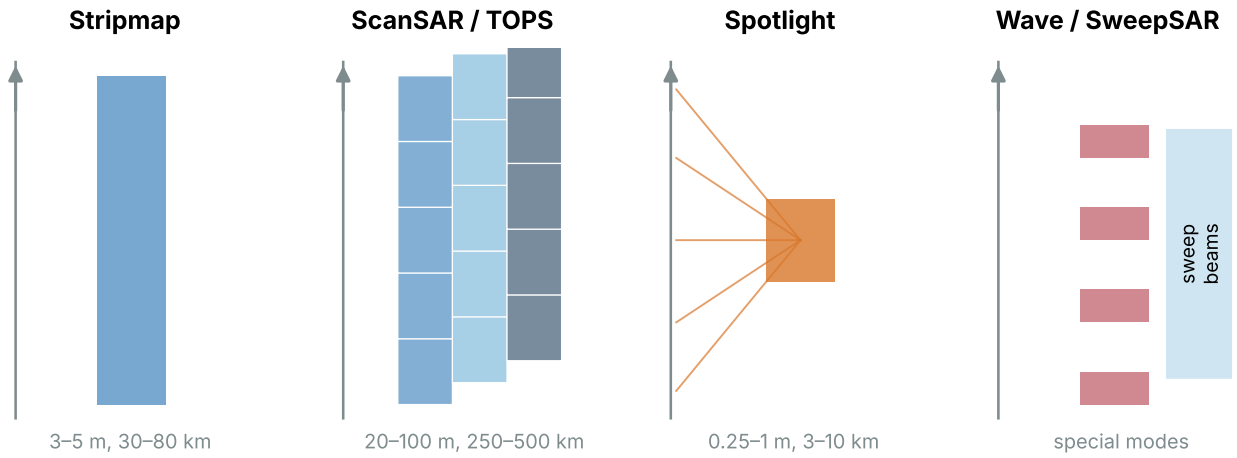


FIGURE 15.14 An early NISAR image of farmland and wetlands in North Dakota. Fields with different crops, moisture and roughness show distinct brightness; centre-pivot fields appear as circles. Image: NASA/JPL (NASA/JPL-Caltech).

15.5 Imaging Modes

By steering its beam electronically (with a phased-array antenna) or mechanically (by rotating the spacecraft), a SAR can trade resolution against swath width in several **imaging modes** (Figure 15.15):

- **Stripmap**: the beam points at a fixed angle and sweeps a continuous strip; resolution about half the antenna length (typically 3–5 m), swath 30–80 km.
- **ScanSAR**: the beam switches periodically between several adjacent sub-swaths, illuminating each in short bursts. The swath becomes several times wider (250–500 km) at the cost of coarser azimuth resolution. ESA's **TOPS** (Terrain Observation by Progressive Scans) mode, used by Sentinel-1's Interferometric Wide-swath (IW) mode, also steers the beam forward-to-backward during each burst, giving uniform image quality across the swath — important for interferometry.
- **Spotlight**: the beam is steered backwards during the pass to keep illuminating the same small area for longer, synthesising a longer aperture and achieving resolutions of 0.25–1 m over a few kilometres. **Staring spotlight** and **dwell** modes extend this further.
- **Wave mode** collects small vignettes over the open ocean to measure wave spectra, and specialised modes such as NISAR's **SweepSAR** combine wide swaths with fine resolution.



Grey arrow: satellite ground track and direction of flight. The radar looks to one side.

FIGURE 15.15 Schematic SAR imaging modes. Grey arrows show the ground track; the radar looks to the right.

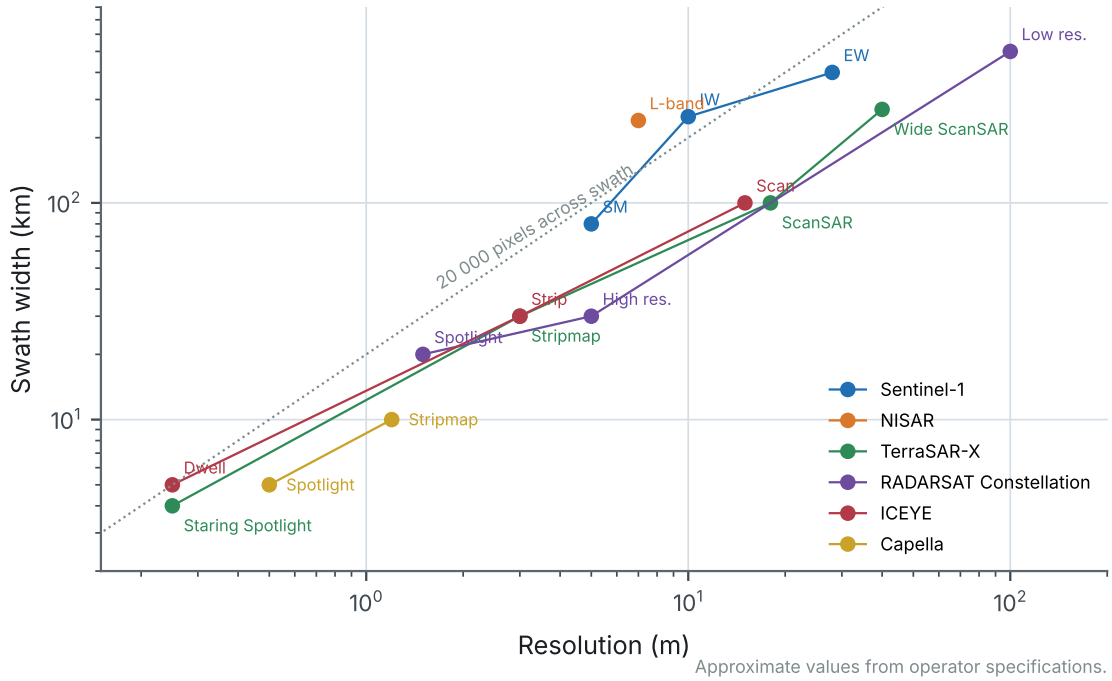


FIGURE 15.16 Resolution versus swath width for the imaging modes of representative SAR missions. Each system offers a family of modes along a line of roughly constant number of pixels across the swath.

15.6 SAR Interferometry

15.6.1 The Principle

A SAR image is a matrix of complex numbers. Each pixel has an **amplitude** (brightness) and a **phase**. The phase depends on the exact two-way distance between the antenna and the scatterers in the pixel, measured in fractions of a wavelength. The phase of a single image looks random, because it includes the unknown arrangement of scatterers within each pixel. But if two images are acquired from nearly the same position, this random part is almost identical in both. The **phase difference** between them — the **interferogram** — then depends only on the difference in distance (Figure 15.17):

$$\Delta\phi = \frac{4\pi}{\lambda}\,\Delta R$$

One complete cycle of phase (a **fringe**) corresponds to a change in distance of half a wavelength — 2.8 cm at C-band and 12 cm at L-band. Since phase can be measured to a small fraction of a cycle, InSAR measures distance changes of **millimetres** from hundreds of kilometres away.

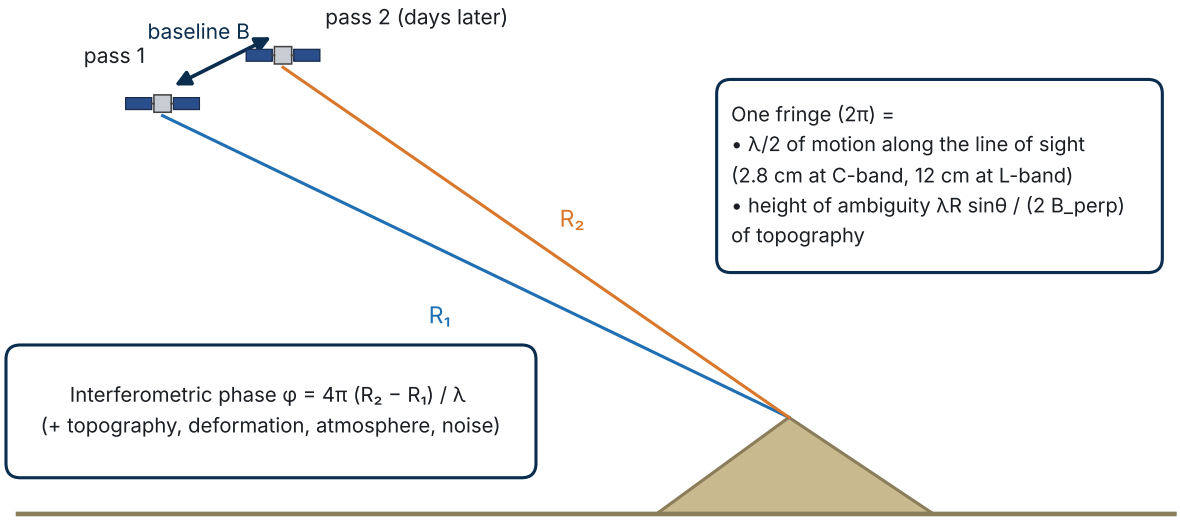


FIGURE 15.17 Geometry of SAR interferometry. The phase difference between two acquisitions depends on topography (through the baseline geometry) and on any movement of the ground between them.

The interferometric phase has several contributions. One is the **topography** (because the two antennas view the terrain from slightly different positions separated by the **baseline**). Others are any **deformation** of the ground along the line of sight between acquisitions, differences in **atmospheric delay** (mainly water vapour in the troposphere and electrons in the ionosphere), orbit errors, and noise. Different applications isolate different components.

15.6.2 Measuring Topography

The two images can also be acquired at the same time from two antennas separated by a known baseline — **single-pass** or **bistatic** interferometry. Then there is no deformation or atmospheric change, and the phase directly measures height. The **height of ambiguity**, the height change that produces one fringe, is $h_a = \lambda R \sin\theta / (2B_\perp)$, where $B_\perp$ is the perpendicular baseline; longer baselines give greater height sensitivity.

The **Shuttle Radar Topography Mission** (February 2000) used a C-band antenna in the Shuttle's payload bay and a second receive antenna on the end of a 60 m mast. It mapped 80 % of the Earth's land surface between 60° N and 56° S, producing the first near-global high-resolution DEM (Figure 15.18). Germany's **TanDEM-X** mission (2010) flew a second satellite in close formation, a few hundred metres from TerraSAR-X. It produced the global **WorldDEM** / Copernicus DEM with 12 m posting and a vertical accuracy of a few metres — the most accurate consistent global elevation model.

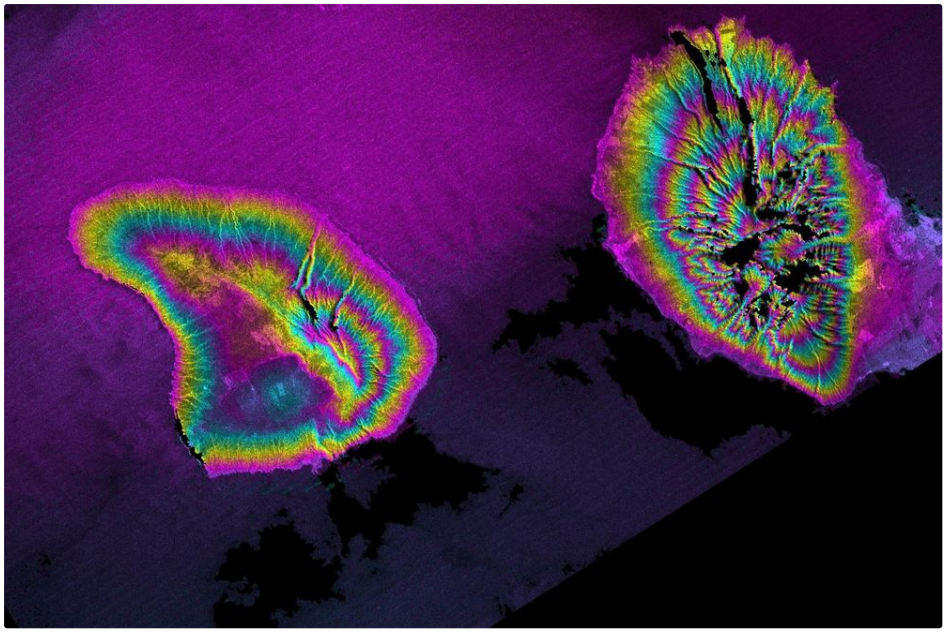


FIGURE 15.18 SRTM interferometric height of Lanai and West Maui, Hawaii. Each cycle of colours represents a fixed increment of elevation. Image: NASA/JPL (NASA/JPL).

15.6.3 Measuring Deformation

Differential InSAR (DInSAR) uses two images taken days to years apart by the same satellite from nearly the same orbit. After the topographic phase is removed using a DEM, the remaining fringes show how the ground moved between the acquisitions. The first spectacular demonstration was the 1992 Landers, California, earthquake, imaged by ERS-1 (published in *Nature* in 1993). Since then InSAR has mapped the deformation of almost every major earthquake (Figure 15.19), volcanic inflation and deflation (Figure 15.20), landslides, and glacier and ice-sheet flow. It has also mapped subsidence caused by groundwater pumping, mining, oil and gas extraction and tunnelling.

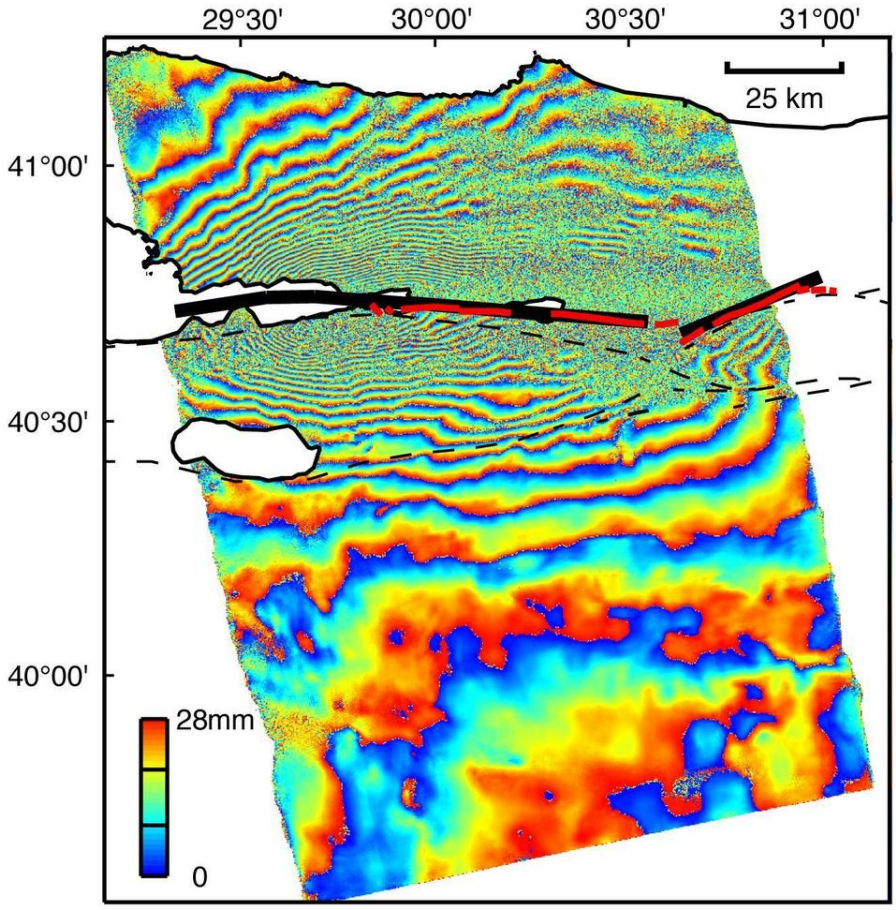


FIGURE 15.19 Wrapped interferogram from ERS-1 and ERS-2 data showing ground displacement from the magnitude 7.6 Izmit, Turkey, earthquake of 17 August 1999. Each colour cycle represents 28 mm of motion towards or away from the satellite. Image: NASA/JPL (NASA/JPL).

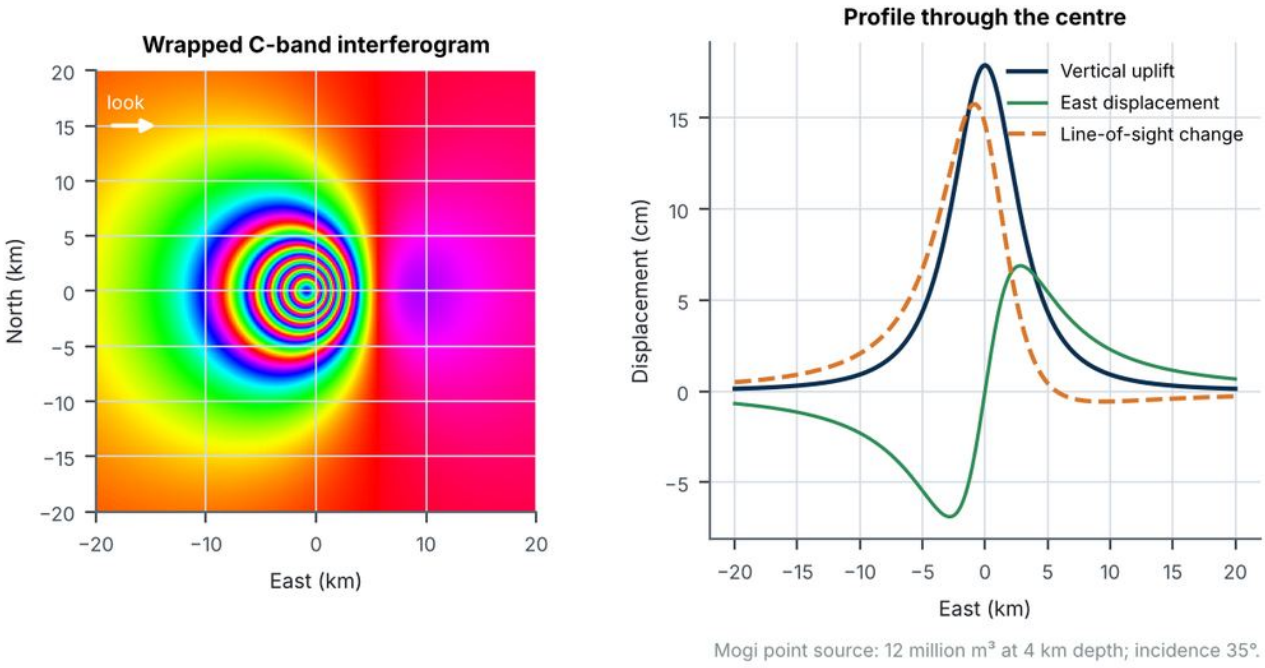


FIGURE 15.20 Synthetic interferogram (left) and displacement profile (right) for a volcano inflating because of 12 million m³ of magma injected at 4 km depth (Mogi model). The radar measures only the component of motion along its line of sight, so the fringe pattern is asymmetric.

15.6.4 Coherence, Time Series and Accuracy

InSAR works only if the scatterers within each pixel remain arranged in the same way between acquisitions — that is, if the images remain **coherent**. **Coherence** (0 to 1) measures this. It is high over cities and bare rock, but decays over vegetation as leaves move and plants grow, and is destroyed by snowfall, ploughing or flooding (Figure 15.21). Longer wavelengths stay coherent longer, which is why L-band missions such as ALOS-2 and NISAR are preferred over vegetated land, and why short revisit intervals (6–12 days) are valuable. Loss of coherence is itself useful: **coherence change detection** reveals damaged buildings after earthquakes and disturbed ground.

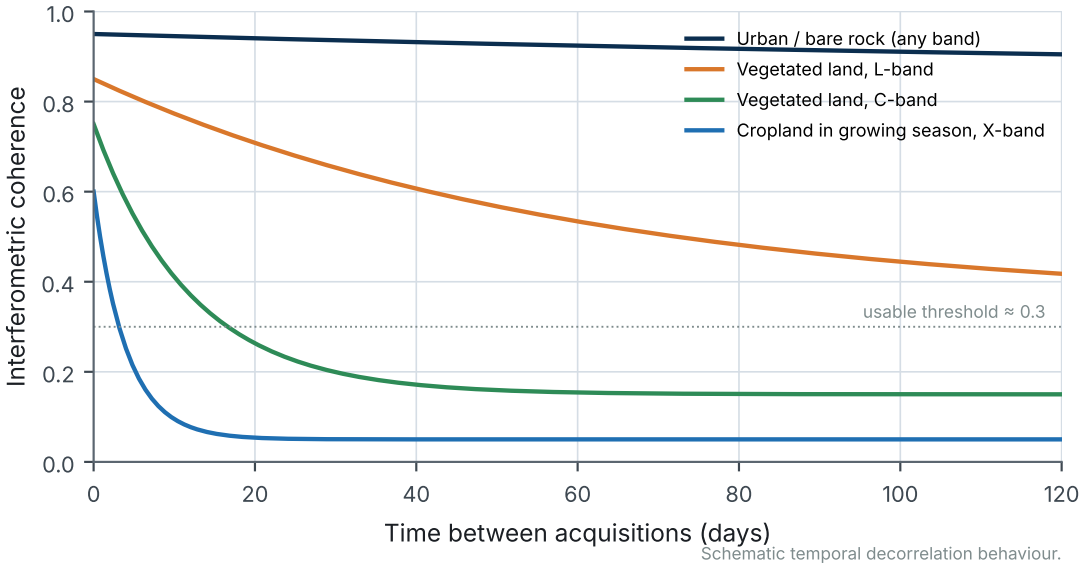


FIGURE 15.21 Schematic decay of interferometric coherence with time between acquisitions for different surfaces and wavelengths.

The biggest error source for deformation measurements is the **atmosphere**: variations in water vapour delay the radar signal by amounts equivalent to several centimetres. **Multi-temporal InSAR** techniques overcome this by analysing stacks of dozens to hundreds of images. One method, **persistent scatterer interferometry (PSI)**, identifies stable point-like targets (buildings, rocks, infrastructure) and estimates their motion history with millimetre-per-year precision. By contrast, **small baseline subset (SBAS)** methods analyse distributed targets. The **European Ground Motion Service (EGMS)**, launched by Copernicus in 2022, provides Sentinel-1 PSI ground-motion measurements for almost all of Europe, updated annually. Its billions of measurement points show subsidence of cities, motion of landslides and the stability of dams, bridges and railways.

Other interferometric techniques include **along-track interferometry**, which uses two antennas separated in the flight direction. It measures the velocity of moving targets such as ocean currents and vehicles (ground moving target indication, Chapter 19). Another is **SAR tomography**, which combines many baselines to resolve the vertical structure of forests and buildings — a key technique of ESA's BIOMASS mission. Where coherence is low, **offset tracking** measures large displacements (metres) by correlating image patches, for example to map glacier velocities.

15.7 Applications

SAR has become essential in many fields (Table 15.2).

TABLE 15.2 Major applications of spaceborne SAR.

Field	Applications	Preferred band and technique
Disasters	Flood mapping (water appears dark), earthquake and landslide deformation, building damage, volcanic monitoring	All bands; amplitude change, coherence change, DInSAR
Oceans	Ship detection, oil-spill detection, ocean wind and waves, currents, internal waves	C and X; wave mode; along-track interferometry
Sea ice and polar regions	Ice type and concentration charts, iceberg tracking, ice-sheet velocity and grounding lines	C and L; ScanSAR; offset tracking
Geohazards and infrastructure	Subsidence of cities, mines and oilfields; stability of dams, bridges, railways, pipelines	C, L, X; PSI, SBAS time series
Forestry and carbon	Deforestation alerts in cloudy tropics, forest biomass and height	L, P; cross-polarisation, tomography
Agriculture	Crop type mapping, crop growth, soil moisture, rice paddies (flooded fields)	C, L; time series of polarimetric backscatter
Hydrology	Soil moisture, wetlands, snow water equivalent, river and lake levels	L, C; Ka-band interferometry (SWOT)
Defence and security	All-weather imaging, change detection, moving targets, maritime surveillance	X (high resolution), all modes

EXAMPLE: MAPPING A FLOOD WITH SENTINEL-1

A major river floods during the monsoon, under persistent cloud.

1. Select a Sentinel-1 IW image acquired during the flood and a reference image from the same orbit before the flood.
2. Calibrate both to σ^0, apply terrain correction with the Copernicus DEM, and reduce speckle with a multi-temporal filter.
3. Open floodwater appears very dark (below about −18 dB in VV). Pixels that were bright before and dark now are classified as newly flooded; a threshold chosen automatically from the image histogram (Otsu's method) or a trained classifier separates water from land.
4. Flooded vegetation and flooded urban areas can appear *brighter* because of double bounce between water and trunks or walls; these are detected as increases in backscatter or drops in coherence.
5. Remove areas of radar shadow and permanent water using a height-above-nearest-drainage map and a reference water mask.

Free Sentinel-1 data are typically available within a few hours of acquisition. With these data and services such as the Copernicus Emergency Management Service and the Global Flood Monitoring system, flood maps can be produced the same day — regardless of clouds.

15.8 SAR Missions in 2026

15.8.1 Government Missions

TABLE 15.3 Major government SAR missions (selected).

Mission	Agency / country	Band	Launch	Key features
Sentinel-1A, 1C, 1D	EU Copernicus / ESA	C	2014, 2024, 2025	693 km SSO, 12-day repeat per satellite; IW mode 250 km at 5 × 20 m; systematic global acquisitions, free data
RADARSAT Constellation Mission	Canada (CSA)	C	2019	Three satellites; 4-day repeat; compact polarimetry; maritime surveillance
TerraSAR-X / TanDEM-X / PAZ	Germany (DLR, Airbus) / Spain	X	2007, 2010, 2018	High resolution; bistatic interferometry (WorldDEM)
COSMO-SkyMed (1st and 2nd gen.)	Italy (ASI, Defence)	X	2007–	Dual-use constellation; spotlight to 0.35 m class
ALOS-2 / ALOS-4	Japan (JAXA)	L	2014, 2024	PALSAR-3 on ALOS-4 images a 200 km swath at 3 m
SAOCOM 1A/1B	Argentina (CONAE)	L	2018, 2020	Full polarimetry; soil moisture; cooperation with Italy
Gaofen-3 series	China	C	2016–	Multi-mode, up to 1 m
EOS-04 (RISAT-1A)	India (ISRO)	C	2022	Agriculture, flood, forestry
Kondor-FKA	Russia	S	2023, 2024	Up to 1 m class
NISAR	NASA / ISRO	L + S	2025	12 m reflector, SweepSAR, 242 km swath, 12-day repeat; global systematic coverage, free data
BIOMASS	ESA	P	2025	First spaceborne P-band SAR; forest biomass and height by tomography
ROSE-L	EU Copernicus / ESA	L	≈ 2029	Copernicus Expansion mission
Harmony	ESA	C (receive-only)	≈ 2029	Two satellites flying with a Sentinel-1 for multi-static measurements

Two missions launched in 2025 stand out. **NISAR** is a partnership between NASA (L-band radar, 12 m deployable mesh reflector, data systems) and ISRO (S-band radar, spacecraft bus, GSLV launch). It is the first radar mission to map nearly all of the Earth's land and ice surfaces twice every 12 days at two frequencies. Launched on 30 July 2025, it began its science phase in November 2025; its free data — tens of terabytes per day — will measure ground deformation, ice-

sheet dynamics, forest biomass and agriculture globally. ESA's **BIOMASS**, launched on 29 April 2025, carries the first spaceborne P-band SAR. At 70 cm wavelength the radar penetrates to tree trunks, allowing the mass of forests — and hence their carbon content — to be estimated. Because P-band frequencies are shared with powerful military space-surveillance radars, BIOMASS is not permitted to operate over Europe and North America.

15.8.2 The Commercial SAR Revolution

Until 2018, all SAR satellites were large government systems weighing 1–3 tonnes. In January 2018 the Finnish start-up **ICEYE** launched ICEYE-X1, the first SAR satellite under 100 kg. Its innovations were a compact X-band phased array, high-power gallium-nitride amplifiers, batteries that supply short bursts of several kilowatts, and agile spacecraft that rotate to perform spotlight imaging. They showed that useful SAR imaging was possible from smallsats costing a few million dollars. By July 2026 ICEYE had launched 76 satellites, offered resolutions down to 16 cm with its fourth-generation design, and was producing about 50 satellites per year. Other commercial operators include **Capella Space** (USA, acquired in July 2025 by the quantum-computing company IonQ), **Umbra** (USA), **Synspective** (StriX, Japan), **iQPS** (QPS-SAR, Japan) and a growing number of Chinese commercial SAR companies (Figure 15.22).

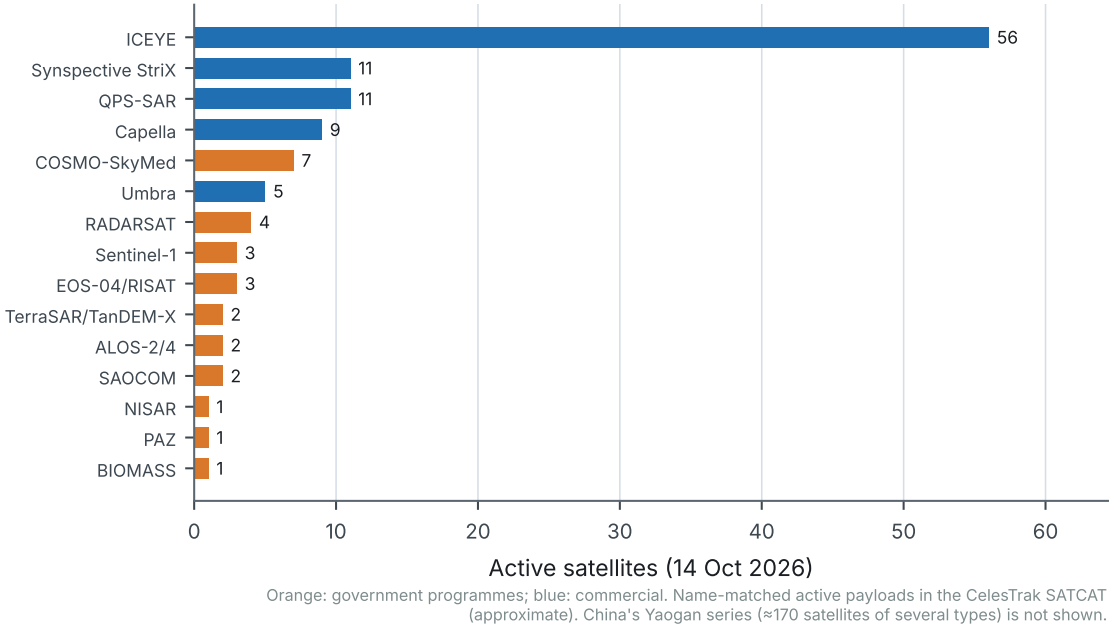


FIGURE 15.22 Active SAR satellites by operator or series in October 2026 (name-matched in the CelesTrak SATCAT, approximate).

Commercial SAR constellations offer frequent revisit (several images per day of a target), fast tasking and delivery, and very high resolution. They have found strong markets in defence and intelligence (the war in Ukraine was a turning point), insurance (rapid flood extent after events), maritime monitoring and infrastructure. Government missions provide the systematic, well-calibrated, free global coverage needed for science and public services; the two are increasingly complementary.

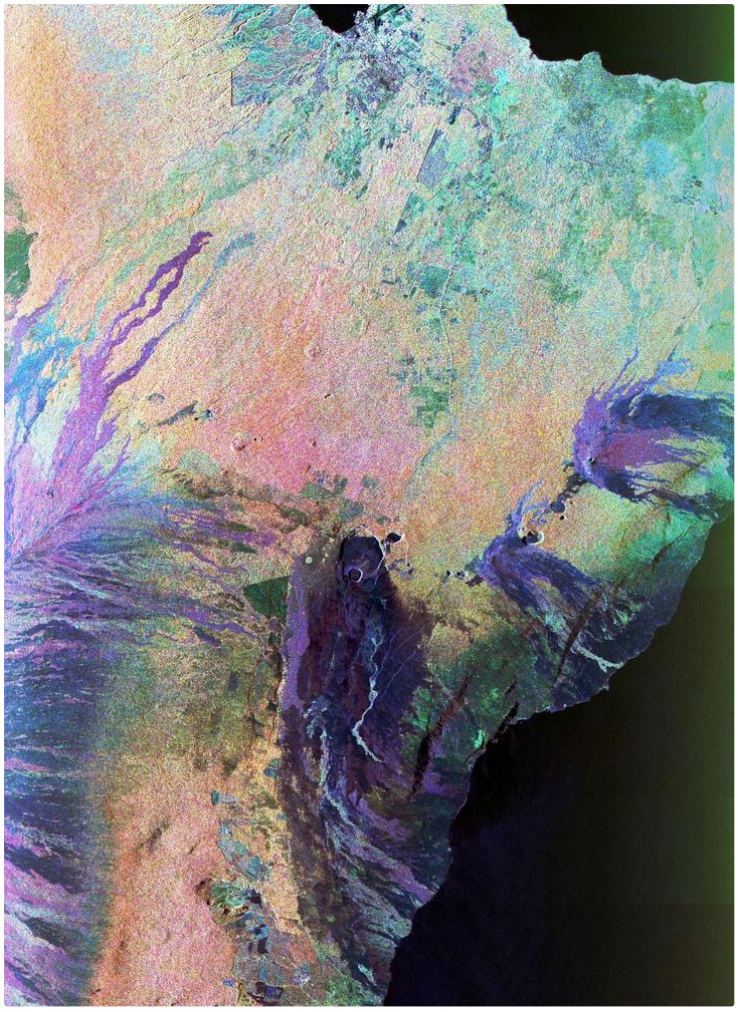


FIGURE 15.23 SIR-C/X-SAR multi-frequency image of the Kilauea volcano, Hawaii, in 1994. Colours combine L-band and C-band channels in different polarisations: rough young lava flows appear bright, smooth older surfaces darker. Image: NASA/JPL (NASA/JPL/NGA).

15.9 Design Example: An X-band SAR Smallsat

EXAMPLE: SIZING A 1 M-CLASS SAR MICROSATELLITE

Requirements: ≈ 1 m stripmap resolution, NESZ better than −15 dB, 550 km orbit, 30° incidence.

1. **Wavelength.** X-band (9.65 GHz, λ = 3.1 cm) allows a small antenna.
2. **Antenna.** A 3.2 m × 0.4 m phased array (1.28 m^2) exceeds the minimum area of about 1.2 m^2 for these conditions (Figure 15.7). Stripmap azimuth resolution ≈ L/2 = 1.6 m; spotlight mode can do better.
3. **Bandwidth.** 300 MHz gives 0.5 m slant-range and 1.0 m ground-range resolution at 30° incidence.
4. **Gain.** With 60 % aperture efficiency, $G = 4\pi A\eta/\lambda^2 \approx 10\,000$ (40 dBi).
5. **Sensitivity.** Assume 800 W average radiated power (4 kW peak at 20 % duty cycle), slant range 650 km, 3 dB noise figure and 3 dB losses. The radar equation for distributed targets then gives NESZ ≈ −15.6 dB — just meeting the requirement.
6. **Power.** 4 kW peak cannot come from a smallsat's solar arrays (a few hundred watts); a high-rate battery supplies imaging bursts of one to two minutes per orbit, recharged between collections.
7. **Data.** Sampling 300 MHz of bandwidth with 8-bit complex samples (reduced to about 4 bits by block-adaptive quantisation) produces a raw data rate of roughly 2–3 Gbit/s during imaging. A 20-second collection produces about 6 GB, downlinked over X-band or Ka-band.

The result — a satellite of about 100 kg with a deployable antenna — is close to the design of the first commercial SAR microsatellites.

AI AND SAR

SAR images are difficult for humans to interpret but well suited to machine learning. Machine learning is now used for ship and aircraft detection, sea-ice classification, despeckling, flood and building-damage mapping, and translating SAR into optical-like images. Deep learning also helps unwrap interferometric phase, separate atmospheric noise from deformation in InSAR time series, and detect anomalies in millions of persistent scatterers — for example, early warning of landslides or structural failures. Chapter 46 discusses these methods.

KEY TAKEAWAYS

- SAR provides its own microwave illumination and images day and night through clouds, smoke and rain.
- Range resolution comes from pulse bandwidth ($c/2B$) through pulse compression; azimuth resolution comes from aperture synthesis and is about half the antenna length, independent of range.
- The PRF constraints imply a minimum antenna area; resolution and swath trade off in stripmap, ScanSAR/TOPS and spotlight modes, and digital beamforming and SweepSAR relax the trade.
- Wavelength determines penetration and interferometric coherence: X-band for high resolution, C-band for all-round use, L- and P-band for forests and long-term interferometry.
- SAR brightness depends on roughness, geometry and moisture; images contain speckle and geometric distortions (foreshortening, layover, shadow).
- Interferometry measures topography (SRTM, TanDEM-X) and ground deformation with millimetre precision; persistent-scatterer time series monitor cities and infrastructure across whole continents.
- The SAR fleet now combines systematic government missions (Sentinel-1, NISAR, BIOMASS, ALOS-4) with fast-growing commercial constellations of 100 kg smallsats.

REVIEW QUESTIONS

1. Why can a SAR image through clouds, smoke and darkness while an optical imager cannot?
2. A SAR transmits a chirp of 300 MHz bandwidth. What is its slant-range resolution, and what is its ground-range resolution at an incidence angle of 35°?
3. A stripmap SAR has a 3.2 m long antenna. What is its approximate azimuth resolution, and why does it not depend on range?
4. Explain how the pulse repetition frequency constrains the minimum antenna area of a SAR.
5. Compare X-, C- and L-band SAR for forest mapping, sea-ice monitoring and high-resolution imaging of ports.
6. Describe foreshortening, layover and shadow, and explain what causes speckle.
7. In C-band differential interferometry ($\lambda = 5.55$ cm), how much line-of-sight ground motion does one full interferometric fringe represent?
8. How have commercial SAR constellations of 100 kg satellites changed what SAR can deliver, compared with large government missions?

CHAPTER

16 Weather and Environmental Satellites



Image: NASA/JSC

Every weather forecast you see depends on satellites. Geostationary satellites watch storms develop minute by minute over a third of the planet, while polar-orbiting satellites carry sounders that measure the temperature and humidity of the atmosphere layer by layer across the whole globe twice a day. Their observations are fed into numerical weather prediction models that now forecast as well five days ahead as they did three days ahead in the 1980s. This chapter explains the global satellite observing system, the instruments that make it work and the physics of atmospheric sounding, and then turns to environmental and climate monitoring — atmospheric composition and air quality, clouds, precipitation and radiation, oceans, sea level, sea ice and ice sheets — illustrated with real satellite records. It ends with the arrival of AI weather models.

IN THIS CHAPTER

16.1 Satellites and the Weather
16.2 Geostationary Weather Satellites
16.3 Polar-Orbiting Weather Satellites
16.4 From Observations to Forecasts
16.5 Atmospheric Composition and Air Quality
16.6 Clouds, Aerosols, Precipitation and the Energy Budget
16.7 Oceans
16.8 The Cryosphere and Land

16.1 Satellites and the Weather

On 1 April 1960, NASA's **TIROS-1** returned the first television picture of the Earth's clouds from orbit. Forecasters immediately saw storms and weather systems that had been invisible to the scattered network of ground stations, especially over the oceans. Within a few years, the **Nimbus**

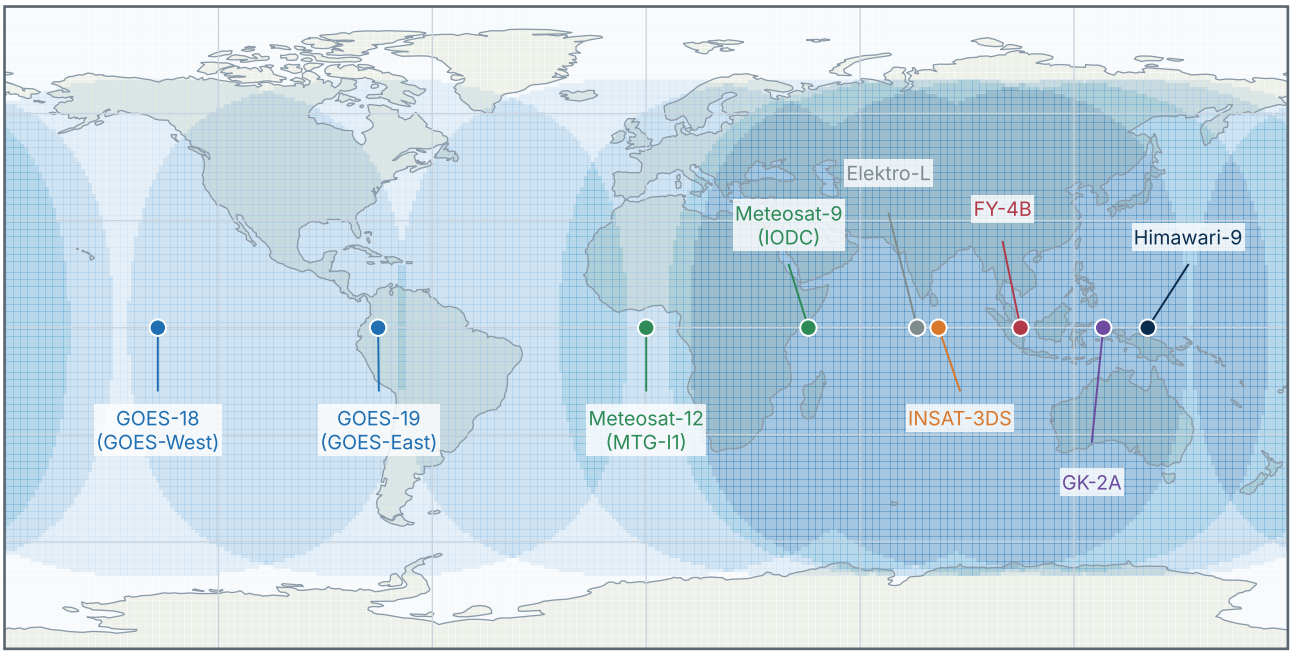
research satellites (from 1964) had demonstrated the first instruments to measure atmospheric temperature profiles from space. **ATS-1** (1966) then returned the first images of the full Earth disc from geostationary orbit. Operational geostationary satellites followed — the US **SMS/GOES** series (1974–75), Europe's **Meteosat** and Japan's **GMS** (both 1977) — and polar-orbiting satellites became the backbone of global forecasting.

Today, meteorological satellites are coordinated internationally through the World Meteorological Organization (WMO) and the **Coordination Group for Meteorological Satellites (CGMS)**. Their data are exchanged freely between nations under WMO resolutions — one of the most successful examples of international cooperation in space. More than 90 % of the observations assimilated into modern global weather forecasts come from satellites, and they account for most of the improvement in forecast skill over the past four decades.

16.1.1 The Space-Based Global Observing System

The WMO's baseline architecture combines two orbital regimes (Chapter 11):

- **Geostationary (GEO) satellites** at 35 786 km provide continuous observation of about a third of the Earth from a fixed position over the equator. A ring of five or six satellites covers the globe between about 60° N and 60° S (Figure 16.1). They are ideal for watching rapidly developing weather — thunderstorms, hurricanes, fog, fires, volcanic ash — and for deriving winds by tracking cloud and water-vapour features between successive images (**atmospheric motion vectors**). Their weakness is poor coverage of polar regions and the coarse resolution imposed by their great distance.
- **Low-Earth-orbit (LEO) polar satellites** in sun-synchronous orbits at 800–850 km cover the entire globe twice a day from close range, carrying the **sounders** that are the most important data source for numerical weather prediction. The WMO baseline calls for three orbital planes — early morning, mid-morning and afternoon — so that every place on Earth is observed at least every four hours (Figure 16.2). Europe (Metop) provides the mid-morning orbit, the USA (JPSS) the afternoon orbit, and China (FY-3E) has filled the previously empty early-morning orbit since 2021.



Approximate 2026 positions of primary meteorological geostationary satellites. Shading: number of satellites viewing each point within 70° Earth-central angle (the practical limit for quantitative imagery).

FIGURE 16.1 Primary meteorological geostationary satellites in 2026. Shading shows how many satellites view each point at a usable angle; overlap provides backup and stereo views of clouds.

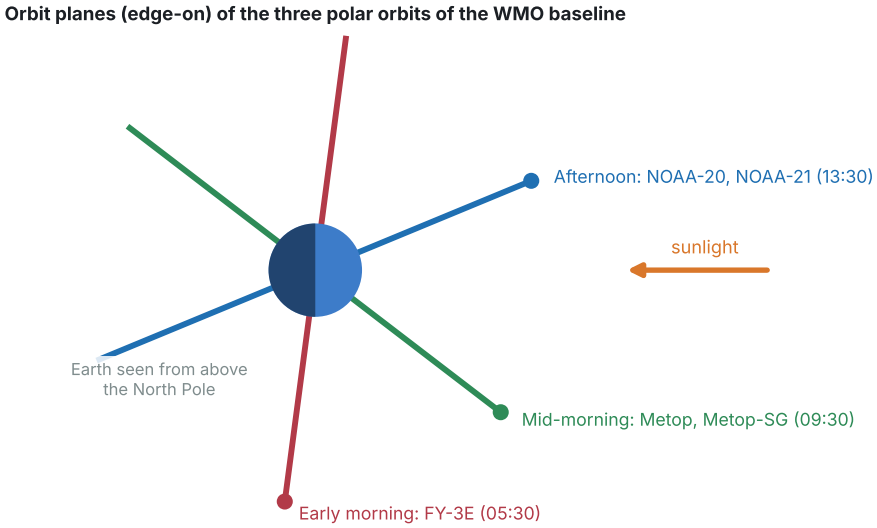


FIGURE 16.2 The three sun-synchronous orbit planes of the WMO polar-orbiting baseline, viewed from above the North Pole. Each plane keeps a fixed orientation relative to the Sun, so a satellite always crosses the equator at the same local solar times.

These are complemented by a growing number of other missions: **GNSS radio-occultation** constellations, **scatterometers** and **microwave imagers**, research missions with radars and lidars, and, increasingly, commercial providers selling weather data to governments.

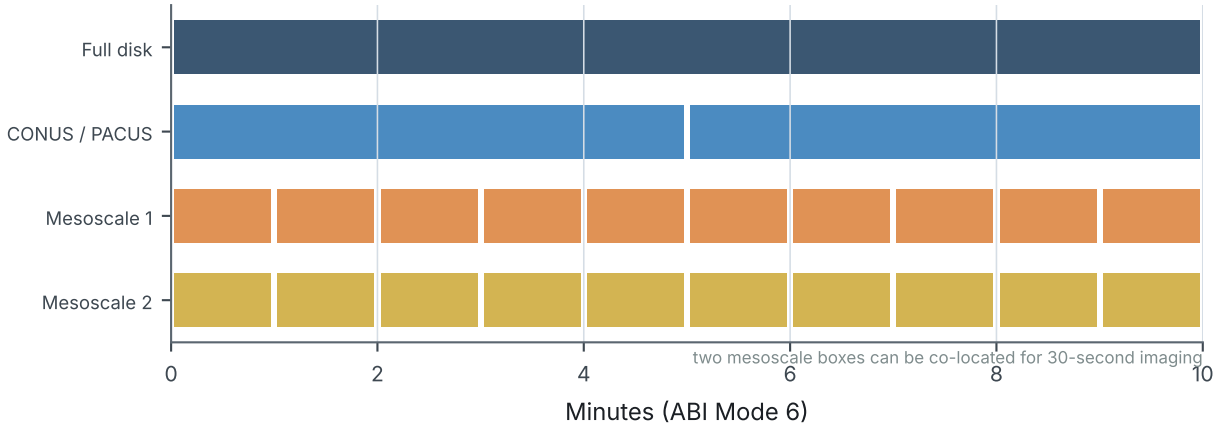
16.2 Geostationary Weather Satellites

16.2.1 The Current Generation

Since 2015 every geostationary weather operator has introduced a new generation of satellites with dramatically improved imagers (Table 16.1). The common design is represented by the US **Advanced Baseline Imager (ABI)**, Japan's **Advanced Himawari Imager (AHI)** and Europe's **Flexible Combined Imager (FCI)**. It scans the full disc in about 10 minutes in 16 spectral channels, with 0.5–1 km resolution in visible channels and 2 km in the infrared. At the same time it scans smaller regions every few minutes or even every 30 seconds (Figure 16.3).

TABLE 16.1 Operational geostationary meteorological satellites in 2026 (selected).

Operator	Series (current satellites)	Positions	Imager	Other instruments
USA (NOAA/NASA)	GOES-R: GOES-18 (West), GOES-19 (East)	137° W, 75° W	ABI, 16 channels	GLM lightning mapper; space weather sensors; CCOR-1 coronagraph on GOES-19
Europe (EUMETSAT/ESA)	Meteosat Third Generation: MTG-I1 (Meteosat-12), MTG-I2, MTG-S1; plus MSG	0°, 9.5° E, 45.5° E	FCI, 16 channels	LI lightning imager; IRS hyperspectral sounder and Sentinel-4 UVN on MTG-S1
Japan (JMA)	Himawari-9 (Himawari-8 backup)	140.7° E	AHI, 16 channels	Himawari-10 (≈ 2029) to add a hyperspectral sounder
China (CMA)	Fengyun-4A, 4B	≈ 86.5° E, 105° E	AGRI	GIIRS hyperspectral sounder; LMI lightning imager
South Korea (KMA)	GEO-KOMPSAT-2A	128.2° E	AMI, 16 channels	Space weather; GEMS air quality on GK-2B
India (IMD/ISRO)	INSAT-3D, 3DR, 3DS	74°–82° E	6-channel imager	19-channel infrared sounder
Russia (Roshydromet)	Elektro-L N2–N4	76° E, 165.8° E, etc.	MSU-GS	—



The scans are interleaved; each bar shows how often a complete image of that domain is produced.

FIGURE 16.3 Interleaved scanning of the GOES Advanced Baseline Imager in its standard mode: the full disc every 10 minutes, the continental USA every 5 minutes, and two 1 000 km mesoscale boxes every minute.

The GOES-R series illustrates the state of the art. Each satellite weighs about 5 200 kg at launch and carries the ABI, the **Geostationary Lightning Mapper (GLM)**, solar and space-environment instruments and a data-relay payload. ABI's 16 channels include two visible, four near-infrared and ten infrared bands; three water-vapour channels at 6.2, 6.9 and 7.3 µm map moisture at different heights, and the 3.9 µm channel detects fires and fog. GOES-16 was launched in 2016, GOES-17 in 2018, GOES-18 in 2022 and GOES-19 in June 2024. The ABI on GOES-17 suffered a cooling-system problem that degraded some infrared channels for several hours per night, and GOES-17 was replaced by GOES-18 in 2023. GOES-19 became the operational GOES-East satellite in April 2025. The follow-on **GeoXO** programme plans its first launch in 2032, adding a hyperspectral infrared sounder and an atmospheric-composition instrument.



FIGURE 16.4 GOES-U (now GOES-19) on its Falcon Heavy rocket before launch in June 2024. The satellite became the operational GOES-East in April 2025. Image: NASA/KSC (SpaceX).

16.2.2 Lightning from Orbit

Lightning mappers fly on the newest geostationary satellites: GLM on GOES, the **Lightning Imager (LI)** on MTG and the **Lightning Mapping Imager (LMI)** on Fengyun-4. They detect the brief optical flashes of lightning at the oxygen emission line of 777.4 nm, day and night. They use staring detector arrays that take about 500 frames per second, and on-board processing that subtracts the bright background of sunlit clouds. Rapid increases in total lightning ("lightning jumps") often precede severe weather such as hail and tornadoes, and lightning data help pilots and air-traffic controllers avoid storms over the oceans, where radar coverage is absent.

16.2.3 Geostationary Sounding

In 2016 China's **Fengyun-4A** carried the first hyperspectral infrared sounder in geostationary orbit (GIIRS). In July 2025 Europe launched **MTG-S1** with the **Infrared Sounder (IRS)**, which measures about 1 700 spectral channels over Europe every 30 minutes. Geostationary sounders track the evolution of temperature and humidity in the hours before storms form — precisely the information needed for forecasting severe convection. MTG-S1 also carries the Copernicus **Sentinel-4** ultraviolet–visible–near-infrared spectrometer for hourly air-quality monitoring over Europe.

16.3 Polar-Orbiting Weather Satellites

16.3.1 The Main Systems

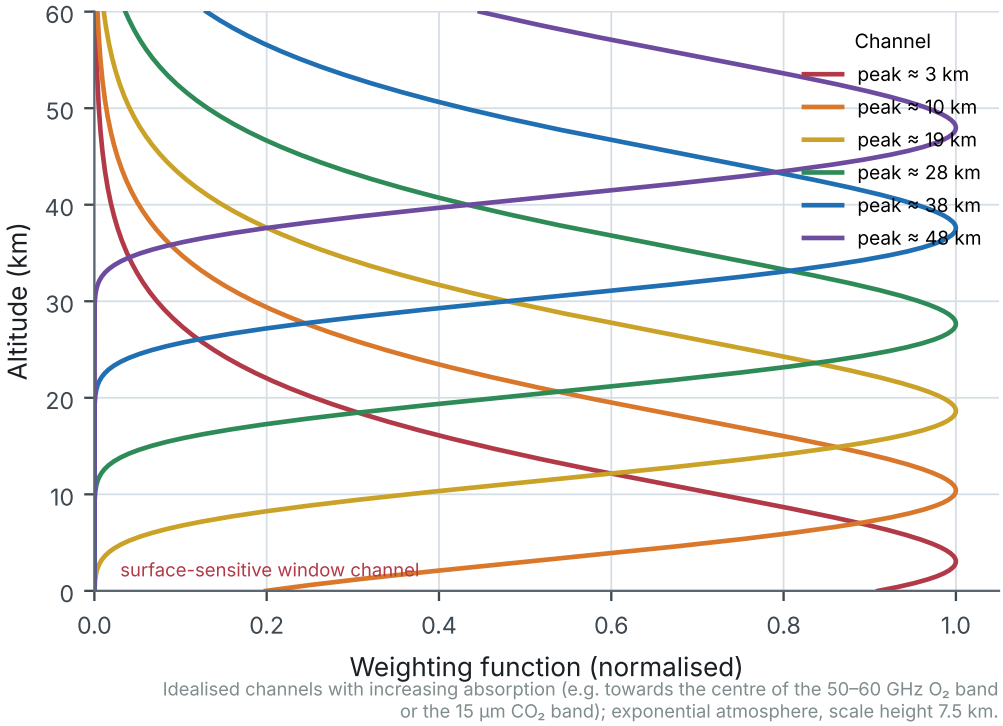
TABLE 16.2 Operational polar-orbiting meteorological satellites (selected).

System	Operator	Equator crossing	Satellites (2026)	Key instruments
JPSS	NOAA / NASA (USA)	13:30 (afternoon)	Suomi NPP (2011), NOAA-20 (2017), NOAA-21 (2022); JPSS-4 from 2027	VIIRS, CrIS, ATMS, OMPS, CERES
EPS / EPS-SG	EUMETSAT / ESA (Europe)	09:30 (mid-morning)	Metop-B (2012), Metop-C (2018), Metop-SG A1 (2025); Metop-SG B1 (late 2026)	IASI and IASI-NG, AMSU-A/MHS and MWS, ASCAT and SCA, METimage, radio occultation, Sentinel-5
Fengyun-3	CMA (China)	early morning, morning, afternoon	FY-3C to FY-3G (FY-3E in the 05:30 orbit, FY-3G for precipitation radar)	MERSI, HIRAS, MWTS, MWHS, GNOS
Meteor-M	Roshydromet (Russia)	morning	Meteor-M N2-2 to N2-4	MSU-MR, IKFS-2, MTVZA
DMSP	US Space Force	various	Last satellites near end of life	SSMIS microwave imager/sounder

The European **Metop Second Generation (EPS-SG)** programme illustrates the trend to specialisation. Instead of one large satellite, it uses two series flying together. **Metop-SG A** (sounding and imaging) carries METimage, IASI-NG, MWS, the 3MI multi-viewing polarimeter, radio occultation and the Copernicus Sentinel-5 atmospheric-composition spectrometer. **Metop-SG B** (microwave imaging and scatterometry) carries SCA, MWI, the ICI ice-cloud imager, radio occultation and Argos. The first A satellite was launched on 13 August 2025 on an Ariane 6.

16.3.2 Sounders: Measuring the Atmosphere Layer by Layer

The most valuable instruments for weather forecasting are **sounders**, which measure the vertical profiles of temperature and humidity. They exploit a simple idea (Figure 16.5). In a spectral band where an atmospheric gas of known concentration absorbs strongly, the radiation that reaches space comes only from high in the atmosphere, because radiation from below is absorbed. Where it absorbs weakly, radiation from lower down escapes. A sounder measures many channels from the centre to the edge of an absorption band, such as the 15 µm band of CO_2 or the 50–60 GHz band of oxygen. The concentrations of these gases are well known. In this way the sounder samples the temperature at a range of heights. Each channel's sensitivity to different heights is described by its **weighting function**. Channels in water-vapour bands (6.3 µm and 183 GHz) similarly sample humidity.



Idealised channels with increasing absorption (e.g. towards the centre of the 50–60 GHz O_2 band or the 15 µm CO_2 band); exponential atmosphere, scale height 7.5 km.

FIGURE 16.5 Idealised weighting functions of six sounding channels. Each channel senses a layer of the atmosphere a few kilometres thick; together they provide a temperature profile.

Infrared sounders offer fine vertical resolution. The hyperspectral sounders now flying are NASA's **AIRS** (2002, 2 378 channels), Europe's **IASI** (2006, 8 461 channels), the US **CrIS** (2011, 2 211 channels) and China's HIRAS. They measure the infrared spectrum in thousands of narrow channels and provide temperature profiles with about 1 K accuracy in 1 km layers. **IASI-NG** on Metop-SG doubles IASI's spectral resolution and halves its noise. Infrared sounders cannot see through clouds, however.

Microwave sounders (AMSU-A, MHS, ATMS, MWHS, and the new MWS) have coarser resolution but see through all but the thickest precipitating clouds. So they observe the atmosphere in exactly the stormy regions that matter most for forecasts. Figure 16.6 shows the absorption lines they use. Microwave sounders have consistently been found to have the largest single impact on global forecast skill.

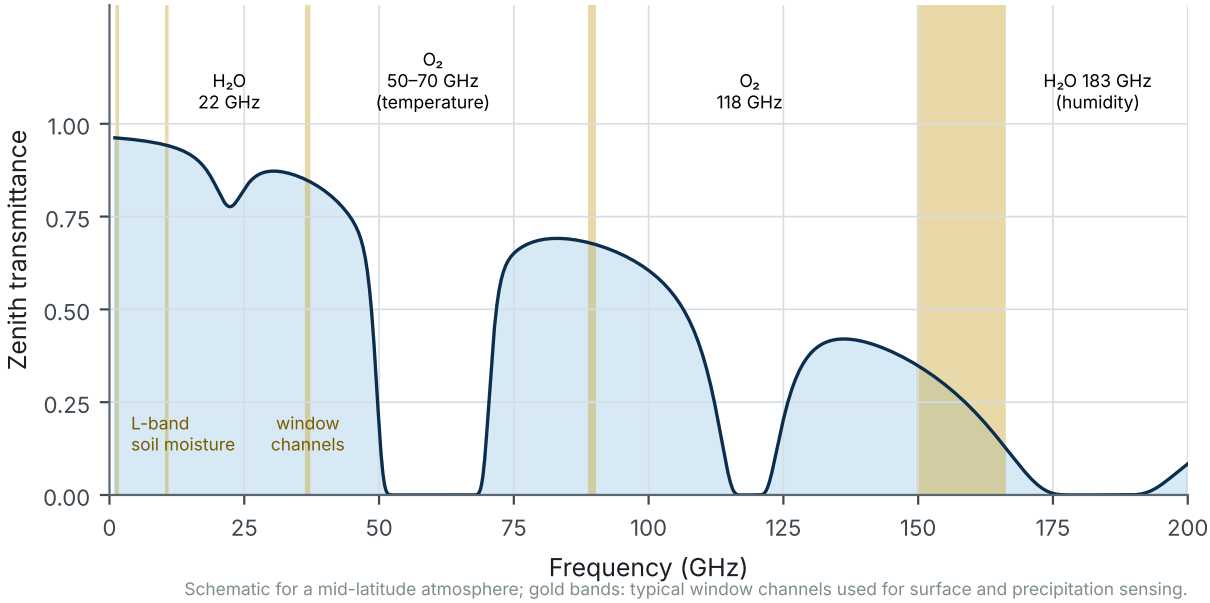


FIGURE 16.6 Schematic transmittance of the atmosphere at microwave frequencies. Sounders use the oxygen bands near 60 and 118 GHz for temperature and the water-vapour line at 183 GHz for humidity; window channels see the surface and precipitation.

Microwave imagers (SSMIS, AMSR2, GMI, MWI) measure emission from the surface and the atmosphere at window frequencies, from which they derive precipitation, total water vapour, cloud liquid water, sea-surface wind speed, sea ice and snow. The SSMIS instruments on the ageing US military DMSP satellites provided about half of the world's microwave imagery. The planned end of these satellites — and NOAA's decision in 2025 to stop processing their data publicly — raised concerns among hurricane forecasters about gaps in observations of storm structure. This also increased the importance of newer instruments and commercial microwave small satellites.

16.3.3 GNSS Radio Occultation

A remarkably powerful technique uses signals from navigation satellites (Figure 16.7). As a GPS, Galileo, GLONASS or BeiDou satellite sets or rises behind the Earth's limb as seen from a LEO satellite, its signals pass through the atmosphere and are bent by refraction. A receiver on the LEO satellite measures the extra delay with millimetre precision. From this, the **bending angle** — and hence the refractivity, temperature, pressure and humidity of the atmosphere — can be computed as a function of height. **Radio occultation (RO)** measurements are self-calibrating (based on time measurements traceable to atomic clocks), unaffected by clouds and highly precise in the upper troposphere and stratosphere. They are used to "anchor" the bias corrections of other satellite instruments.

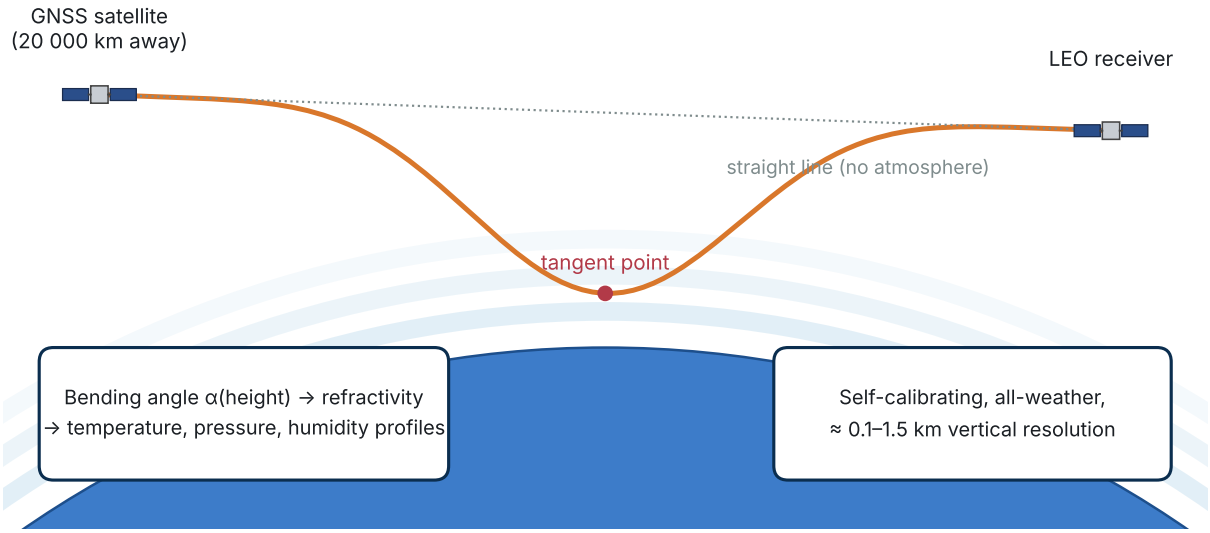


FIGURE 16.7 GNSS radio occultation. A signal from a navigation satellite is bent as it passes through the atmosphere on its way to a receiver in low Earth orbit.

The US–Taiwan **COSMIC-2** constellation (2019) provides about 5 000 tropical occultations per day, and Metop carries the GRAS receiver. Since about 2020, commercial companies such as **Spire** and **PlanetiQ** have sold radio-occultation data from constellations of small satellites to NOAA, EUMETSAT and other agencies — one of the first successful commercial weather-data markets. Spire also measures reflected GNSS signals (GNSS-R) for ocean winds and soil moisture.

16.3.4 Winds

Wind is the hardest weather variable to observe from space. Most wind information comes from **atmospheric motion vectors**, found by tracking clouds and water-vapour features in successive geostationary images. It also comes from **scatterometers** such as ASCAT on Metop. These measure ocean-surface wind speed and direction from the radar backscatter of wind-driven ripples observed from several directions. ESA's **Aeolus** (2018–2023) was the first satellite to carry a Doppler wind lidar, measuring wind profiles directly by the Doppler shift of laser light backscattered by molecules and aerosols. Despite technical difficulties with the laser, Aeolus clearly improved forecasts, particularly in the tropics, and an operational follow-on (EPS-Aeolus) is planned for the 2030s.

16.4 From Observations to Forecasts

16.4.1 Numerical Weather Prediction and Data Assimilation

A weather forecast is produced by a **numerical weather prediction (NWP)** model that solves the equations of fluid dynamics and thermodynamics for the atmosphere on a three-dimensional grid. For the global model of the European Centre for Medium-Range Weather Forecasts (ECMWF), this grid is today about 9 km horizontally and 137 levels vertically. Because the atmosphere is chaotic, a forecast can only be as good as its **initial conditions** — the best estimate of the current state of the whole atmosphere, called the **analysis**.

Producing the analysis is the job of **data assimilation** (Figure 16.8). Every 6 or 12 hours, tens of millions of observations — over 90 % of them from satellites — are compared with a short-range forecast (the "first guess"). Modern systems assimilate satellite **radiances** directly, rather than retrieved temperatures. A fast **radiative-transfer model** (such as RTTOV or CRTM) computes what each instrument channel would measure given the model atmosphere. An optimisation algorithm (**4D-Var**, ensemble Kalman filters or hybrids) then adjusts the model state to fit all the observations over a time window as well as possible, weighted by their expected errors. Instrument biases are corrected continuously within the system, using unbiased "anchor" observations such as radiosondes and GNSS radio occultation.

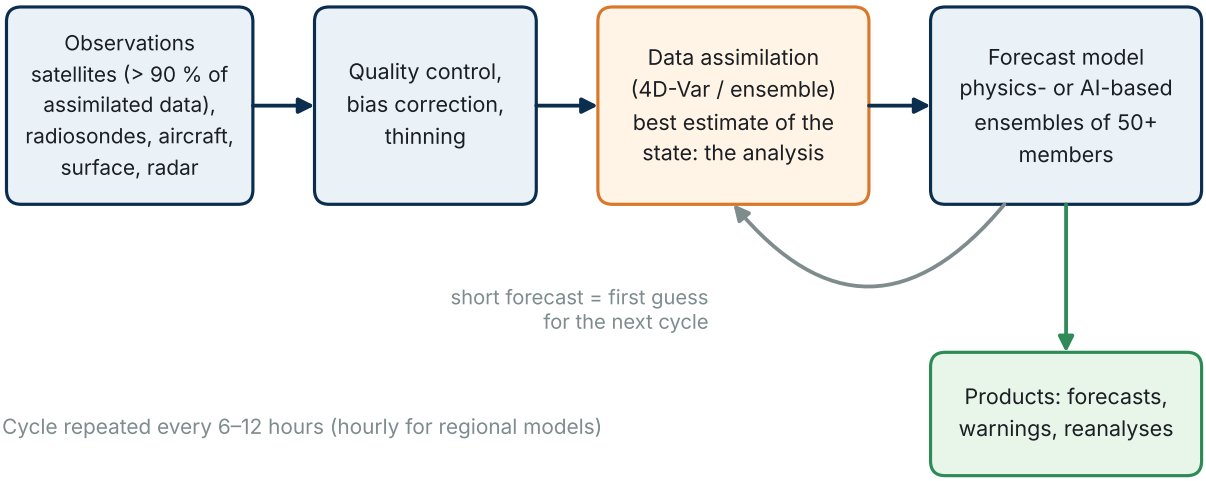
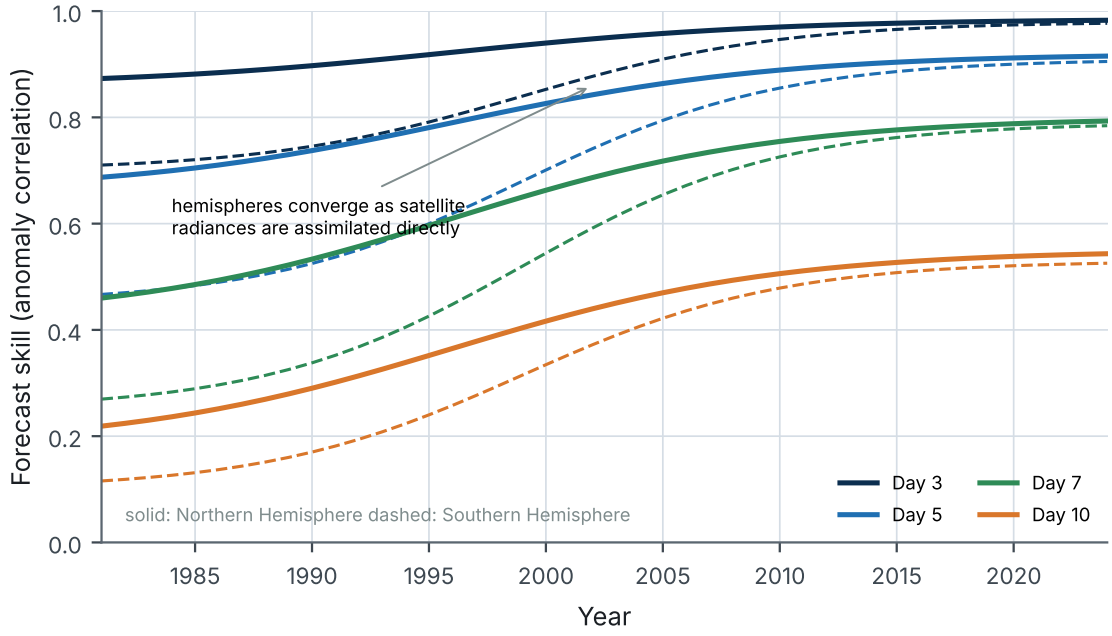


FIGURE 16.8 The data-assimilation and forecasting cycle used by operational weather centres.

The result has been called a "quiet revolution": forecast skill has improved by about one day per decade, so that a five-day forecast today is as accurate as a three-day forecast in the 1980s (Figure 16.9). One of the clearest signs of the satellites' contribution is the closing of the gap between the Northern and Southern Hemispheres around 2000. The Southern Hemisphere has few ground observations; once satellite radiances were assimilated directly, its forecasts became nearly as good as those of the well-observed north.



Schematic redrawn after published ECMWF 500 hPa height anomaly-correlation records (Bauer et al., 2015).

FIGURE 16.9 Schematic evolution of medium-range forecast skill. Skill has risen steadily, and the Southern Hemisphere caught up with the Northern Hemisphere as satellite data assimilation matured.

Studies measure each observing system's contribution, either by denying data in experiments or by adjoint-based "forecast sensitivity to observation impact" methods. They consistently find that microwave sounders, hyperspectral infrared sounders, GNSS radio occultation and atmospheric motion vectors are among the most valuable observations, together with conventional radiosondes and aircraft reports.

16.4.2 AI Weather Models

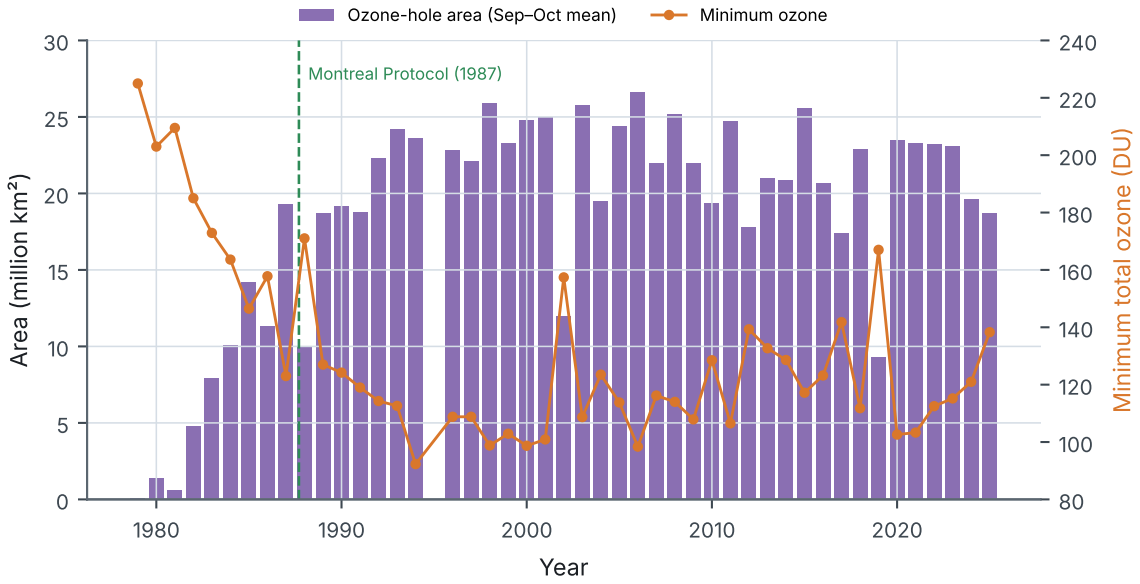
Machine-learning models can be trained on decades of ECMWF's **ERA5 reanalysis** — itself a product of assimilating satellite data. Since 2022, such models have matched or beaten physics-based models on many forecast scores, while running thousands of times faster. Notable models include NVIDIA's FourCastNet (2022), Huawei's Pangu-Weather and Google DeepMind's GraphCast (2023) and GenCast (2024). In February 2025 ECMWF made its own **Artificial Intelligence Forecasting System (AIFS)** operational. It was the first data-driven global model in an operational service at a major international centre, and was followed by an AI ensemble (AIFS ENS) in July 2025. These models still depend on conventional data assimilation for their initial conditions, and therefore on satellite observations; research models that learn to forecast directly from raw observations are under development. Part XIII discusses these advances in detail.

16.5 Atmospheric Composition and Air Quality

16.5.1 Ozone

Satellites played a central role in one of the great environmental success stories. The ozone layer, at 15–35 km, absorbs harmful solar ultraviolet radiation. After British Antarctic Survey scientists reported the **Antarctic ozone hole** in 1985 from ground measurements, NASA's Total Ozone

Mapping Spectrometer (TOMS) on Nimbus-7 revealed its full extent. The **Montreal Protocol** (1987) and its amendments phased out the chlorofluorocarbons responsible, and satellites have monitored the result ever since (Figure 16.10). The ozone hole still forms every southern spring but has begun to shrink; full recovery is expected around 2066 over the Antarctic.



Data: NASA Ozone Watch (TOMS, OMI, OMPS; gaps filled with MERRA-2). No satellite data in 1995.

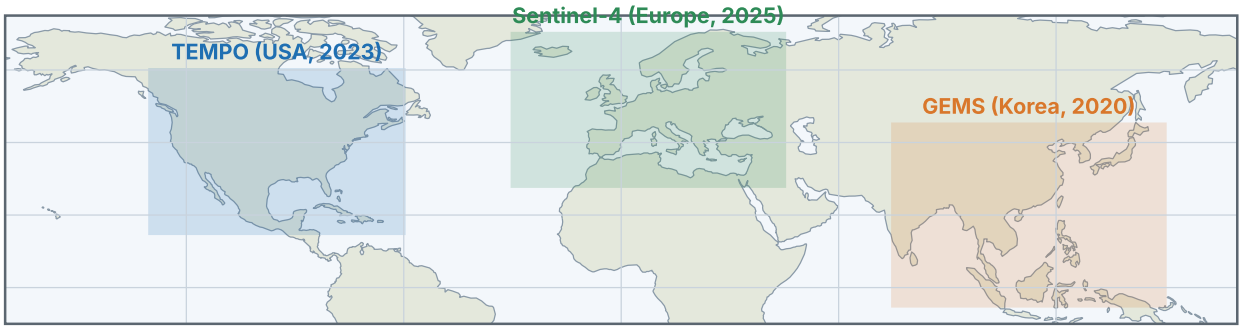
FIGURE 16.10 Size of the Antarctic ozone hole and minimum total ozone, 1979–2025, from satellite measurements.

FIGURE 16.11 The Antarctic ozone hole on 1 October 2010 seen by the Ozone Monitoring Instrument on NASA's Aura satellite. Blue and purple show the lowest ozone. Image: NASA/GSFC.

Ozone and other trace gases are measured by **ultraviolet–visible spectrometers** that observe sunlight scattered by the atmosphere and identify gases by their absorption features. Examples are TOMS (1978–2005), the Dutch–Finnish **OMI** on Aura (2004), **OMPS** on Suomi NPP and NOAA-20/21, and GOME-2 on Metop. Since 2017, **TROPOMI** on the Copernicus **Sentinel-5 Precursor** has mapped nitrogen dioxide, ozone, sulphur dioxide, formaldehyde, carbon monoxide and methane daily, with pixels of 5.5 × 3.5 km.

16.5.2 Air Quality from Geostationary Orbit

Air pollution changes hour by hour with traffic, industry and weather. A **geostationary air-quality constellation** now observes the world's three most populated northern regions every daylight hour (Figure 16.12). Korea's **GEMS** on GEO-KOMPSAT-2B (2020) covers Asia, and NASA's **TEMPO** (2023, hosted on a commercial communications satellite) covers North America. Copernicus **Sentinel-4** on MTG-S1 (2025) covers Europe and North Africa. These instruments map pollutants such as nitrogen dioxide at neighbourhood scales and show their daily cycles — rush-hour peaks, ship tracks, wildfire smoke — for the first time.



Approximate fields of regard of the geostationary air-quality constellation (hourly daytime NO_2, O_3, HCHO, aerosols).

FIGURE 16.12 Fields of regard of the geostationary air-quality constellation.

16.5.3 Greenhouse Gases

Measuring carbon dioxide from space is extremely demanding. Adding all of humanity's emissions for a year changes the atmospheric concentration by only about 0.5 %. A large power plant raises it locally by a fraction of a percent. Dedicated missions use high-resolution spectrometers in the near-infrared and short-wave infrared absorption bands of CO_2 (1.6 and 2.06 µm) and of oxygen (0.76 µm, to measure the total air column). They aim for a precision of better than 0.25 %. Missions include Japan's **GOSAT** (2009), **GOSAT-2** (2018) and **GOSAT-GW** (2025), NASA's **OCO-2** (2014) and **OCO-3** on the ISS (2019), China's TanSat (2016) and the French **MicroCarb** (2025). Europe's **CO2M** constellation, the first Copernicus Expansion mission, is designed to monitor anthropogenic CO_2 emissions at the scale of countries and large cities for verification of the Paris Agreement. Methane — whose super-emitters are mapped by imaging spectrometers (Chapter 14) — is measured globally by TROPOMI and GOSAT.

16.6 Clouds, Aerosols, Precipitation and the Energy Budget

16.6.1 Profiling Radars and Lidars

Passive sensors see mostly the tops of clouds. Active instruments reveal their vertical structure. From 2006 until 2023, NASA's **CloudSat** (94 GHz cloud radar) and the NASA–CNES **CALIPSO** (lidar) flew in the "A-Train" formation with Aqua and Aura, revolutionising understanding of clouds and aerosols. Their successor, the ESA–JAXA **EarthCARE** (launched May 2024), combines an atmospheric lidar (ATLID), the first **Doppler cloud-profiling radar** in space (provided by Japan), a multispectral imager and a broadband radiometer. Its aim is to understand how clouds and aerosols affect the Earth's energy balance — the largest uncertainty in climate projections.

16.6.2 Precipitation

The US–Japan **Tropical Rainfall Measuring Mission (TRMM, 1997–2015)** carried the first precipitation radar in space. Its successor, the **Global Precipitation Measurement (GPM)** Core Observatory (2014, Figure 16.13), carries a dual-frequency precipitation radar (Ku- and Ka-band) and a 13-channel microwave imager. GPM serves as a calibration reference for a constellation of about ten microwave imagers and sounders from several nations. Together with geostationary infrared data, they feed the **IMERG** product, which maps rainfall worldwide every 30 minutes on a 0.1° grid. Small satellites are joining: NASA's **TROPICS** CubeSats (2023) observe tropical cyclones with microwave sounders roughly every hour.



FIGURE 16.13 The GPM Core Observatory during testing at NASA Goddard. Its dual-frequency precipitation radar and microwave imager calibrate a whole constellation of precipitation-sensing satellites. Image: NASA/GSFC.

16.6.3 The Earth's Energy Budget

The climate is governed by the balance between absorbed sunlight and emitted thermal radiation. The **CERES** instruments, flying since 1997 on TRMM, Terra, Aqua, Suomi NPP and NOAA-20, measure reflected solar and emitted longwave radiation. Combined with ocean heat-content measurements, they show that the Earth's **energy imbalance** roughly doubled between 2005 and 2019, to about 1 W/m^2 — energy that is warming the oceans. Total solar irradiance has been measured from space since 1978, most recently by NASA's TSIS-1 on the ISS, showing that changes in the Sun are far too small to explain recent warming. Future missions such as the UK–ESA TRUTHS aim to set SI-traceable climate benchmarks (Chapter 13).

16.7 Oceans

16.7.1 Sea-Surface Temperature and Ocean Colour

Sea-surface temperature (SST) has been measured from space since the 1970s with infrared radiometers (AVHRR, and today VIIRS and Sentinel-3's dual-view SLSTR). SLSTR uses two viewing angles to correct for the atmosphere. Microwave radiometers (AMSR2) are also used, and these see through clouds. SST drives the weather and climate — it fuels hurricanes, defines El Niño, and is a key input to forecasts.

Ocean colour sensors measure the subtle colour of the sea to derive chlorophyll concentration, sediments and harmful algal blooms. Chlorophyll concentration shows the abundance of phytoplankton, which produce about half of the world's oxygen and form the base of the marine food web. After SeaWiFS (1997–2010), MODIS and Sentinel-3's OLCI, NASA's **PACE** mission (launched February 2024, Figure 16.14) carries the first hyperspectral ocean-colour instrument (OCI). OCI measures the full spectrum from ultraviolet to near-infrared in 5 nm steps, so that different types of phytoplankton can be distinguished. PACE also carries two polarimeters for aerosols and clouds.



FIGURE 16.14 NASA's PACE spacecraft being prepared for launch in December 2023. Its Ocean Color Instrument measures more than 200 wavelengths. Image: NASA/KSC (NASA/Kim Shiflett).

16.7.2 Satellite Altimetry and Sea Level

A radar **altimeter** measures the height of the sea surface by timing radar pulses reflected vertically from the ocean (Figure 16.15). The satellite's orbit must be known with centimetre accuracy, which is achieved with GNSS receivers, the French DORIS system and laser ranging. Corrections are also applied for the ionosphere, water vapour, sea state and tides. The sea-surface height can then be measured to about 2–3 cm along each ground track. Ocean currents, eddies and El Niño show up as bumps and dips of tens of centimetres in the sea surface.

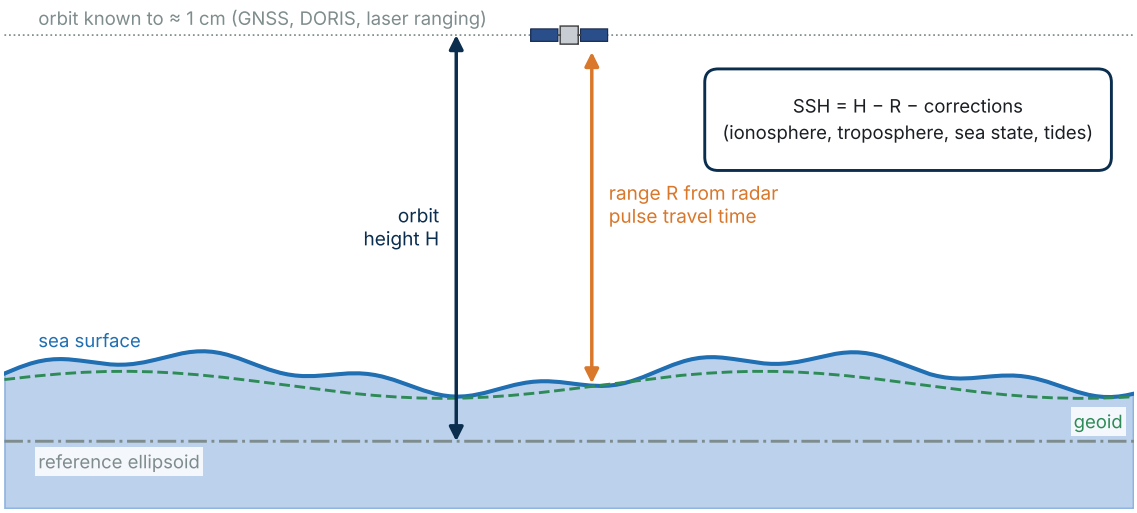
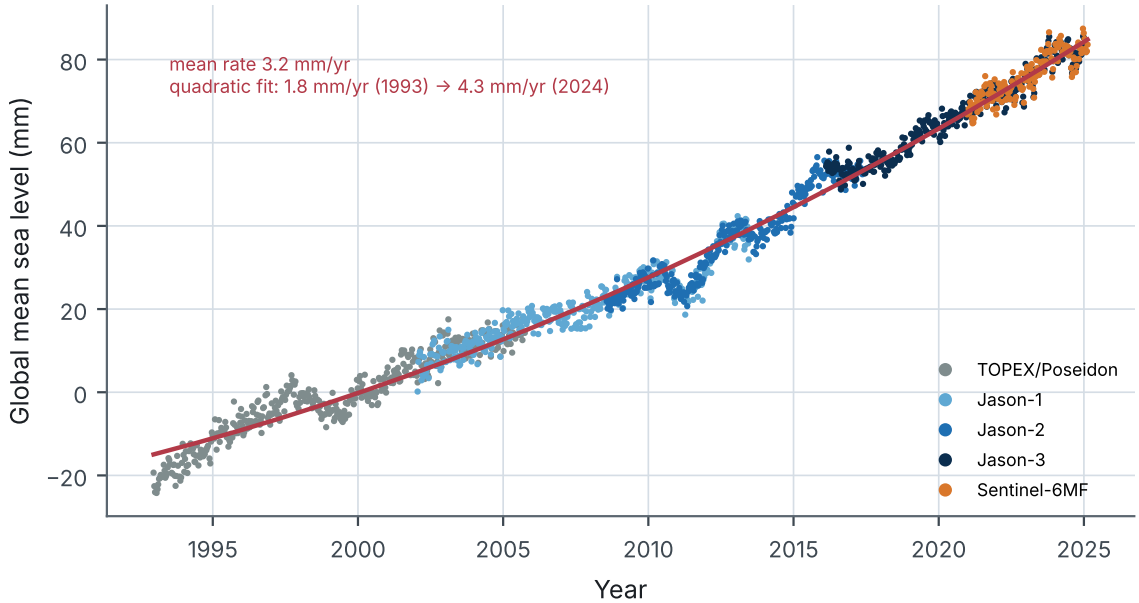


FIGURE 16.15 Principle of satellite radar altimetry. Sea-surface height is the orbit height minus the measured range, corrected for atmospheric and sea-state effects.

Since the US–French **TOPEX/Poseidon** mission (1992), a series of reference altimetry missions has measured global mean sea level with continuous, cross-calibrated precision (Figure 16.16). These are Jason-1 (2001), Jason-2 (2008), Jason-3 (2016), **Sentinel-6 Michael Freilich** (2020) and **Sentinel-6B** (launched November 2025). The record shows a rise of about 10 cm since 1993, at a rate that has roughly doubled, driven by ocean warming (thermal expansion) and by melting glaciers and ice sheets.



Altimetry data are provided by NOAA Laboratory for Satellite Altimetry (seasonal signals removed, no GIA correction).

FIGURE 16.16 Global mean sea level from the reference altimetry missions, 1993–2025, with a quadratic fit showing acceleration.

The NASA–CNES **SWOT** mission (2022) introduced a new technique: a Ka-band radar interferometer (KaRIn) with two antennas on a 10 m boom. It measures water-surface height across two 50 km swaths rather than along a single line. SWOT resolves ocean eddies ten times smaller than before and measures the height and slope of rivers and lakes worldwide — the first global survey of fresh water from space (Figure 16.17).

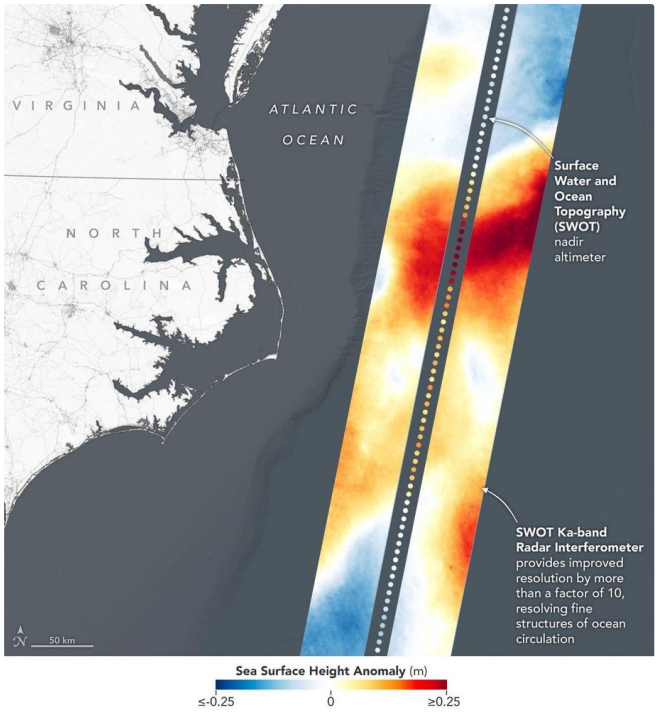
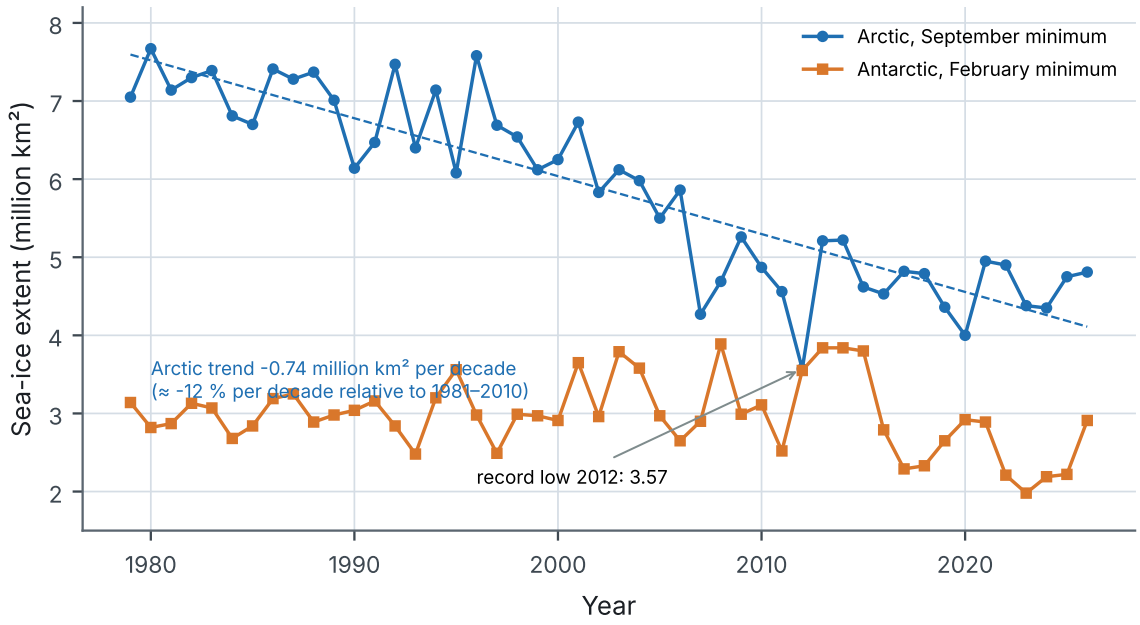


FIGURE 16.17 SWOT's first-light sea-surface height measurements off the US East Coast (2023), showing fine-scale ocean features within the wide swath compared with conventional altimetry. Image: NASA/JPL (NASA/JPL-Caltech).

16.8 The Cryosphere and Land

16.8.1 Sea Ice and Ice Sheets

Passive microwave radiometers have mapped polar sea ice every day since 1978, regardless of clouds and polar darkness, because ice and open water emit microwaves very differently. The record shows a dramatic decline in Arctic sea ice, especially at the end of summer (Figure 16.18): the September minimum extent has fallen by about 12 % per decade, with a record low in 2012. Antarctic sea ice changed little until 2016, then fell sharply, with record lows in 2023.



Data: NSIDC Sea Ice Index (passive microwave, SMMR, SSM/I, SSMIS, AMSR2).

FIGURE 16.18 Minimum sea-ice extent in the Arctic (September) and Antarctic (February) from the passive-microwave record, 1979–2026.

Radar altimeters (ESA's **CryoSat-2**, 2010) and laser altimeters (NASA's **ICESat-2**, 2018) measure the thickness of sea ice and the changing height of ice sheets and glaciers. The US–German **GRACE** (2002–2017) and **GRACE Follow-On** (2018) missions measure tiny changes in the Earth's gravity field to "weigh" ice sheets and groundwater. They do this by tracking the distance between two satellites 220 km apart to within a fraction of a micrometre, with microwave and laser ranging. They show Greenland losing about 270 billion tonnes of ice per year and Antarctica about 150 billion tonnes, and reveal groundwater depletion in regions such as northern India and California's Central Valley.

16.8.2 Soil Moisture and Land

ESA's **SMOS** (2009) and NASA's **SMAP** (2015) measure soil moisture and ocean salinity with L-band (1.4 GHz) radiometers, exploiting a protected frequency band where the emission of the surface depends strongly on its water content. SMOS uses a Y-shaped interferometric array of 69 small antennas to synthesise a large aperture; SMAP uses a 6 m rotating mesh antenna. Land-surface monitoring by optical, thermal and radar missions is described in Chapters 13–15.

DATA BOX: ESSENTIAL CLIMATE VARIABLES FROM SPACE

Of the 55 Essential Climate Variables defined by the Global Climate Observing System, more than half can only be observed globally from space. Long, stable, cross-calibrated satellite records are a foundation of the assessments of the Intergovernmental Panel on Climate Change (IPCC). Examples are those produced by ESA's Climate Change Initiative, the EUMETSAT Climate Monitoring Satellite Application Facility and NOAA's climate data records. Producing them requires overlap between successive satellites, careful recalibration of old instruments and the reprocessing of decades of data.

EXAMPLE: WHY A SOUNDER CHANNEL "SEES" A PARTICULAR HEIGHT

Consider a channel in which the absorption optical depth of the whole atmosphere is $\tau_0 = 40$, in an atmosphere where the absorbing gas density falls with a scale height H = 7.5 km. The optical depth above height z is $\tau(z) = \tau_0 e^{-z/H}$, and the radiation from height z reaching space is attenuated by $e^{-\tau(z)}$. The weighting function $W(z) = d(e^{-\tau(z)})/dz$ peaks where $\tau(z) = 1$:

$$z_{peak} = H \ln \tau_0 = 7.5 \times \ln 40 \approx 28 \text{ km}$$

A channel with $\tau_0 = 4$ peaks at about 10 km; one with $\tau_0 = 600$ at about 48 km (Figure 16.5). The weighting functions are each about 10–15 km wide, which is why sounders need many overlapping channels and why the retrieved profiles are smooth. Hyperspectral sounders with thousands of channels sharpen the vertical resolution considerably.

AI IN WEATHER AND ENVIRONMENTAL MONITORING

Artificial intelligence is transforming every stage of the chain. Uses include on-board cloud detection and data compression, fast neural-network radiative-transfer models for data assimilation, and nowcasting of thunderstorms from geostationary imagery. Others are retrieval of precipitation and atmospheric composition, and global AI forecast models such as ECMWF's AIFS. Chapter 47 discusses AI for weather, climate and Earth-system science in depth.

KEY TAKEAWAYS

- Weather satellites are coordinated globally through WMO and CGMS; satellites supply more than 90 % of the observations used in global forecasts.
- Geostationary imagers (ABI, AHI, FCI) scan the full disc every 10 minutes in 16 channels and small regions every 30–60 seconds; lightning mappers and geostationary sounders are the newest additions.
- Polar orbiters in early-morning, mid-morning and afternoon orbits carry infrared and microwave sounders, the most valuable data for numerical weather prediction; GNSS radio occultation provides unbiased anchor measurements.
- Sounding works because channels with different absorption strengths sense different heights (weighting functions).
- Data assimilation combines observations with models; forecast skill has improved by about a day per decade, and AI forecast models became operational in 2025.
- Satellites monitor the ozone layer, air quality (now hourly from geostationary orbit), greenhouse gases, clouds, precipitation and the Earth's energy budget.
- Altimetry shows global sea level rising about 10 cm since 1993 at an accelerating rate. Passive microwave records show Arctic summer sea ice declining about 12 % per decade, and gravity missions weigh the ice sheets' losses.

REVIEW QUESTIONS

1. Why are both geostationary and polar-orbiting satellites needed for weather forecasting? What does each do best?
2. Explain how infrared and microwave sounders measure temperature at different heights using weighting functions.
3. Why are GNSS radio-occultation measurements described as "unbiased", and why are they valuable for anchoring other observations?
4. What is data assimilation? Why has forecast skill improved by about one day per decade?

5. How do AI weather-forecast models differ from physics-based numerical models, and what are their present limitations?
6. Why is air-quality monitoring from geostationary orbit an important advance over polar-orbiting measurements?
7. Global mean sea level has risen about 10 cm since 1993. What is the average rate, and why is the present rate higher than the average?
8. Which satellite techniques measure sea ice, ice-sheet mass and soil moisture? Give one mission for each.